Characterization of Hadronic Shower in the Belle II Electromagnetic Calorimeter

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Abstract

The reconstruction of neutral hadrons and the measurement of their associated quantities represent one of the most complex challenges in HEP experiments. This is particularly true for experiments lacking a dedicated hadronic calorimeter, which must rely solely on the electromagnetic calorimeter for neutral hadron detection, as is the case in Belle II. Anti-neutrons with GeV-scale energies favor annihilation processes in matter, producing multiple final-state particles that escape the detector volume. Since anti-neutrons are copiously produced in contemporary physics processes -from hyperon decays to cosmological dark matter studies- their accurate reconstruction would enable a more complete analysis of diverse fundamental processes. The PRIN2022 MANTRA project develops an algorithm for direct anti-neutron energy measurement up to 10 GeV, thereby improving their identification. This approach requires detailed integration of information from surrounding detectors, overcoming limitations imposed by the lack of a hadronic calorimeter.
Furthermore, the GEANT4 physical models simulation of anti-neutron hadronic interactions -and especially the annihilation process- rely on isospin symmetry with anti-protons. This may potentially introduce discrepancies between Monte Carlo simulations and experimental reality.

This thesis, developed within the PRIN2022 MANTRA project, presents a study on the accuracy of Belle II software simulation of anti-neutron hadronic interactions. To this end, the channel $e^+e^- \to p\pi^-\bar{n}$ is analyzed, measuring anti-neutron associated quantities through its calorimeter clusters. The anti-neutron is identified via a semi-exclusive technique based on the recoil mass of the other particles, with or without Initial State Radiation.
The first part analysis on a pure signal Monte Carlo production demonstrates that recoil mass correctly reconstructs anti-neutron kinematics. However, its momentum is not properly reconstructed from ECL clusters, mainly due to fragmentation of the primary annihilation cluster. This also affects shower shape variables, which exhibit structures typical of electromagnetic showers.
Then, the full $q\bar{q}$ cocktail Monte Carlo production, corresponding to an integrated luminosity of $1429.23 fb^{-1}$ of data, is analyzed. Using a performance metrics based approach, a more stringent selection on reconstructed anti-neutrons is obtained via the figure of merit $\frac{S}{\sqrt{S+B}}$. Additionally, the sidebands technique on recoil mass extracts shower shape variables in the signal region, subtracting residual background due to other $q\bar{q}$ channels.
Finally, Data/MC agreement studies are performed on $\bar{n}$ candidate cluster variables.

Abstract

La ricostruzione degli adroni neutri e la misurazione delle relative grandezze rappresentano una delle sfide più complesse negli esperimenti di fisica delle alte energie. Questo è particolarmente vero per esperimenti sprovvisti di calorimetro adronico dedicato, che sfruttano il calorimetro elettromagnetico come unico strumento per rilevare determinati adroni neutri, come nel caso di Belle II. Gli anti-neutroni con energia dell'ordine del GeV prediligono processi di annichilazione nella materia, producendo particelle multiple nello stato finale che sfuggono al volume del rivelatore. Poiché l'anti-neutrone è prodotto in numerosi processi di fisica contemporanea -dai decadimenti degli iperoni allo studio della materia oscura su scala cosmologica- una sua ricostruzione accurata consentirebbe l'analisi completa di diversi processi fondamentali. Il progetto PRIN2022 MANTRA sviluppa un algoritmo per misurare direttamente l'energia degli anti-neutroni fino a 10 GeV, migliorandone l'identificazione. Tale approccio richiede l'integrazione dettagliata delle informazioni dai rivelatori circostanti il calorimetro elettromagnetico, superando le limitazioni dovute all'assenza di un calorimetro adronico.

Inoltre, i modelli fisici alla base della simulazione GEANT4 dell'interazione adronica degli anti-neutroni, e soprattutto del processo di annichilazione, si basano sulla simmetria di isospin con gli anti-protoni. Questo potrebbe potenzialmente comportare una discrepanza tra i modelli utilizzati dalle simulazioni Monte Carlo e la realtà.

Questa tesi, che si inserisce all'interno del progetto PRIN2022 MANTRA, presenta uno studio sulla correttezza della simulazione dell'interazione adronica degli anti-neutroni da parte del software di Belle II. A tal fine, si studia il canale $e^+e^- \to p\pi^-\bar{n}$, misurando le grandezze associate all'anti-neutrone tramite i cluster da esso prodotti nel calorimetro. L'anti-neutrone viene identificato con una tecnica semi-esclusiva basata sul recoil mass delle altre particelle, considerando o meno l'emissione di Initial State Radiation.

La prima parte di analisi svolta su una produzione Monte Carlo di solo segnale dimostra come il recoil descriva propriamente la cinematica dell'anti-neutrone, evidenziando, tuttavia, come il momento di quest'ultimo non sia correttamente ricostruito a partire dalla misura ottenuta dal calorimetro, principalmente a causa dalla frammentazione del cluster primario prodotto dall'annichilazione dell'anti-neutrone. Questo si riversa anche sulle grandezze di sciame ricostruite, che presentano diverse strutture tipiche degli sciami elettromagnetici.

Dopodiché si studia l'intera produzione Monte Carlo del cocktail $q\bar{q}$, corrispondente a una luminosità integrata di $1429.23 fb^{-1}$. Utilizzando delle metriche di performance, si decreta un selezione più stringente sull'anti-neutrone ricostruito, basato sull'ottimizzazione della figura di merito $\frac{S}{\sqrt{S+B}}$.

Inoltre, viene impiegata la tecnica delle sidebands sulla recoil mass, per ottenere le grandezze di sciame relative alla regione di segnale, annullando in buona misura il contributo residuo proveniente del background prodotto dagli altri canali in $q\bar{q}$.

Si conclude svolgendo un confronto dati e Monte Carlo per quanto riguarda le variabili di sciame ricostruite per i candidati anti-neutroni.

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Chapters subdivision

Introduction: the Chapter 1 describes the properties of the anti-neutron and its detection techniques in the high-energy physics (HEP) context. Its primary interactions -such as annihilation and hadronic shower production- are then exploited, discussing some hadronic models very quickly. The MANTRA (Measuring Anti-Neutron: Tagging and Reconstruction Algorithm) project is then described as a method to measure anti-neutron energies up to 10 GeV using an electromagnetic calorimeter.

The Belle II experiment: the Chapter 2 provides an overview of the SuperKEKB accelerator and Belle II detector. In the context of $\bar{n}$ studies, the Electromagnetic Calorimeter (ECL) plays a key role in hadron detection; a detailed discussion about the ECL is thus provided, covering the description of its inorganic crystals and the explanation of the clustering algorithm. The ECL performance are then discussed.

The Belle II Analysis Software Framework: the Chapter 3 introduces the the Belle II Analysis Software Framework (basf2), providing an overview of its operation method and its role in the Monte Carlo simulation and reconstruction processes at Belle II experiment. Then, the Topology Analysis (TopoAna) independent software is described, concluding with the introduction of the Belle II Grid.

The ECL reconstruction: the Chapter 4 provides a detailed description of the ECL clustering algorithm and analyzes ECL energy performance, followed by the definition of shower cluster variables.

Study of the Monte Carlo signal channel: the Chapter 5 is devoted to the study of the kinematic variables of anti-neutrons $\bar{n}$, including momentum, angular distributions and related observables. The analysis follows a semi-exclusive approach over the selected channel $e^+e^- \rightarrow p\pi^-\bar{n}$, studying both cases with and without Initial State Radiation (ISR) reconstruction, where the recoil system is made of $p\pi^-(\gamma_{ISR})$. Therefore, the recoil system variables are compared to those of the anti-neutrons candidates, studying the response of a Kinematic Fit as well.
Cluster split-off and pre-showering effects are investigated studying anti-neutrons relatives chain. Several neutral cluster variables associated to anti-neutron candidates are then described and studied for both ISR and NO ISR cases, including the analysis of their relatives contributions.

Study of the Monte Carlo cocktail production: the Chapter 6 introduces the study of the channel of interest within the entire $q\bar{q}$ cocktail production.

First, the possible Belle II processes are introduced, along with the typical cross sections that characterize them, and their contributions to the recoil mass distribution are discussed. Further selections from the previous analysis are then discussed, including a stricter selection on the reconstructed $\bar{n}$ direction based on a Performance Metric figure-of-merit study. Subsequently, the sideband method is applied to the reconstructed $\bar{n}$ cluster variables, evaluating the feasibility of subtracting the background contribution from the signal.

Data - Monte Carlo comparison: the Chapter 7 finalizes the analysis by performing a Data/MC agreement between the cluster variables obtained from the $q\bar{q}$ cocktail and those from real data.

Chapter 1
Introduction

The Standard Model is one of the greatest achievements of modern science, especially in the field of high-energy particle physics. Its reliability is the result of a long process that began in the 20th century and received its definitive confirmation in 2012 with the discovery of the Higgs boson.
However, there are some topics which remain unsolved, like the non-trivial neutrino mass or the presence of dark matter on cosmological scales. The Standard Model still represents an excellent phenomenological description of subatomic processes, up to the TeV energy scale.
In this framework, the Belle II experiment, located at the SuperKEKB facility, plays a fundamental role in exploring scenarios that go beyond the standard physics, thus entering the so-called *new physics*.
In parallel with experiments that maximize the energy, the aim of Belle II is to explore the intensity frontier, investigating scenarios of high instantaneous luminosity up to $8 \times 10^{35} cm^{-2} s^{-1}$, around (and not limited to) the $\Upsilon(4S)$ resonance energy.
This approach enables flavour physics studies of heavy mesons such as B mesons, searching for very rare processes and facilitating the identification of potential discrepancies with Standard Model predictions and opening the door to Beyond Standard Model physics.

1.1 The B Factories

Since the end of the XX century, two asymmetric-energy e^+e^- B factories - KEKB for the Belle experiment at KEK and PEP-II for the BaBar experiment at SLAC- have achieved huge success, confirming the Standard Model in the quark flavour sector, establishing the presence of CP violation in the B meson system through the decay process $B^0(\bar{B}^0) \rightarrow J/\Psi K_S^0$ [1, 2, 3, 4]. This result fixed the main target of the present B factories. Experimental results confirm that the Kobayashi–Maskawa mechanism, a key component of the Standard Model of elementary particles, is the dominant source of CP violation observed in nature.
The Belle experiment proved its ability to measure a number of decay modes of the B meson and to extract Cabibbo-Kobayashi-Maskawa (CKM) matrix elements and other interesting observables. Precise measurements and new observations from Belle have overconstrained the quark flavor sector of the Standard Model. This sector remains self-consistent within the current experimental precision.
However, several fundamental questions persist in the quark and lepton flavor sectors; the Standard Model requires numerous parameters, including quark and lepton masses and mixing angles that must be determined experimentally.
Furthermore, from a cosmological perspective, the matter-antimatter asymmetry of the Universe poses a great challenge. While the CP violation constitutes

one of the necessary conditions for the evolution of a matter-dominated Universe [5], its magnitude cannot be explained solely by the CP violation within the Standard Model, which originates from quark mixing.
Exploring such scenarios requires substantial luminosity at present B factories. The SuperKEKB B Factory has been designed to investigate the physics of B mesons processes. Due to the very large production cross section of b-quarks in hadron collisions, certain measurements such as $B_s \to \mu\mu$ can be performed more effectively at hadron colliders, such as LHC. However, e^+e^- machines provide a much cleaner environment, essential for precision observables involving photons, π^0's, K_L^0's, or neutrinos in the final state.
At the $\Upsilon(4S)$ resonance, $B\bar{B}$ pairs are produced near threshold without associated particles. By fully reconstructing the four-momentum of one B ($\bar{B}$) meson from its decay products (full reconstruction technique), the missing momentum of the other $\bar{B}$ (B) meson decay can be inferred. This technique is essential for measuring channels with neutrinos in the final state. Therefore, SuperKEKB undoubtedly complements hadron collider experiments from both theoretical motivation and experimental perspectives.

1.2 Anti-neutron in HEP physics

The anti-neutron $\bar{n}$ ($\bar{u}\bar{d}\bar{d}$) is the anti-particle of the neutron n (udd), neutral hadron of mass 0.939565 GeV/c^2 and spin $\frac{1}{2}$, with magnetic moment of $1.91\mu_N$, where μ_N is the nuclear magneton. Its existence was expected by the principle of invariance under charge conjugation and experimentally proved in 1956 [6], one year after the anti-proton $\bar{p}$ [7].
The anti-neutron plays a key role in several physics measurements, both in collider physics and in astrophysical contexts. Since it is electrically neutral, it cannot be easily observed directly, allowing its detection via the products of its annihilation with ordinary matter.
For example, the measurement of the neutron electromagnetic form factor in the time-like region is strongly affected by the performance of the $\bar{n}$ detection. The BESIII experiment has shown [8] an oscillating behaviour in the electromagnetic structure of the neutron in the study of the $e^+e^- \to n\bar{n}$ annihilation process. Direct detection of the $\bar{n}$ would further refine this analysis by enabling a more proper identification of the final state.
Another example, where an improved direct measurement of the $\bar{n}$ would significantly enhance the studies, is the analysis of hyperon decay channels at B factories (and beyond). Some possible decay channels where the $\bar{n}$ is the only unreconstructed particle in the final state are shown below:

$$\bar{\Lambda}^0 \to \pi^0 + \bar{n}, \qquad \bar{\Sigma}^- \to \pi^- + \bar{n}, \qquad \bar{\Lambda}_c \to K_s^0 + \pi^0 + \bar{n}$$

Furthermore, the anti-deuteron $\bar{D}$ hadronization is commonly described by a coalescence mechanism [9], where a $\bar{p}$ and a $\bar{n}$ sufficiently close in phase space bind to form a $\bar{D}$. The latter is expected among the products of dark matter annihilation [10]. Thus, improved $\bar{n}$ detection would enhance the simultaneous

measurement of the $\bar{n}$–$\bar{p}$ correlation function in $\bar{D}$ production, providing deeper insight into dark matter distributions.
GEANT4 is the main toolkit designed for simulating the passage of particles through matter [11]. It finds applications across high-energy, nuclear and accelerator physics, as well as in medical and space science research. A deeper knowledge on anti-neutrons kinematic and interaction would improve also Monte Carlo GEANT4 models of the hadronization process. This requires better $\bar{n}$ reconstruction, which remains challenging in standard HEP experiments.

1.2.1 Anti-neutron interactions in physics

Due to isospin symmetry, before the 1980s it was believed that anti-neutron strong interactions with nucleons (N) and matter would provide no additional information beyond anti-proton results. For anti-protons obtain proper beams was much easier.
The experimental landscape underwent a significant transformation in the mid-1980s with the development of dedicated anti-neutron beams, which enabled the observation of unique $\bar{n}$ characteristics, such as:

- The anti-nucleon nucleon $\bar{N}N$ interaction, in $\bar{p}p$ interactions, is a mixture of isospin $I = 0$ and $I = 1$ amplitudes, while the $\bar{n}p$ is a pure $I = 1$ eigenstate. The same difference between $\bar{p}p$ and $\bar{p}n$ interactions could be obtained by means of the charge-symmetric system $\bar{p}n$ copiously produced at deuterium target experiments. However, the use of deuterium introduces several drawbacks, such as re-scattering effects that may largely distort the final spectra [12]. This effect is absent if the $\bar{N}N$ system with pure $I = 1$ is formed by $\bar{n}$ interactions on protons of the material.

- The $\bar{n}p$ is a pure strong interaction, since it is free from Coulomb corrections, that are necessarily present in the $\bar{p}p$ interaction and may introduce further complications and ambiguities in the analysis of the data, especially at low-momenta.

- At low momentum, typically below 1 GeV/c, only S- and P-waves are present in the elementary $\bar{n}p$ interaction, described by the Dover-Richard parametrization [13]. Fig. 1.1 [14, 15] shows the trends of S-wave and P-wave annihilation cross sections according to the Dover-Richard parametrization, where in $\bar{n}p$ interactions the S-wave contribution diminishes with increasing energy, whereas the P-wave contribution grows.

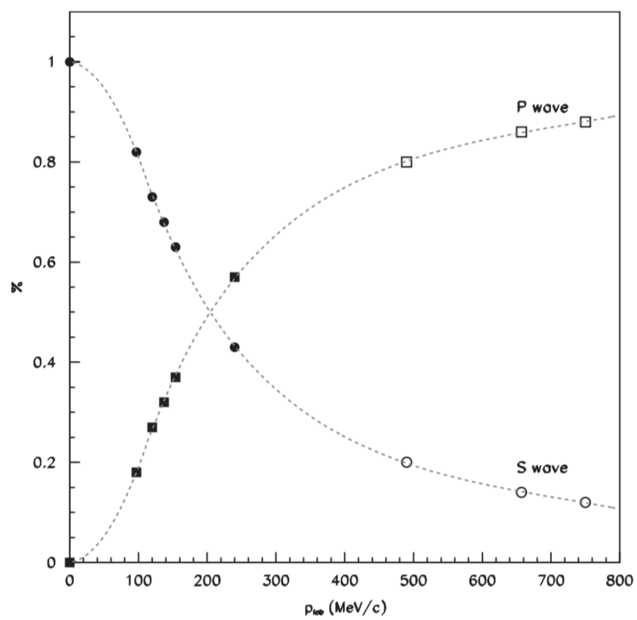


Figure 1.1: Trends of S- and P-wave annihilation cross sections according to annihilation models: the full markers (circles for S-wave, squares for P-wave) are from the Dover–Richard model, while the open ones have been obtained by using a version of the optical model.

Therefore, the $\bar{n}p$ system can be represented as an isospin eigenstate $|I, I_3\rangle = |1, +1\rangle$. It is not a C-parity eigenstate, then only J (total angular momentum), P (spatial parity) and G-parity are good quantum numbers, where G-parity is defined as the combination of charge conjugation and a rotation of 180° in isospin space as $G = C\, e^{i\pi I_2}$.
Since $\bar{n}p$ is an antifermion-fermion system, the following equations can be obtained:

$$P = (-1)^{L+1} \qquad G = (-1)^{L+S+1}$$

where L is the relative angular momentum between the anti-neutron and the proton, while S is the spin of the system, namely 0 or 1. The $I^G(J^P)$ quantum numbers allowed for the lowest initial states are shown in table 1.
Even if the total cross section σ_T is the first piece of information on the interaction between to particles, for the case $\sigma_T(\bar{n}p)$ could be tough to measure it. The final value of σ_T published in [16] shows how the data could be fitted to a simple parametrization [17], obtaining the results showed in 1.2.

Wave	$^{2S+1}L_J$	$I^G(J^P)$
S	$^1S_0$	$1^-(0^-)$
	$^3S_1$	$1^+(1^-)$
P	$^1P_1$	$1^+(1^+)$
	$^3P_0$	$1^-(0^+)$
	$^3P_1$	$1^-(1^+)$
	$^3P_2$	$1^-(2^+)$
D	$^1D_2$	$1^-(2^-)$
	$^3D_1$	$1^+(1^-)$
	$^3D_2$	$1^+(2^-)$
	$^3D_3$	$1^+(3^-)$

Table 1: Quantum numbers $I^G(J^P)$ on S, P e D waves.

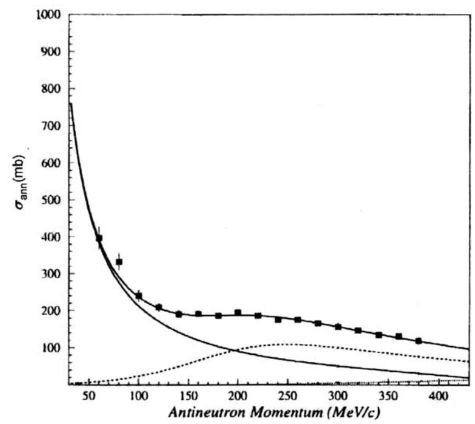


Figure 1.2: Fit of the annihilation cross section $\sigma_{ann}(\bar{n}p)$, measured by OBELIX up to 400 MeV/c, including S (solid line), P (dashed line) and D (dotted line) wave contribution.

1.2.2 The anti-nucleon–nucleon interaction

The first approach to describe the antinucleon-nucleon ($\bar{N}N$) interaction relies on interaction models. Unlike Quantum ElectroDynamics (QED), where the coupling constant α grows with energy and diverges at finite energy, asymptotic freedom characterizes the Quantum ChromoDynamics (QCD), the quantum field theory of the strong force between quarks and gluons. Quarks interact weakly at high energies, enabling perturbative calculations, but at low energies the interaction strengthens dramatically, confining quarks and gluons within composite hadrons. Here, the divergent coupling constant α_s renders perturbative approaches invalid, making interaction models essential for describing strong interactions in the $\lesssim 1$ GeV energy range.
The core of this approach is that, following the $\bar{N}N$ annihilation, a *fireball*

(which represents the $\bar{N}N$ system) with definite energy and volume $\Omega = \frac{4}{3}\pi R_0^3$ is formed, where $R_0 \sim 1$ fm is the nucleon radius. All the possible $\bar{N}N$ annihilation final states -primarily light mesons- may be reached through the decay of the fireball, according to the statistical Fermi model, where the probability of a final state is determined solely by the available phase space [18].
The use of interaction models in $\bar{N}N$ interaction generally allows a correct description of annihilation, even if it doesn't reproduce the whole experimental scenario, such as differences between nucleon and anti-nucleon scattering and stopping power (Barkas effect) [19]. The reason why it happens is the that the potential formulation used to describe the short range interactions (where annihilation plays the most important role) is not sensitive to the details of the internal dynamics, namely the interactions between quarks and gluons.
The second approach is the Effective Range (ER) expansion one, which is independent on any hypothesis concerning the interaction potential and it is based on the obtained fit of experimental cross sections [20]. The annihilation cross section for two interacting particles may be expressed as a partial waves expansion as follows:

$$\sigma_{\text{ann}} = \sum_L \sigma_L^{\text{ann}} = \frac{4\pi}{k^2} \sum_L (2L+1) \left(\Im m f_L - |f_L|^2\right),$$

where k is the center-of-mass momentum of the incident particle, and the partial amplitude f_L may be written as a function of the δ_L wave phase shifts induced by the interaction in the system wave function:

$$f_L = \frac{1}{cot\delta_L - i},$$

The partial waves may be expanded as a series in k, following the so-called *Effective Range (ER) expansion formalism*:

$$k^{2L+1} \cdot cot\delta_L = \frac{1}{A} + \rho k^2 + O(k^4),$$

where A is the scattering length, and ρ is the interaction range. For S and P partial waves can be obtained:

$$k \cdot cot\delta_0 = \frac{1}{a} + \frac{1}{2} r k^2$$
$$k^3 \cdot cot\delta_1 = \frac{1}{b} - \frac{3}{2}\frac{1}{R} k^2$$

a and b are, respectively, the scattering length and the nucleon volume, while r and R are the two effective ranges for the two partial waves. These parameters must be determined by a phenomenological fit to the annihilation cross section data. For example, for the $I = 1$ case, relevant to the $\bar{n}p$ interaction case, it was found that [17]:

$$a_1 = (0.4 + 0.5i) \text{ fm and } r_1 = (-1.4 + 1.8i) \text{ fm for S-wave,}$$

$$b_1 = (0.8 + 0.1i) \text{ fm}^3 \text{ and } R_1 = (0.2 - 0.4i) \text{ fm for P-wave}$$

1.2.3 Electromagnetic and hadronic showers

In the energy scenario (beyond 100 MeV), bremsstrahlung and pair production are the main processes which respectively involve electrons/positrons and photons, leading to the development of electromagnetic showers within the material. Therefore, to detect such phenomena, HEP experiments employ electromagnetic calorimeters. The electromagnetic calorimeter is a device whose size and density are designed to fully contain the electromagnetic shower, producing a signal proportional -or at least correlated- to the energy of the incident particle that initiated the shower. Since the radiation length X_0 (defined as the distance over which a high-energy electron loses $1/e$ of its energy through bremsstrahlung) determines the size of the electromagnetic shower, the typical longitudinal size for these detectors is of the order of 10-20 X_0 ($\sim 20-40$cm).
Furthermore, hadronic showers can be originated from strong interactions of high energy (beyond $\sim GeV$) hadrons, such as protons, neutrons and pions, with the material. To detect this kind of showers, a hadronic calorimeter may be employed; since the interaction length λ determines the hadronic shower size and $\lambda >> X_0$, hadronic showers are larger and longer than electromagnetic ones, requiring hadronic calorimeters to be significantly bigger than electromagnetic calorimeters. Therefore, homogeneous calorimeters cannot be used for hadronic calorimetry as they are for electromagnetic calorimetry. Instead, the sampling calorimeter technique is mandatory, employing alternating layers of passive, dense absorber material and active detection material.
For example, approximately 95% of a hadronic shower is contained within a cylinder of radius equal to the hadronic interaction length ($\lambda_{had} \simeq 44.1$ cm, in CsI(Tl)). In contrast, about 90% of an electromagnetic shower is contained within a cylinder defined by the Molière radius ($R_M \simeq 3.6$ cm in CsI(Tl)). Therefore, there is an order of magnitude difference between the two values. A shape comparison between these two type of shower can be observed in Fig. 1.3.

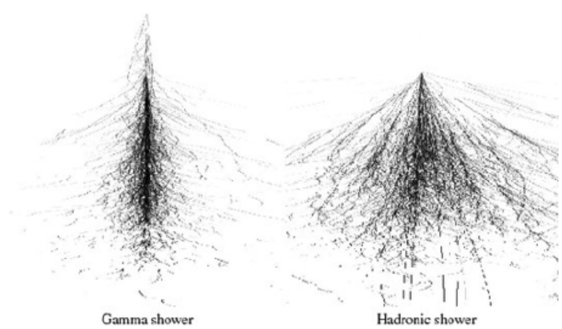


Figure 1.3: Electromagnetic shower vs hadronic shower comparison.

The $\bar{n}$ primarily interacts via the strong nuclear force, annihilating with nucleons in the material ($\sigma \sim 100$ mb), mainly producing light mesons such as pions which subsequently interact within the calorimeter:

$$\bar{n} + p/n \rightarrow \pi^{\pm}, \pi^{0}..$$

The π^0 meson mainly ($\sim 98.8\%$) decays into a couple $\gamma\gamma$, producing electromagnetic showers that are fully contained in the calorimeter.
While charged pions π^+ and π^- undergo both electromagnetic and hadronic interactions, producing final products that are not fully contained within the calorimeter volume, this leads to a lack in the energy deposited in the detector. Both forward and backward directions are then involved, according to the reaction kinematics.

1.3 The MANTRA Project

It may be challenging for general-purpose experiments to directly identify and reconstruct anti-neutrons produced in e^+e^- collisions, such as those at Belle II and BESIII, which lack a hadronic calorimeter. Anti-neutrons generally annihilate with detector material nucleons such as that of an electromagnetic calorimeter, producing hadronic showers which only allows a loose measurement of the anti-neutron energy with extremely large uncertainty.
Since anti-neutron kinematics have so far been measured via missing mass methods or isospin symmetry with anti-protons, a direct anti-neutron reconstruction would be highly desirable, especially in studying processes with neutral hadrons, such as $B^+ \rightarrow K^+ n\bar{n}$ or $J/\Psi \rightarrow p\pi^-\bar{n}$.
This thesis fits in the larger context of the MANTRA (Measuring Anti-Neutron: Tagging and Reconstruction Algorithm) PRIN2022 project, an inter-university (University of Turin and University of Ferrara) and INFN effort aimed at developing a method to improve the reconstruction of anti-neutron in existing high-energy physics experiments. It provides a general approach to measure the energy of anti-neutrons $\bar{n}$ up to 10 GeV using an electromagnetic calorimeter. The main idea is to combine the information from a high time-resolution time of flight detector (TOP and TOF in the Belle II and in the BESIII experiments, respectively) with that from an electromagnetic calorimeter (ECL and EMC in the Belle II and in the BESIII experiments, respectively), where the anti-neutrons annihilation occurs.
Due to its neutral charge and its immediate annihilation, tagging $\bar{n}$ can be very challenging. However, the annihilation can produce self-contained light mesons, mostly made of high-energy pions (π^-, π^0, π^+) and soft nuclear fragments. These can be detected in the Electromagnetic Calorimeter (detailed examination on the next chapter), where charged pions can produce hadronic showers, while neutral pions immediately decay into two photons which produce electromagnetic showers in the calorimeter volume. The comparison of the cluster energy deposition patterns, the so called shower shapes variables, left in the ECL by anti-neutrons and other neutral particles, such as photons, can improve the characterization of neutral particles.
A crucial aspect of the analysis process is the role of simulation, where GEANT4 [11] (more details in the next chapter) plays a central role, modeling the hadronic interactions in the calorimeter; realistic simulation is essential to accurately re-

produce real data. For $\bar{n}$ annihilation, parameterizations are typically derived from $\bar{p}$ measurements, necessitating validation against data.
This thesis aims to study the $\bar{n}$ hadronic interactions modeled by simulation software and compare them with those observed in Belle II data, focusing exclusively on signals from the Belle II electromagnetic calorimeter. To achieve this, $\bar{n}$ variables are first studied using a signal sample, where only the specifically selected processes with $\bar{n}$ in the final state are generated. Several background channels are then included to simulate the complete Belle II environment, concluding with a comparison between the results obtained from simulations and those from real data, with a Data/MC agreement.
Therefore, this work addresses the following question:

Are $\bar{n}$ hadronic showers correctly simulated in the Belle II software?

Chapter 2
The Belle II experiment

The following sections introduce the SuperKEKB [21] collider, describing its technical setup briefly. Then the Belle II [22] experiment is presented, discussing it detector by detector. Afterwards, the Belle II Analysis Software Framework (basf2) and TopoAna package are briefly described, with attention to the role of Monte Carlo methods in collider experiment simulation.

2.1 The SuperKEKB collider

SuperKEKB [21] is an asymmetric e^+e^- double-ring collider, located at KEK (The High Energy Accelerator Research Organization), in Tsukuba, Japan. It is the upgrade of the previous collider KEKB. It has a circumference of 3016 m and collides bunches of electrons with bunches of positrons, respectively of energy 7 GeV (High Energy Ring -HER-) and 4 GeV (Low Energy Ring -LER-). Fig. 2.1 shows a schematic image of the SuperKEKB design.

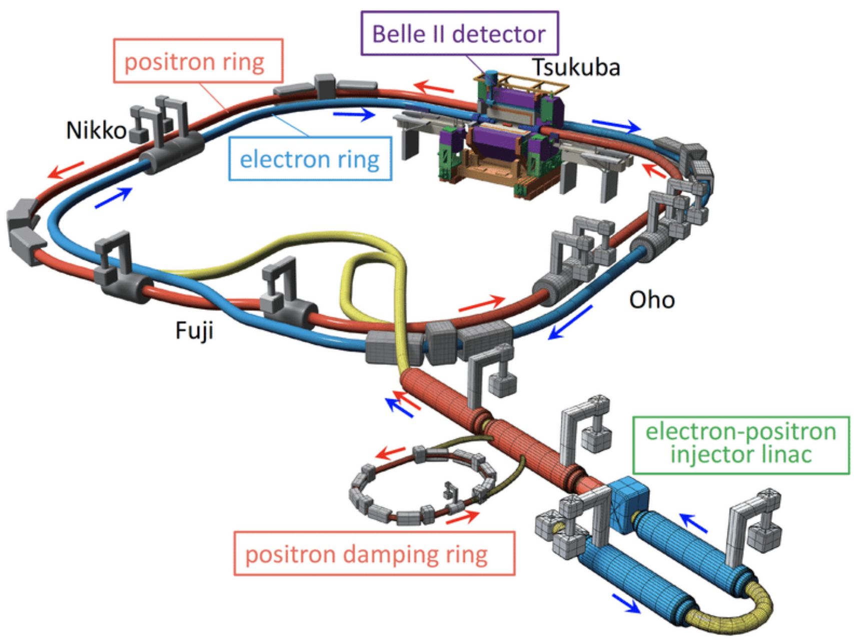


Figure 2.1: SuperKEKB design, where the Interaction Point (IP) is located in Tsukuba Hall, surrounded by the Belle II detector.

It is noteworthy that the positron damping ring reduces the e^+ beam emittance, as positrons are obtained from a rotating tungsten target target bombarded by

an electron beam. Electrons, by contrast, are injected directly from the linac since a controlled e^- beam can be easily obtained with a Linear Accelerator (LINAC) [23].
The resulting center-of-mass energy $\sqrt{s}$ in the Interaction Point (IP) can be modulated, but most of the data are acquired at the $\Upsilon(4S)$ resonance (10.58 GeV), an excited quarkonium state of $b\bar{b}$, which mainly decays into an entangled couple of $B\bar{B}$ ($\sim$ 100%). The $\Upsilon(4S)$ mass is just 20 MeV above the $B\bar{B}$ threshold, then the mesons are produced nearly in the rest frame. Since the lifetime of neutral B-mesons is short ($\tau = 1.510$ ps) and they are produced with low velocities, their path before decaying is short, making tough to separate the decays of the two produced mesons. The explicit asymmetry of bunches results in a relativistic forward Lorentz boost of the center-of-mass, where the decay length of the mesons is longer in the LAB frame, allowing the observation of the B mesons separately. Due to the huge amount of produced B mesons, SuperKEKB is by definition an example of so-called (super) B-factory.
SuperKEKB has been running since 2018 and it achieved the world's highest instantaneous luminosity ($5.244 \times 10^{34} cm^{-2}s^{-1}$) in March 19, 2026. The design luminosity of SuperKEKB is $8 \times 10^{35} cm^{-2}s^{-1}$, 40 times higher than the Belle's one, with the goal to accumulate data corresponding to an integrated luminosity of 50 ab^{-1}, 50 times the Belle data sample. The March 19, 2026 SuperKEKB 24-Hour Operation Summary is shown in Fig. 2.2.

2.2 The Belle II detector

The only interaction point of SuperKEKB is surrounded by the Belle II detector [22], a general purpose high-energy particle layered spectrometer with almost 4π of solid angle coverage. It presents a cylindrical shape to cover the e^+e^- collisions, composed of a barrel and two end-caps located along the z-axis.
The Belle II spectrometer measures 8 m in both width and height, with a total weight of approximately 1400 tons. With the exception of the KLM, all its subsystems are immersed in a 1.5 T magnetic field.
The spectrometer is composed of several detectors arranged cylindrically, reflecting the asymmetric distribution of the final state particles. The cylindrical structure of the detector is the *barrel*, while the caps are made of the forward end-cap (along the direction of the e^- beam) and the backward end-cap (along the direction of the e^+ beam).
The Belle II spectrometer consists of the following detectors, arranged radially outward from the IP:

(a) The Beam Pipe made of Beryllium (Be),

(b) Two layers of silicon Pixel Detector (PXD),

(c) Four layers of double-sided silicon strip detector (SVD),

(d) The Central Drift Chamber for tracking the charged particle (CDC),

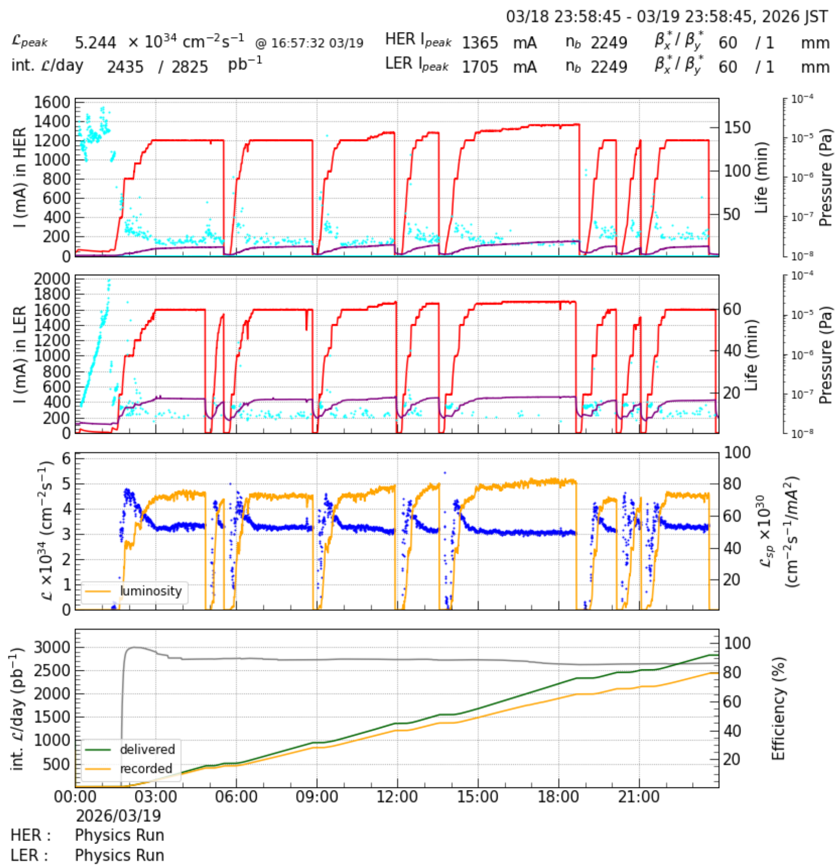


Figure 2.2: SuperKEKB operating parameters on March 18-19, 2026. From top to bottom: HER and LER ring currents (first two panels), instantaneous and integrated luminosity (bottom two panels).

(e) The Time of Propagation (TOP) Cherenkov counter in the barrel and the Cherenkov ring imaging (ARICH) in the front end-cap region,

(f) The Electromagnetic Calorimeter (ECL) to detect photons and to identify charged particles,

(g) The superconductive magnet which provides 1.5 T axial magnetic field in order to bend charged particle,

(h) The K_L and Muon detector (KLM) to track muons and K_L.

The electromagnetic calorimeter (ECL) has a central role in this elaborate since neutral ECL clusters associated to anti-neutron candidates are studied.
A Belle II scheme is shown in Fig. 2.3.

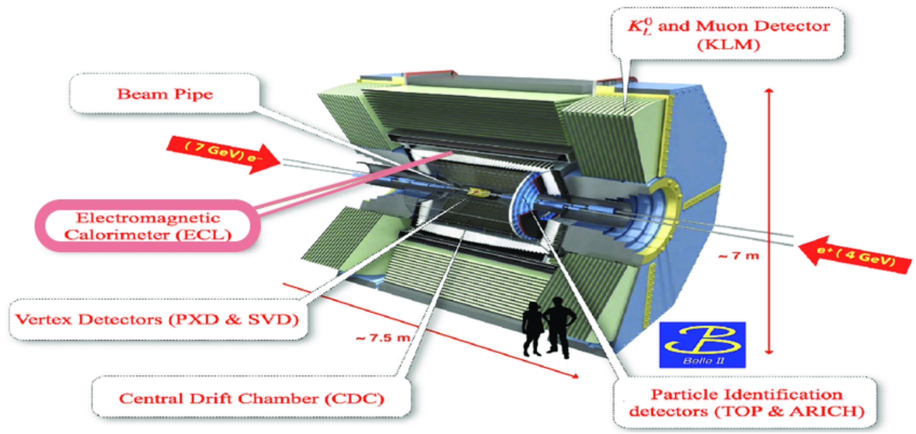


Figure 2.3: An artistic view of the Belle II detector.

2.2.1 The vertex detector VXD

Immediately following the beryllium beam pipe, which does not play a detection role, the innermost layer is the Vertex Detector (VXD).
It consists of two devices: the Silicon Pixel Detector (PXD - 2 layers) [24] and the Silicon Vertex Detector (SVD - 4 layers) [25] . Overall there are six layers, of whose the first two ($r_1 = 14mm,\ r_2 = 22mm$) are pixelated sensors of the DEPFET type and the remaining four ($r_3 = 38$ mm, $r_4 = 88$ mm, $r_5 = 115$ mm, $r_6 = 140$ mm) are double-sided silicon strip sensors.
The VXD provides coverage of $17° < \theta < 150°$ and full azimuthal coverage in ϕ, where θ is the polar angle relative to the z-axis -aligned with the e^- beam direction- and ϕ is the azimuthal angle in the xy plane using cylindrical coordinates. A scheme of the VXD is shown in Fig. 2.4.

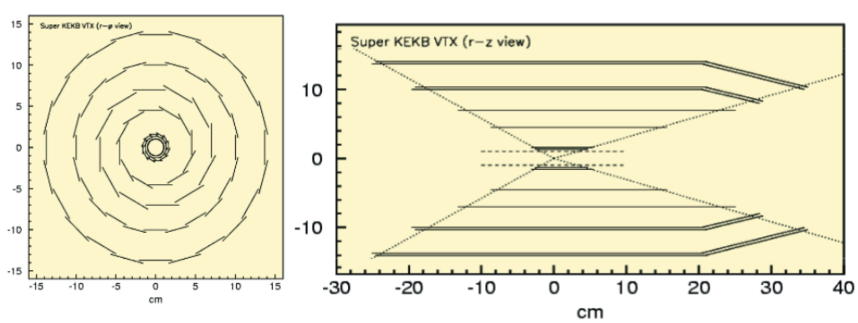


Figure 2.4: A scheme of the Belle II VXD.

The VXD purpose is to reconstruct the position of the primary vertex, the vertex of the B-mesons decay which are directly produced by the decay of the $\Upsilon(4S)$, improving the resolution of the tracking system as well.

2.2.2 The Central Drift Chamber (CDC)

The Central Drift Chamber (CDC) [26] is a large volume composed of small drift cells and it is the main tracking system of the Belle II detector. The CDC reconstructs charged-particle tracks by measuring the hit positions, allowing the determination of their momenta. Furthermore, the specific energy loss dE/dx measurement provides PID (Particle IDentification) information, complemented by the 1.5 T magnetic field that enables charge separation.
The CDC has an internal radius of 16 cm and an external radius of 113 cm and it contains 14339 wires in total distributed in 56 layers, either parallel (axial orientation) or skewed (stereo orientation) to the magnetic field direction. Combining their information, it is possible to reconstruct a full tridimensional helix track. Inside the cell, the average drift velocity is 3.3 $cm/\mu s$.
The CDC has the same VXD angular (θ, ϕ) coverage and, in order to increase the acceptance at forward angles, it is implemented asymmetrically along the z axis. It is filled with a mixture of helium (He) and ethane (C_2H_6), which represents a proper trade-off between momentum resolution for tracking and dE/dx resolution for the PID. A schematic view of the CDC is shown in Fig. 2.5.

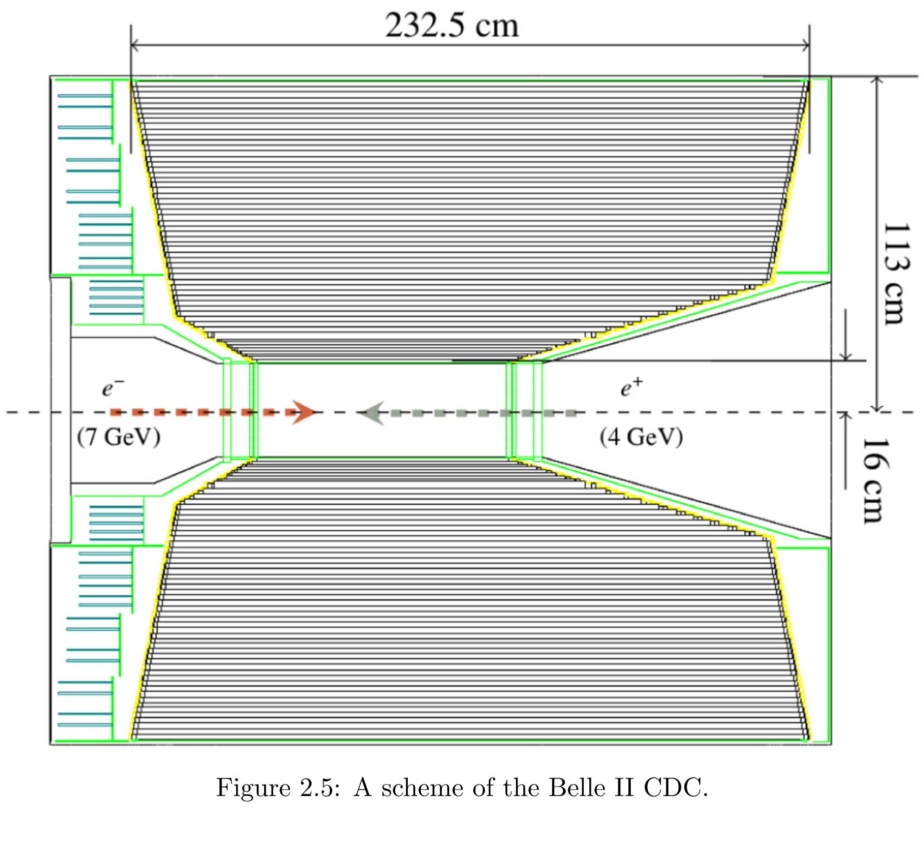


Figure 2.5: A scheme of the Belle II CDC.

2.2.3 PID systems: TOP and ARICH

Particles crossing a material at speeds exceeding the phase velocity of light in that medium ($\beta > 1/n$, where n is the refractive index) emit photons at the Cherenkov angle $\theta_c = \arccos(1/\beta n)$. Particles with identical momentum but different masses produce Cherenkov radiation at distinct angles, enabling Particle IDentification (PID).
For particle identification in the barrel region, a Time of Propagation (TOP) [27] counter -a novel Cherenkov detector- is employed. In total, there are 16 modules, each consisting of a 45 cm wide and 2 cm thick quartz bar with a small expansion volume of about 10 cm long at the sensor end. There is an expansion wedge that behaves as a lens and improves the spatial resolution of the Cherenkov photon imaging. At the exit window of the wedge, two rows of sixteen fast multi-anode photon detectors are mounted.
Within the 2.6 m-long quartz, the particle produces Cherenkov radiation that travels via total internal reflection to the mirror. Then the radiation is reflected towards the photon detectors at the backward end, where a small prism is located, acting as a volume expansion and a light guide. The time of arrival and the impact position of the Cherenkov photons provide the PID information.
The hit distributions differ according to the particle species and their masses: different particles exhibit distinct Cherenkov angles, resulting in different photon paths. Consequently, they are detected at different positions within the PhotoMulTiplier (PMT) array and at different times.
A scheme of the TOP counter module is illustrated in Fig. 2.6 and a side view of the TOP counter in Fig. 2.7.

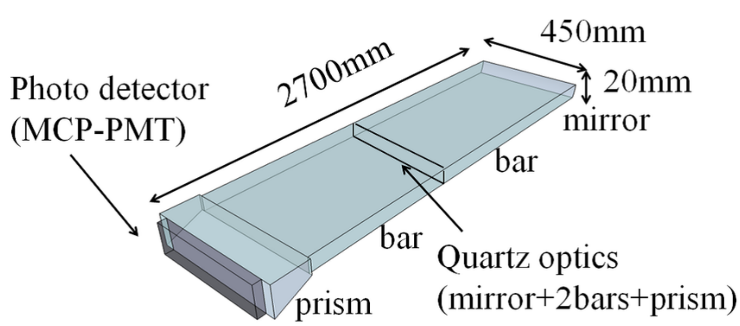


Figure 2.6: Scheme of the TOP counter module.

The TOP counters are equipped with photosensors, that provide a time resolution of about 100 ps, made of 16-channel Micro Channel Plate PhotoMultiplier (MCP-PMT).
The Aerogel Ring Imaging CHerenkov (ARICH) [28] is located in the forward end-cap region of the Belle II detector and thus complements the aforementioned TOP. Its main goal is to separate kaons K from pions π across the full kinematic range of the experiment, from 0.5 to 4.0 GeV/c.
It consists of two 2 cm thick aerogel layers ($n_{up} = 1.045$, $n_{down} = 1.055$), which

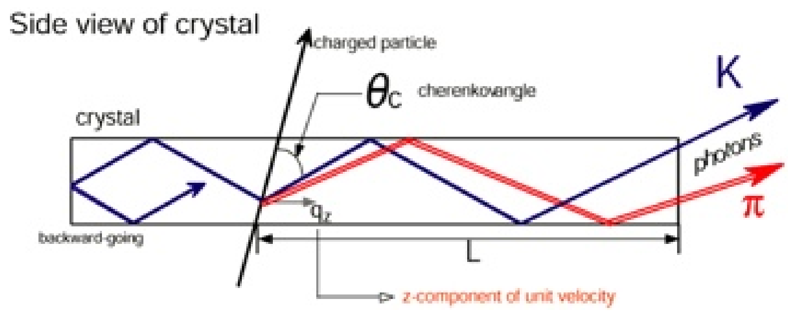


Figure 2.7: Representation of the side view of the TOP counter with internal reflection of the Cherenkov radiation.

increase the photon yield without degrading the Cherenkov angle resolution compared to using a single aerogel type [29]. A scheme of the ARICH working principle and an example of a ring produced by a cosmic-ray muon are shown in Fig. 2.8.

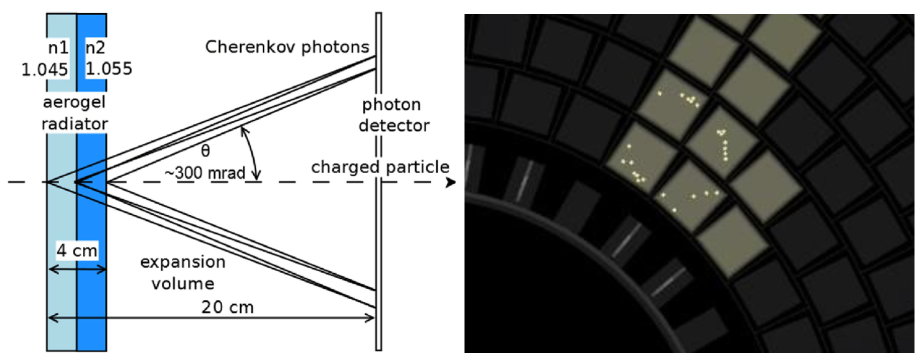


Figure 2.8: (left) Working principle of the ARICH. (right) Ring produced by a cosmic-ray muon.

2.2.4 The Electromagnetic Calorimeter (ECL)

The Electromagnetic CaLorimeter (ECL) [30] plays a central role in this thesis, since it is aimed at the study of the reconstruction clusters associated to anti-neutron candidates.
The primary purpose of the ECL is to detect photons and electrons, providing precise measurements of their energy and the position the produced electromagnetic shower (E, $\vec{x}$). The ECL contributes to identify neutral particles -such as photons and neutral hadrons- as well as to separate the charged particles, primarily electrons and pions. Anti-neutrons can be detected through its annihilation products (mainly pions). A typical shape of an ECL shower caused by a 7 GeV photon is shown in Fig. 2.9.

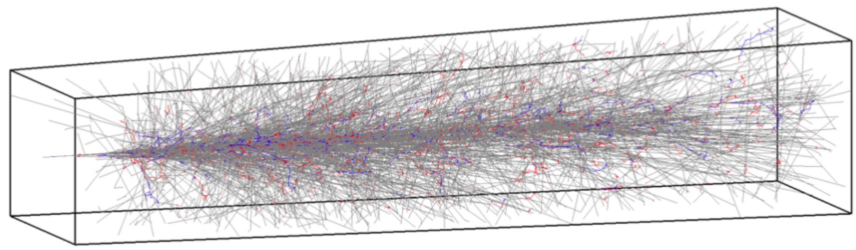


Figure 2.9: Typical ECL shower caused by a 7 GeV photon.

The Molière radius R_M is a characteristic constant of a material that defines the transverse scale of fully contained electromagnetic showers initiated by high-energy electrons or photons. Its definition, for an electromagnetic calorimeter, is that $\sim 90\%$ of electromagnetic shower is contained in a cylinder with radius R_m, while the $\sim 95\%$ is contained in a cylinder with radius $2R_m$.
The radiation length X_0 is defined as the mean length (in cm) over which an electron's energy is reduced by a factor of $1/e$. The critical energy E_c is defined as the energy at which radiation losses equal ionization losses for electrons/photons. X_0 and E_c determine the Molière radius R_M as follows:

$$R_m = \frac{21MeV}{E_c} \times X_0$$

CsI(Tl) crystals
The thallium-doped caesium iodide CsI(Tl) crystals have $X_0 = 1.860$ cm and $R_m = 3.53$ cm. This material is one of the brightest known scintillators, providing a light yield Y of ~ 52000 photons/MeV, with a photon emission spectrum peaked at ~ 550 nm, ideal for photodiode readout. Properties of typical scintillators used in calorimetry [31] are shown in Fig. 2.10.

Material	Density (g/cc)	Wavelength of Maximum Emission	Refractive Index	Principal Decay Constant (µ-Sec)	Total Light Yield	Pulse Rise Time (µ-Sec)	Absolute Scintillation Efficiency	Relative Pulse Height
NaI(Tl)	3.67	415	1.85	0.23	38000	0.5	11.3	1.00
CsI(Tl)	4.51	540	1.80	1.0	52000	4.0	11.9	0.49
LiI(Eu)	4.08	470	1.96	1.4	11000		2.8	0.23
BGO	7.13	505	2.15	0.30	8200	0.8	2.1	0.13
BC-400	1.032	423	1.58	0.002	10000		3.0	0.25
BC-454	1.026	425	1.58	0.0022	10000		3.0	0.25

Figure 2.10: Properties of scintillators used in calorimetry.

The ECL is composed of 8736 array of CsI(Tl) crystals. Each is a truncated pyramid with dimension 6x6x30 cm^3, where the radial length is around $16.2X_0$. Each crystal is wrapped with a layer of 200 μm thick Gore-Tex porous teflon and then covered with a 50 μm thick aluminized polyethylene.
Assuming the z-axis is aligned with the e^- beam momentum direction, with $z = 0$ coinciding with the nominal IP position, the ECL is made of a 3 m long barrel section and the end-caps at $z = 2.0$ m (forward) and $z = -1.0$ m (backward). An ECL scheme is shown in Fig. 2.11:

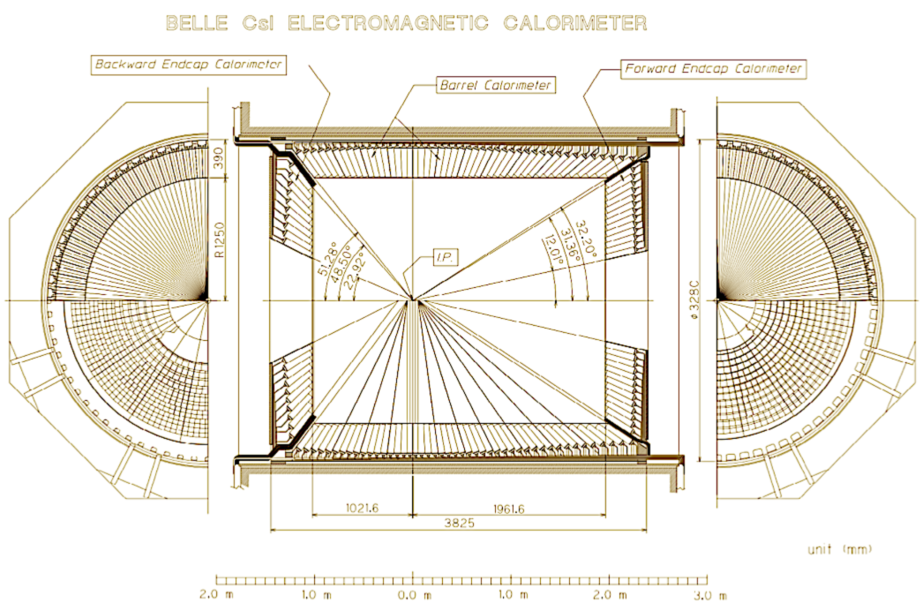


Figure 2.11: A scheme of the Belle II ECL detector.

It covers about 90% of the solid angle in the center-of-mass system (polar coverage: $12° \leq \theta \leq 155°$); there is a $\sim 1°$ gap between the barrel and each end-cap.
A table showing the ECL θ coverage and segmentation for each region is shown in Fig. 2.12.

	θ coverage	θ segments	ϕ segments	# of counters
Forward Endcap	12.01° - 31.36°	13	48 - 144	1152
Barrel	32.2° - 128.7°	46	144	6624
Backward Endcap	131.5° - 155.03°	10	64 - 144	960

Figure 2.12: ECL coverage and segmentation for each ECL region.

Light from each crystal is detected by two $10 \times 20\,\mathrm{mm}^2$ Hamamatsu S2744-08 photodiodes [32], glued to the rear surface of the crystal and coupled to charge-sensitive preamplifiers.

2.2.5 The solenoid and iron structure

Externally to the ECL are the spires of the 23-ton superconducting solenoid that generates the 1.5 T axial magnetic field. The curvature of charged-particle trajectories (Larmor radius) allows the determination of their momentum and charge, enabling particle identification when combined with additional detector information.
The magnet coils are made of niobium-titanium-copper (NbTi/Cu) alloy and is cooled by a liquid helium cryostat to achieve superconducting conditions.

2.2.6 The K_L and Muon Detector (KLM)

The K_L and Muon Detector (KLM) [33] represents the outermost layer of Belle II and it is dedicated to the detection of K_L mesons and muons, which are very penetrating particles. It is composed of alternating layers of 4.7 cm thick iron plates and active detector elements; the former acts as a return yoke for the magnetic flux. The alternating layers provide at least 3.9 interaction lengths (compared to 0.8 in the ECL), allowing the K_L to interact and produce hadronic showers.
The KLM covers the barrel region (15 layers) and the end-caps regions (forward 14 layers and backward 12 layers). The two inner barrel layers and the entire end-caps are instrumented with scintillator strips, while the 13 outer barrel layers are instrumented with Resistive Plate Chambers (RPC).
As compared to charged hadrons, muons travel longer distances and with smaller deflections inside the KLM, which allows discrimination between them.

Chapter 3
The Belle II Analysis Software Framework

To perform the simulations, reconstruction and the analyses of this thesis, the Belle II Analysis Software Framework (basf2) [34] has been used. It is the primary software used in all aspects of the data-processing chain at Belle II, such as generating simulated data, unpacking raw data, performing reconstruction (tracking, clustering) and carrying out high-level analysis. The final analysis steps (as histogramming, fitting distributions, etc.) are usually performed in other frameworks as well, such as ROOT or the Matplotlib Python library.

3.1 Monte Carlo simulation and reconstruction

The Monte Carlo method is a technique used to compute probabilities and related quantities by exploiting sequences of pseudo-random numbers. These sequences are "pseudo-random" because sequences of random numbers possess meaning, not individual ones -though the latter are frequently referenced. Randomness characterizes the sequence, not single numbers.
Monte Carlo methods are widely used in HEP physics for tasks such as event generation and detector simulation. The first one regards the production and the decays of particles with a certain set of properties, while the second one concerns the simulation of the particle interactions during the path in the detector materials.
In Belle II, electron-positron collisions produce a certain amount of particles, according to the adopted physics model, which may be based on the Standard Model or alternatives such as supersymmetry or dark matter production. Example of a physics generator is *PYTHIA* [35] for the generation of high-energy physics collision events, *EvtGen* [36] which simulates the decays of particles and *Phokhara* [37] which simulates the leptonic and hadronic final states at the next-to-leading order (NLO) accuracy, namely simulating Initial State Radiation (ISR), which has a primary role in the analysis, and so forth.
After the generation, the particles are propagated through the detector, accounting for their interactions with material and the effects of the magnetic

field, using GEANT4 [11]. This is an open source software which simulates the interaction between particles and the detector systems.
The reconstruction represents the process in which particles are measured and identified. The closer the reconstructed quantities (such as initial positions and four-momenta) are to the generated-level information, the better the reconstruction performance in simulated MC and real data. Since several products of e^+e^- collisions are short living particles, it is necessary to reconstruct them indirectly via their daughter particles. Furthermore, intrinsic noise must be taken into account, as it may generate spurious random signals unrelated to actual particles. Interactions between beam electrons or positrons, either with each other or with other accelerator components -known as beam background- must be considered. The same reconstruction algorithms are employed in data and MC, in order to obtain the same condition. To test the correct response of the reconstruction, generator-level Monte Carlo truth of the simulated data is used (the so called MC truth). The Monte Carlo matching (MC-matching) procedure associates the generator-level particle responsible for a reconstructed candidate, crucial for understanding reconstruction effects and backgrounds.

3.2 basf2 modules and path

A basf2 module is the smallest building block of the framework and it is a script, usually written in C++, designed to process a specific unit of data. A typical data chain consists of a linear arrangement of modules. All the work is done in modules, including reading and writing data to and from disk or more complex operations such as full detector simulation.
Modules are linearly arranged in a path, which is a container for them. The linear order depends on the specific task defined by the user. Therefore, the modules are executed one at a time, in the same order they are added into the path, from the first to the last. A schematic view of the processing flow is shown in Fig. 3.1.

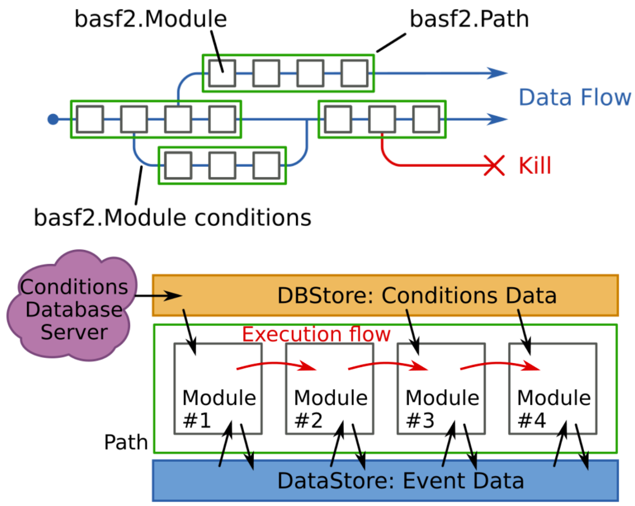


Figure 3.1: Scheme of the processing chain in basf2.

The modules process data stored in the DataStore, with each module having access to read from and write to it. There are also non-event data, the so called conditions, which are loaded from a central conditions database, the DBStore. This database stores the information about the specific detector conditions at each run of the experiment.
A package is a logical collection of code in basf2, consisting of several modules and python scripts that configure paths to do common things. The packages have meaningful names related to their specific operation (tracking, reconstruction, ecl, klm, etc...). All the packages are stored in the Belle II GitLab repository. The most used package in this thesis is the analysis package, which contains python functions and C++ basf2 modules, such as modularAnalysis.
The steering file is the core script in which analysis and data-processing tasks are performed, with a path being declared and modules added to it. Some highlights of the used steering files are provided in the appendix.

3.3 TopoAna: Topology Analysis

In high-energy physics data analysis, a comprehensive understanding of Monte Carlo samples is crucial to select signals with high efficiency while suppressing backgrounds. This is exactly the case of the present elaborate.
Understanding the physics processes enables identification of the main backgrounds. Selection criteria can then be further optimized by analyzing differences between the background and the signal. An example of topology diagram is provided in 3.2.

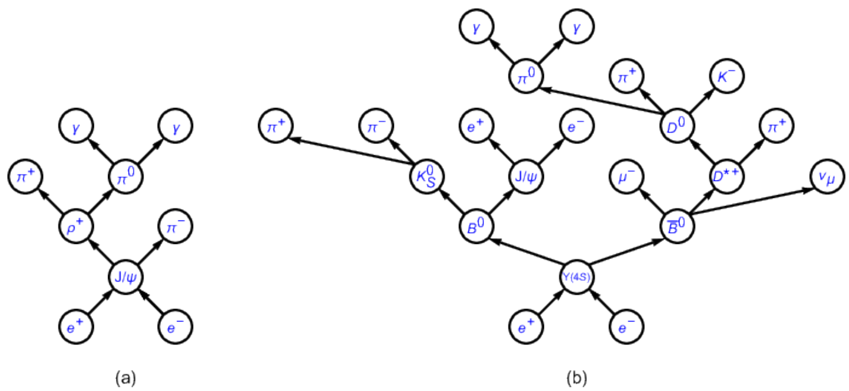


Figure 3.2: Topology diagrams of (a) $e^+e^- \to J/\psi$ and (b) $e^+e^- \to \Upsilon(4S) \to B^0\bar{B}^0$, two typical processes at e^+e^- colliders, such as BESIII and Belle II.

The TopoAna program [38], developed in C++ and ROOT, provides a convenient interface for verifying Monte Carlo topology information beyond scratch inspection.
The program input consists of one or more ROOT files containing a TTree -a column-oriented data structure used to store large sets of events or entries- with raw topology truth information from the MC samples. This TTree stores the processes that occurred in each event, including particle multiplicity (number of final state occurrences), PDG codes and mother indices (parent-daughter relationships).
TopoAna is an offline tool independent of basf2. It extracts human-readable topology information from raw MC truth data. Example of MC information, depicting the topology diagram in Fig. 3.2 (b), is shown in Fig. 3.3.

Number of particles	:	63
PDG codes of particles	:	300553, −511, 511, −433, 421, 211, 22, −413, 111, 111, 113, 211, −431, 22, −323, 213, −421, −211, 22, 22, 22, 22, 211, −211, 333, 11, −12, 22, −311, −211, 211, 111, 221, 331, 321, −321, 310, 22, 22, 111, 111, 111, 111, 111, 221, 111, 111, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22, 22
Mother indices of particles	:	−1, 0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 4, 4, 7, 7, 8, 8, 9, 9, 10, 10, 12, 12, 12, 12, 14, 14, 15, 15, 16, 16, 24, 24, 28, 31, 31, 32, 32, 32, 33, 33, 33, 36, 36, 39, 39, 40, 40, 41, 41, 42, 42, 43, 43, 44, 44, 45, 45, 46, 46

Figure 3.3: Raw MC information example from Fig. 3.2 b.

The information from Fig. 3.3 may seem rather non-intuitive to consult, so the TopoAna human-readable MC information (Fig. 3.4) can be consulted instead.

0	$e^+e^- \to \Upsilon(4S)$	-1	9	$\rho^+ \to \pi^0\pi^+$	6
1	$\Upsilon(4S) \to B^0\bar{B}^0$	0	10	$K^{*-} \to \pi^-\bar{K}^0$	6
2	$B^0 \to \pi^0\pi^0\rho^0\pi^+D^{*-}$	1	11	$D_s^- \to e^-\bar{\nu}_e\phi\gamma$	7
3	$\bar{B}^0 \to \pi^+D^0D_s^{*-}\gamma$	1	12	$\eta \to \pi^0\pi^0\pi^0$	8
4	$\rho^0 \to \pi^+\pi^-$	2	13	$\eta' \to \pi^0\pi^0\eta$	8
5	$D^{*-} \to \pi^-\bar{D}^0$	2	14	$\bar{K}^0 \to K_S^0$	10
6	$D^0 \to \rho^+K^{*-}$	3	15	$\phi \to K^+K^-$	11
7	$D_s^{*-} \to D_s^-\gamma$	3	16	$\eta \to \gamma\gamma$	13
8	$\bar{D}^0 \to \eta\eta'$	5	17	$K_S^0 \to \pi^0\pi^0$	14

Figure 3.4: Highly readable MC information example from Fig. 3.2 b, as presented by the TopoAna program.

3.4 The Belle II Distributed Computing

The Belle II datasets are categorized into two main types: Data and Monte Carlo. The Data are obtained from calibrated collision events processed by the experiment. Monte Carlo samples are generated using the Monte Carlo method according to various event generators, as discussed in Chapter 2. These are further classified into Signal MC -a specific process generated with a dedicated decay file upon analyst request- and Generic MC, or "cocktail", containing all processes expected in data with their correct relative fractions. These are usually in mdst (Mini Data Summary Table) format.

Data and Monte Carlo samples can be filtered by selecting events through analysis-dependent offline pre-selections, commonly referred to as skims (i.e., skimmed samples). Skims reduce data volume by selecting events of interest for specific analyses and retaining only the necessary information, producing smaller, more manageable datasets.

Using these Data and MC collections, analyses are performed via custom steering files with basf2, producing final ROOT ntuples. Subsequent offline analysis is then conducted outside the basf2 framework using ROOT, matplotlib, etc.

A scheme of the Belle II Data Model is shown in Fig. 3.5. The Belle II Data and Monte Carlo samples are distributed across numerous storage sites worldwide. The Distributed Computing System (known as the Grid) consists of a network of loosely coupled computers with varying local conditions (operating systems, resources, etc.).

Two full copies of the raw data are maintained at dedicated data centers. Raw data is processed, skimmed, and distributed over other storage sites. A scheme of the Belle II Distributed Computing is shown in Fig. 3.6.

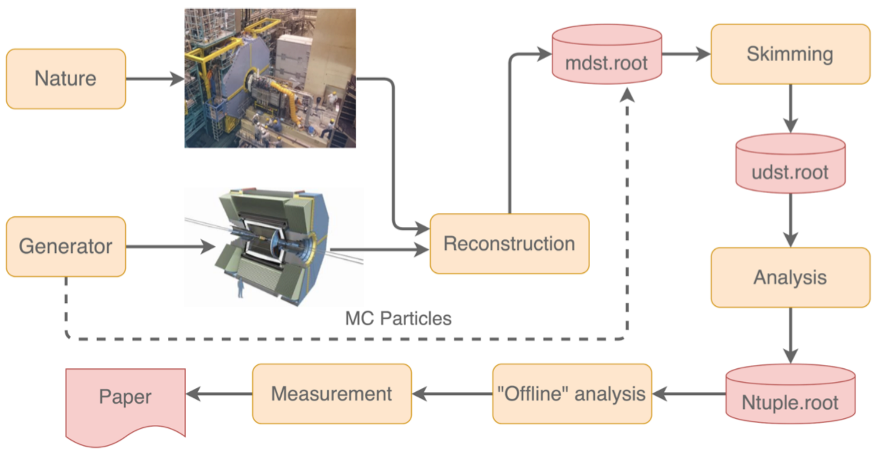


Figure 3.5: A scheme of the Belle II Data Model, starting from the nature processes information to the final results. Monte Carlo truth information bypasses reconstruction, since the information is directly picked at generator level.

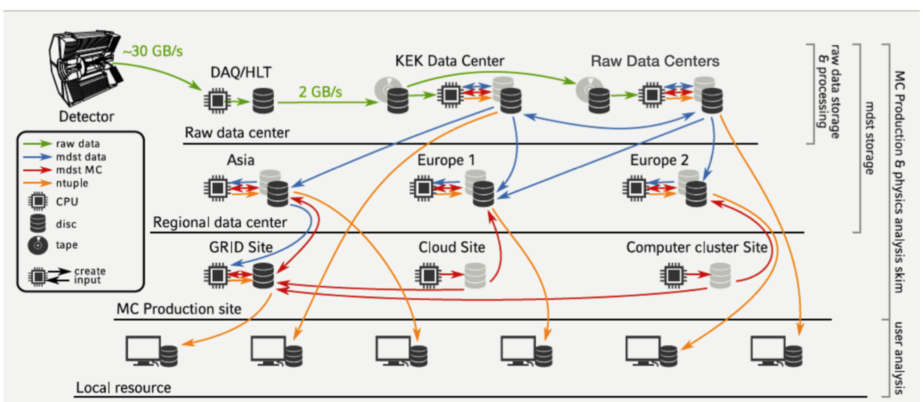


Figure 3.6: A scheme of the Belle II Distributed Computing sites.

Each site interacts with DIRAC, an interware that manages interactions between Belle II and its computing resources. BelleDIRAC is an extension that enables users to interact with DIRAC for job submission, data placement, monitoring, and other tasks.

Chapter 4
The ECL reconstruction

This chapter describes the main software features of the Electromagnetic Calorimeter, including the ECL reconstruction algorithm and its energy performance. Then, the cluster variables are introduced. These play a central role in this work, since the final goal is to assess the Data/MC agreement for the cluster variables.

4.1 The ECL reconstruction algorithm

For each ECL crystal, a 32-bit word is recorded as the digitized output from the electronics. This 32-bit word, referred to as ECLDigit, encodes:

- energy information as Flash Analog-to-Digital Converter (FADC) counts (18 bits),
- time information in clock units (12 bits),
- 2 status bits (signal quality flags).

To reconstruct the shower, an algorithm processes the ECLDigit objects into two new data objects that contain the combined information from adjacent crystals.:

- ECLShowers, an ECL-internal object containing additional debug information,
- ECLClusters, the primary analysis object.

Only ECLShowers with $E > 20$ MeV are converted to ECLClusters.
The first step identifies connected regions, which rapidly reduces the combinatorial complexity. These are specific cell collections determined by a dedicated clustering procedure. An example of this procedure is shown in Fig. 4.1 and in Fig. 4.2.

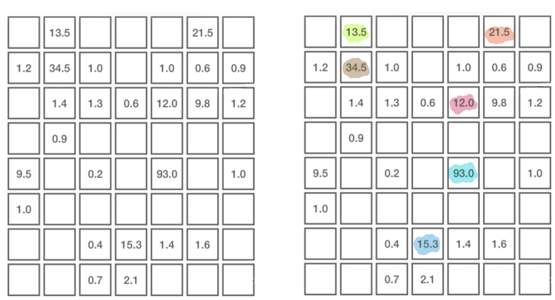


Figure 4.1: Description of ECL clustering process: (left) Step 0; (right) Step 1.

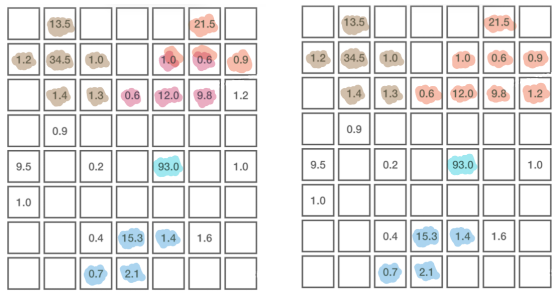


Figure 4.2: Description of ECL clustering process: (left) Step 2; (right) Step 3.

Each cell represents one ECLDigit, with its value corresponding to the deposited energy in MeV. Empty cells indicate ECLDigit values below the readout threshold (0.1 MeV in this example).
The step 0 corresponds to the initial condition. In step 1, every ECLDigit above 10 MeV is selected. In step 2, any neighbors above 0.5 MeV are attached to the highest-energy seed found in step 1. Sometimes two seeds compete for the same neighbor, and sometimes clusters have no neighbors (i.e. 93.0). In step 3, overlapping neighbor regions trigger a merge. Additionally, any cells above 1.5 MeV recruit their next neighbors iteratively.
After this process, the connected regions are obtained, as shown in Fig. 4.3.

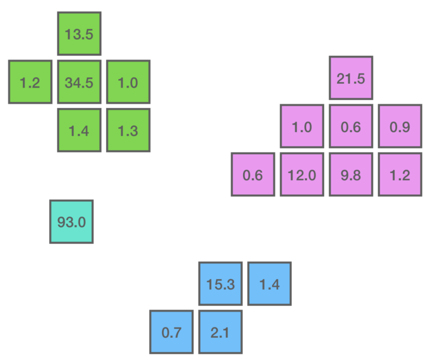


Figure 4.3: Connected regions obtained in the example (threshold: 0.1 MeV).

The final step is to pick the crystals with energy above 10 MeV, namely the Local Maxima (LM). Every connected region has one LM at least. The connected regions are furtherly processed in order to obtain the position and the energy of the cluster, processing the LM with a splitting-weighting process among the connected regions crystals [39].
The main idea is that an ECLShower object is produced from every LM. If a connected region contains only one LM, the total energy of its crystals is assigned to the corresponding ECLShower. Otherwise, if multiple local maxima

are present, an iterative procedure is performed, where the weight for each crystal depends on the distance of the crystal from the center-of-gravity of each ECLShower (center-of-gravity is calculated using weighted energies).
Connected regions are used to minimize the number of times the computationally intensive splitting procedure must be performed.

4.2 ECL performance

The observed ECL energy resolution can be written as follows:

$$\frac{\sigma_E}{E} = \sqrt{\left(\frac{0.81\%}{\sqrt[4]{E}}\right)^2 + \left(\frac{0.066\%}{E}\right)^2 + (1.34\%)^2},$$

where the first term represents stochastic fluctuations (Poissonian statistics) in the shower generation, the second term accounts for electronic noise, and the third is a constant term dependent on calibration.
For example, $\frac{\sigma_E}{E} = 4\%$ at 100 MeV and $\frac{\sigma_E}{E} = 1.4\%$ at 8 GeV [40]). The energy resolution behavior [41] is shown in Fig. 4.4.

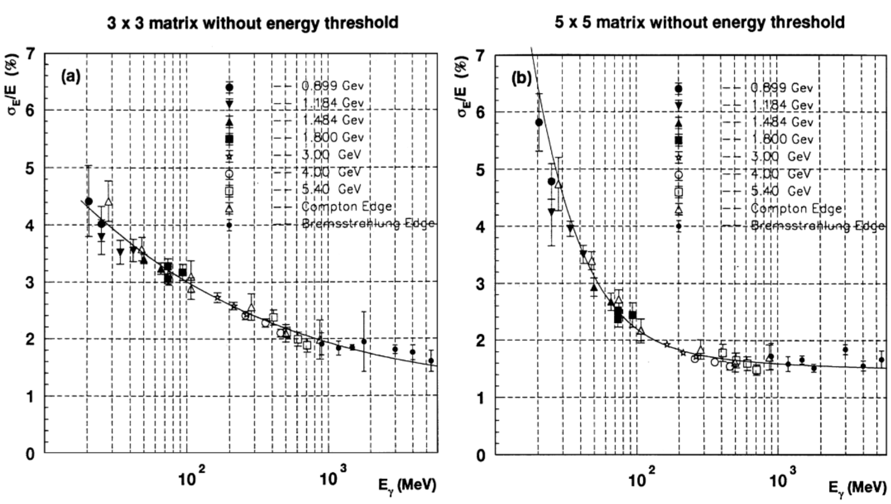


Figure 4.4: The energy resolution as a function of incident photon energy for the (a) 3Y3 and (b) 5Y5 matrix with total energy sum. The error bars are not statistical but are the rms of four measurements taken with the crystals (3,3), (3,4), (4,3) and (4,4) into the photon beam.

4.3 ECL cluster variables

The characteristics of a shower can be identified through various variables derived from the calorimeter information. For example, electrons and charged hadrons exhibit different shower shapes, allowing their distinction. Electromagnetic showers are contained within a small number of crystals, whereas hadronic showers exhibit a broader spatial and energy distribution. Different cluster variables are now examined and discussed.

4.3.1 Cluster Energy

The Cluster Energy (*clusterE* in the basf2 software) represents the total energy within the cluster associated with a given candidate particle. To avoid disk saturation, the basf2 reconstruction software stores only clusters with energy above 20 MeV, producing physics-ready ECLShower objects.

4.3.2 Cluster E1E9 and E9E21

Cluster E1E9 and cluster E9E21 variables (respectively *clusterE1E9* and *clusterE9E21* in the basf2 software) are defined as follows:

$$E1E9 = \frac{E_1}{\sum_{i=1}^{9} E_i} \qquad E9E21 = \frac{\sum_{j=1}^{9} E_j}{\sum_{j=10}^{25} E_j}, \text{ with } j \neq 10, 14, 18, 22$$

Referring to Fig. 4.5, the first variable returns the ratio of the energy in the central crystal (E_1) to the total energy in the surrounding 3×3 crystal grid (E_9). The second one returns the ratio of the energy in the 3x3 crystal grid around the central crystal (E_1) to the total energy in the 5×5 crystal grid around the central crystal, excluding the corners.

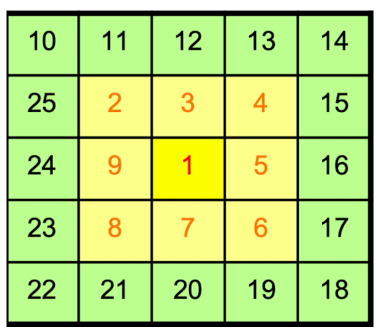
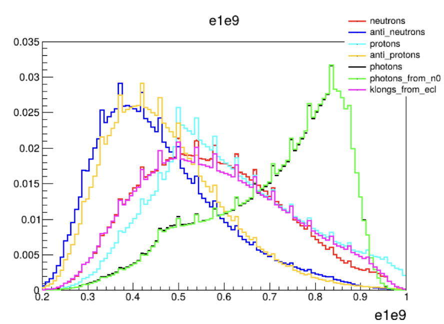


Figure 4.5: Ideal representation of a cluster pattern, consisting of a 5×5 matrix in which each cell corresponds to a single crystal.

These ratios are always less than 1 and tend toward larger values for photons and smaller values for hadrons, since the latter produce more spread-out showers. These variables are dimensionless and are included within the range $[0, 1]$.
An example of these variables distributions for different particles is provided in Fig. 4.6.



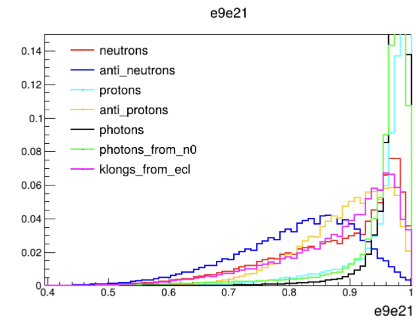


Figure 4.6: Comparison of $E1/E9$ and $E9/E_{21}$ distributions for different particle species.

Photons have higher values of E1E9 compared to hadrons, indicating that photons produce more compact clusters. This difference is more explicit in the E9E21 distribution. It can be also noted that anti-neutrons produce more spread-out clusters compared to other particles.

4.3.3 Lateral Momentum

The cluster Lateral Momentum (or Lateral Energy) distribution (*clusterLAT* in the basf2 software framework) characterizes the spatial energy deposition pattern transverse to the shower longitudinal direction. It is defined as follows:

$$C_{LM} = \frac{\sum_{i=2}^{n} E_i r_i^2}{E_0 r_0^2 + E_1 r_0^2 + \sum_{i=2}^{n} E_i r_i^2},$$

with $r_0 \simeq 5$ cm (average distance between two crystals centres), n is the number of crystals associated to the shower, $E_i = (E_0, E_1, ...)$ are the crystal energies sorted by descending energy and r_i is the distance of the i-th crystal to the shower center projected onto a plane perpendicular to the shower axis. For radially-symmetric electromagnetic showers, this variable peaks at ~ 0.3, where it is larger for hadronic particles. This variable is dimensionless and it is included within the range $[0, 1]$.
An example of Lateral Momentum distribution for electrons and charged pions with $E/p > 0.8$, from BaBar experiment [42], is shown in Fig. 4.7.

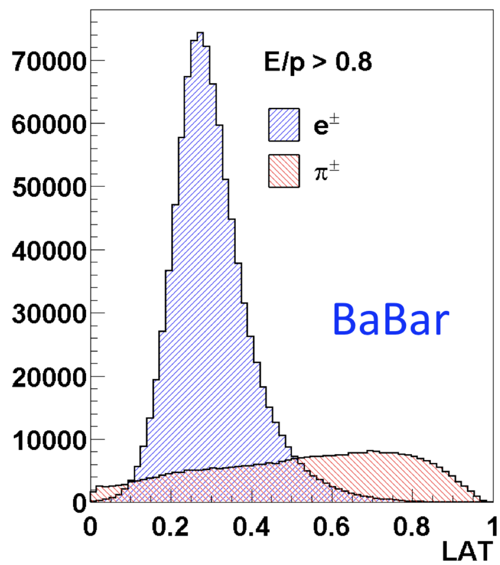


Figure 4.7: Comparison of Lateral Momentum (*clusterLAT*) distributions for electrons and charged pions with $E/p > 0.8$, from BaBar experiment.

4.3.4 Second Moment

The cluster Second Moment (*clusterSecondMoment* in the basf2 software) characterizes the longitudinal development of a shower. It quantifies the "width" or "compactness" of the energy deposit within a cluster. It is defined as follows:

$$C_{SM} = \frac{1}{S_0} \frac{\sum_{i=0}^{n} E_i r_i^2}{\sum_{i=0}^{n} E_i},$$

where E_i and r_i are the same variables as used for the lateral momentum, while S_0 is a normalization factor that normalizes to 1 for isolated photons, making C_{SM} dimensionless. The Second Moment measures the dispersion of cluster size.
An example of Second Moment distribution for different particles is provided in Fig. 4.8.

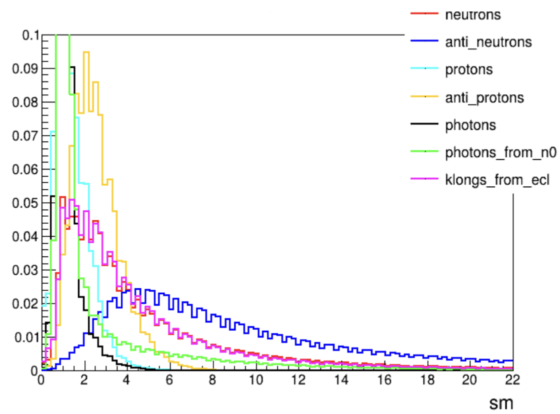


Figure 4.8: Comparison of *Second Moment* distribution for different particle species.

The Second Moment of photons is lower than that of hadrons, suggesting that photons have a smaller longitudinal development. In the example, anti-neutrons have the highest value of the second moment, indicating that their shower is more extended than those of the other particles.

4.3.5 Zernike Moments

The Zernike Moments (Zernike Polynomials) are a set of orthogonal functions which are defined on the unit disk. Since each moment represents a specific pattern shape, they are widely used to represent smooth, two-dimensional functions (especially wavefronts) over circular apertures.
On the unit disk parametrized by polar coordinates (ρ, ϕ) with $0 \leq \rho \leq 1$, the Zernike modes are usually written as:

$$Z_n^m(\rho,\phi) = R_n^m(\rho)\,\cos(m\,\phi)$$

$$Z_n^{-m}(\rho,\phi) = R_n^m(\rho)\,\sin(m\,\phi),$$

where:

$$R_n^m(\rho) = \sum_{k=0}^{\frac{n-m}{2}} \frac{(-1)^k\,(n-k)!}{k!\left(\frac{n+m}{2}-k\right)!\left(\frac{n-m}{2}-k\right)!}\,\rho^{n-2\,k}$$

In the formula, n represents the radial degree and m represents the azimuthal frequency ($m = 0$ for spherical Zernike polynomials). A figure of the Zernike polynomials is given in Fig. 4.9.

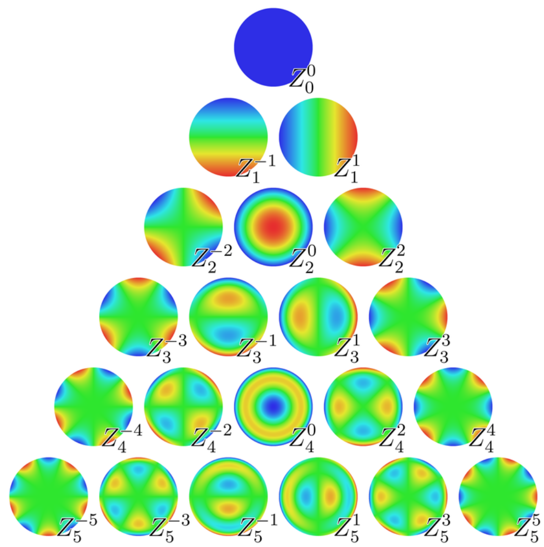


Figure 4.9: Zernike Polynomials representation. Moments Z_4^0 and Z_5^1 are widely discussed in this thesis.

Precisely, Zernike Moments Z_4^0 and Z_5^1 (*clusterAbsZernikeMomen40* and *clusterAbsZernikeMoment51* in the basf2 software) are studied in this thesis, where the first one represents the radial spread distribution structure, while the second one represents the level of dipolar asymmetry.
Z_4^0 and Z_5^1 represent the most effective Zernike moments for electron identification, which is why they are the only ones available in the basf2 software.

4.3.6 Number of Crystals hit

The variable Number of Crystals hit (*clusterNhits* in the basf2 software) indicates the number of crystals that contributed to the cluster. It returns the sum of the weights w_i, with $w_i \leq 1$, of all the crystals in the cluster. For non-overlapping clusters this is equal to the number of crystals in the cluster. However, in the case of overlapping clusters, where individual crystal energies are split among them, this can be a non-integer value.

Chapter 5
Study of the Monte Carlo signal channel

The first part of the present work aims to characterize the kinematics of the $\bar{n}$, using a Monte Carlo signal sample generated specifically for the selected reference channels:

$$e^+ \ + \ e^- \to p \ + \ \pi^- \ + \ \bar{n} \ (+ \ \gamma_{\text{ISR}})$$

This is selected because it features simple kinematics, with the $\bar{n}$ as the only missing particle, enabling a semi-exclusive analysis. The ISR gamma is a pre-interaction radiation emitted by the e^+ and/or e^-. The recoil system is made of $p\pi^-(\gamma_{ISR})$. The Feynman diagrams for these processes, without and with ISR emission, are shown in Fig. 5.1.

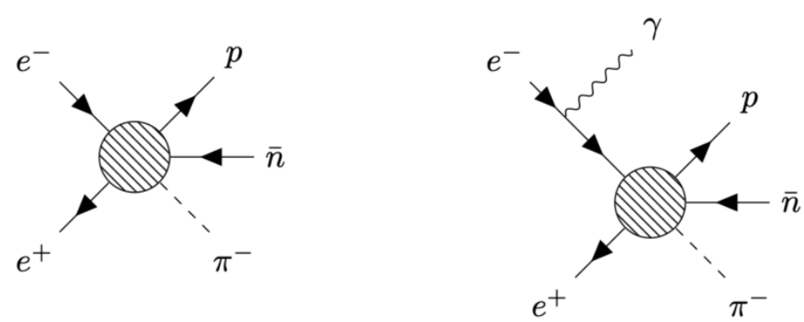


Figure 5.1: Feynman Diagrams of the channels of interest.

In particular, the kinematic relation between the $\bar{n}$ and the recoil system $p\pi^-(\gamma_{ISR})$ will be exploited, considering both configurations: in the presence of Initial State Radiation (ISR) photon emission and in its absence. ISR corresponds to the emission of photons from incoming particles before they collide in high-energy physics experiments. This reduces the effective center-of-mass energy available, which will no longer be $\sim$ 10.58 GeV, in the Belle II case. The ISR energy and angle distributions will be discussed in detail, providing a comparison with a simple phase space model emission.
Then, the effect of a one constraint (1C) kinematic fit over the recoil mass will be discussed, in order to obtain a higher recoil momentum resolution.
In total, 100000 events are generated and to perform the event generation and the analysis, basf2 is employed. Script code highlights for this section are shown in Appendix.
The basf2 Phokhara_evtgen_combination generator has been adopted. It is made

of two generators: Phokhara and EvtGen.
Phokhara is a Monte Carlo event generator that simulates e^+e^- annihilation into hadrons at next-to-leading order (NLO) accuracy. This includes virtual and soft photon corrections to single-photon events, plus double hard photon emission.
EvtGen is a Monte Carlo event generator that simulates decays of heavy-flavor particles, primarily B and D mesons. It includes decay models for intermediate and final states with scalar, vector, and tensor mesons/resonances, as well as leptons, photons, and baryons. Decay amplitudes generate each branch of the full decay tree, accounting for angular and time-dependent correlations to simulate CP-violating processes.
Therefore, these two generators have been adopted as depicted in Fig. 5.2.

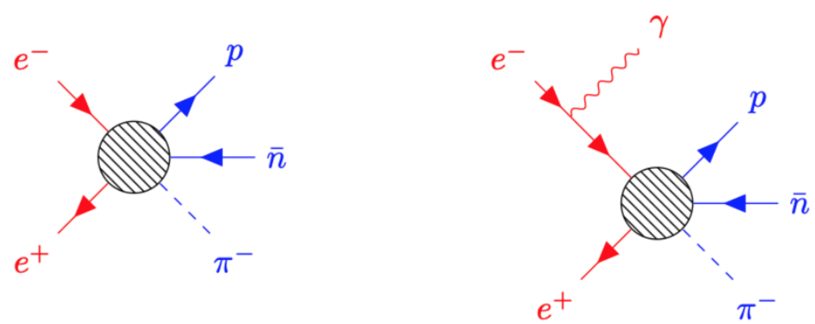


Figure 5.2: Feynman diagrams of the channels of interest. *Phokhara* handles the initial hard photon emission (red), while *EvtGen* simulates the decay to final-state particles (blue).

The Phokhara_evtgen_combination generator produces both events with and without ISR emission, as explicitly illustrated by the TopoAna overview of generated events in Fig. 5.3.
A *vpho* (virtual photon) is now introduced as a fictitious particle used by the generator Phokhara to produce ISR photons before the e^+e^- collisions and then enforce (evt_gen) the selected decay channel, approaching the following scheme:

$$e^+ \ + \ e^- \rightarrow vpho \ (+\gamma_{ISR}) \text{ (Phokhara generator)}$$

$$vpho \rightarrow p \ + \ \pi^- \ + \ \bar{n} \text{ (EvtGen generator)}$$

5.1 Kinematic variables comparison

Two scenarios can be studied. If the reconstructed final state particles are a proton (p), a negative pion (π^-) and an anti-neutron ($\bar{n}$), it is referred to as the NO ISR case. If the ISR photon (γ_{ISR}) is also reconstructed, it is referred to as the ISR case. Therefore, in the NO ISR case, the recoil system is defined as the $p\pi^-$ system, while in the ISR case it is defined as the $p\pi^-\gamma_{\rm ISR}$ system.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$vpho \to \pi^-\bar{n}p$	$\pi^-\bar{n}p$	0	35291	35291
2	$e^+e^- \to vpho\gamma^I\gamma^I, vpho \to \pi^-\bar{n}p$	$\pi^-\bar{n}p\gamma^I\gamma^I$	2	22971	58262
3	$e^+e^- \to vpho\gamma^I, vpho \to \pi^-\bar{n}p$	$\pi^-\bar{n}p\gamma^I$	1	18735	76997
4	$vpho \to \pi^-\bar{n}p\gamma^F$	$\pi^-\bar{n}p\gamma^F$	3	10005	87002
5	$e^+e^- \to vpho\gamma^I\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^F$	6	5274	92276
6	$e^+e^- \to vpho\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^F$	4	4621	96897
7	$vpho \to \pi^-\bar{n}p\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^F\gamma^F$	7	1503	98400
8	$e^+e^- \to vpho\gamma^I\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^F\gamma^F$	8	700	99100
9	$e^+e^- \to vpho\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^F\gamma^F$	5	597	99697
10	$vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^F\gamma^F\gamma^F$	9	167	99864
11	$e^+e^- \to vpho\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^F\gamma^F\gamma^F$	12	63	99927
12	$e^+e^- \to vpho\gamma^I\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^F\gamma^F\gamma^F$	10	61	99988
13	$e^+e^- \to vpho\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^F\gamma^F\gamma^F\gamma^F$	11	4	99992
14	$e^+e^- \to vpho\gamma^I\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^F\gamma^F\gamma^F\gamma^F$	15	4	99996
15	$vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^F\gamma^F\gamma^F\gamma^F$	14	2	99998
16	$vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^F\gamma^F\gamma^F\gamma^F\gamma^F$	13	1	99999
17	$e^+e^- \to vpho\gamma^I\gamma^I, vpho \to \pi^-\bar{n}p\gamma^F\gamma^F\gamma^F\gamma^F\gamma^F$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^F\gamma^F\gamma^F\gamma^F\gamma^F$	16	1	100000

Figure 5.3: Topology Analysis (TopoAna) output of the generated 100k events. iDcyTr is the unambiguous decay tree index associated to the correspondent channel, nEtr is the number of entries per channel and nCEtr is the cumulative number of entries.

As a first step, the kinematic variables are discussed through the recoil system consisting of $p\pi^-(\gamma_{ISR})$. In parallel, the neutral $\bar{n}$ cluster candidate is reconstructed and its properties are compared to those from the recoil, using different kinematic variables. Through the recoil analysis, the antineutron can be identified as the missing particle in the final state.
The first three contributes in Fig. 5.3 shows that events without ISR emission are the most frequently generated ($\sim 35\%$), followed by those with two ($\gamma^I\gamma^I$) ($\sim 23\%$) and one (γ^I) ISR photon ($\sim 19\%$).
Additionally, numerous Final State Radiation (FSR) photons are emitted by the charged final state particles due to next order perturbative QED processes.

5.1.1 ISR distributions

The ISR energy and angular distributions are discussed below. The primary effect of ISR emission is the reduction of the center-of-mass energy $\sqrt{s}$ below the $\Upsilon(4S)$ resonance.
Therefore, the energy E and θ distributions of ISR radiation are shown if Fig. 5.4.

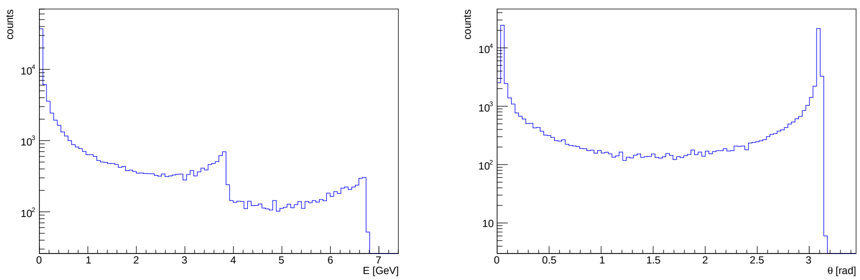


Figure 5.4: (left) ISR energy distribution. (right) ISR θ distribution. Both distributions are shown on a logarithmic y-axis scale.

The 2D energy E and θ distribution of ISR radiation are shown if Fig. 5.5.

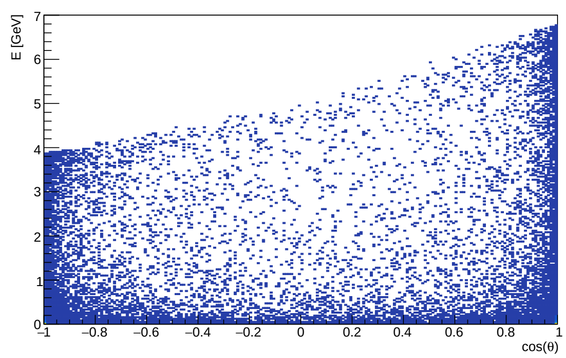


Figure 5.5: 2D energy E and $cos(\theta)$ distribution of ISR radiation.

In the ISR energy distribution, two peaks are observed slightly below 4 GeV and 7 GeV, corresponding to the energies of the e^+ and e^- beams, respectively. The angular distribution present a U-shape, indicating that forward and backward directions are the most probable.
Furthermore, as can be observed in Fig. 5.5, the higher the energy of the ISR radiation, the closer $|cos(\theta)|$ is to 1, indicating that high-energy γ_{ISR} photons are emitted preferentially in the forward and backward directions. In fact, peaks are observed at $\sim 7\,\mathrm{GeV}$ for $cos(\theta) \sim 1$ rad (e^- beam direction) and at $\sim 4\,\mathrm{GeV}$ for $cos(\theta) \sim -1$ rad (e^+ beam direction).

5.1.2 Event selections

To identify the event, the p, π^- (and γ_{ISR} for the ISR case) candidates are first identified in the final state, and their recoil mass (recoil system) is measured. The following selections are applied to p, π^- and recoil system:

- Recoil system: $0\ GeV/c^2 < recoil\ mass < 2\ GeV/c^2$ and in ECL acceptance;
- p: $protonID > 0.9$ and $dr < 1$ cm and $|dz| < 3$ cm;
- π^-: $pionID > 0.1$ and $dr < 1$ cm and $|dz| < 3$ cm.

The first one is a cut that selects a quite broad region of interest in the recoil mass distribution, corresponding to the interval [0,2] GeV/c^2 around the anti-neutron mass ($\sim 939,56\ MeV/c^2$), requiring that the recoil direction lies within the ECL acceptance.
The second and the third criteria are standard selections recommended by the basf2 framework. The ID requirement imposes a constraint on the protonID or pionID based on a maximum likelihood criterion as follows:

$$protonID = \frac{L_p}{L_e+L\mu+L\pi+L_K+L_d}; \quad pionID = \frac{L_\pi}{L_e+L\mu+Lp+L_K+L_d};$$

where L_p is the likelihood for the proton hypothesis, L_π is the likelihood for the pion hypothesis, and so on.
The cuts on dr and $|dz|$ select only tracks originating from the interaction point (IP) area, where dr is defined as the transverse distance with respect to IP and dz is defined as the axial distance, along the z-axis, with respect to IP.
The p, π^- (and γ_{ISR}) candidates can be identified through the recoil mass peak at the anti-neutron mass.
Parallel to the recoil system reconstruction, neutral ECL cluster $\bar{n}$ candidates are identificated. To analyze only $\bar{n}$ signals from the calorimeter, the following selection on $\bar{n}$ candidates is applied:

- $\bar{n}$: neutral candidates from the ECL;

this selection requires that the neutral $\bar{n}$ cluster candidates are reconstructed within the electromagnetic calorimeter (ECL), avoiding the $\bar{n}$ candidates from other detectors (mainly from KLM).
Furthermore, since multiple $\bar{n}$ neutral cluster candidates may exist per event, only the $\bar{n}$ candidate with the smallest angle -defined as α- relative to the recoil system is selected. Therefore, a selection is imposed using the basf2 rankByLowest method. It allows to select only the highest rank candidate, based on a specific variable; in this case, the angle between the neutral cluster and the recoil vector.
Using spherical coordinate system, after some algebraic steps, the 3D angle α between the two vectors can be obtained:

$$\alpha(\theta_1, \theta_2) = arcos[sin(\theta_1)sin(\theta_2)cos(\phi_1 - \phi_2) + cos(\theta_1)cos(\theta_2)]$$

Here vector 1 denotes the recoil vector and the vector 2 denotes the $\bar{n}$ direction vector. The respective α distributions, before and after the aforementioned selection, are shown in Fig. 5.6, in both ISR and NO ISR cases:

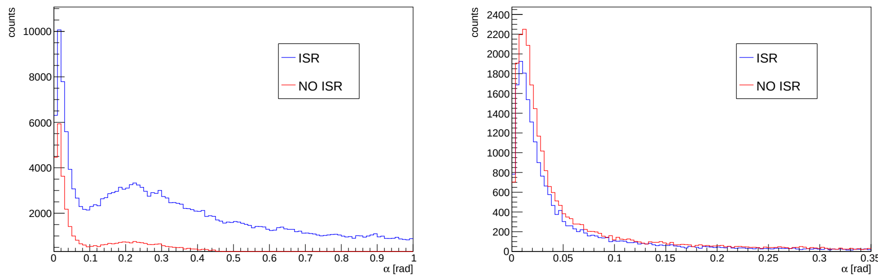


Figure 5.6: (a) α distribution. (b) α distribution after the rank selection.

To select the most probable $\bar{n}$ candidates, only those with $\alpha < 0.35$ ($\sim 20°$) are stored. The event is therefore fully defined after applying the discussed selection criteria.
The efficiency ϵ is defined as follows:

$$\epsilon = \frac{n^\circ \ of \ reconstructed \ candidates}{n^\circ \ of \ generated \ events}$$

Referring to Fig. 5.3, 35291 events are generated without γ emission ($iDcyTr \in 0$)and 41706 with ISR emission ($iDcyTr = 1, 2$; only the first two contributions are considered).
Since 20146 candidates are reconstructed for the ISR case and 25013 for the NO ISR case, an efficiency respectively of $\sim$ 48% and 71% can be observed. Since most of the ISR radiation is emitted in the backward ($\theta \sim 180°$) and in the forward ($\theta \sim 0°$) directions -where detection coverage is absent- the efficiency in the ISR case is lower than in the NO ISR case.

5.1.3 Recoil mass distribution

To select events with anti-neutrons, the recoil system $p\pi^-(\gamma_{ISR})$ is studied through the recoil mass. Specifically, the recoil mass should peak at the nominal anti-neutron mass, since no background channels are present in this sample. The background mainly consists of secondary particles produced during primary particle propagation through the detectors, combinatorial background, and incorrectly reconstructed γ.
The recoil mass is defined as follows:

$$\text{recoil mass} = \sqrt{\left(p_{e^+e^-} - p_{\text{sum}}\right)^2}$$

where $p_{e^+e^-}$ is the total four-momentum of the initial e^+e^- system at the $\Upsilon(4S)$ resonance, and p_{sum} is the sum of the 4-momenta of $p\pi^-(\gamma_{ISR})$.
The recoil mass distribution, for both ISR and NO ISR cases, is shown in Fig. 5.7.

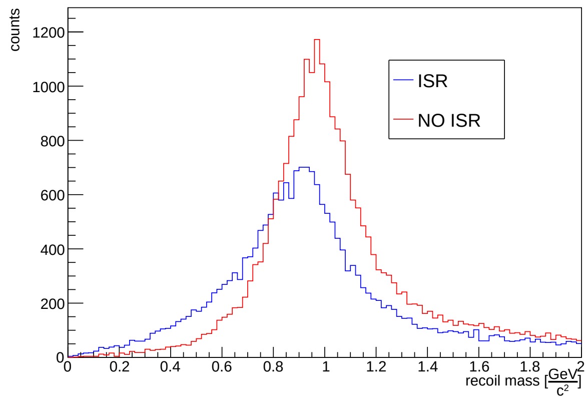


Figure 5.7: Recoil mass distribution, in both ISR and NO ISR cases.

The recoil mass distribution peaks on the $\bar{n}$ mass (~ 939.56 MeV), indicating that the the recoil mass is well reconstructed in both ISR ($p\ \pi^-\ \gamma_{ISR}$) and NO ISR ($p\ \pi^-$) cases. Therefore, the variables associated with the $\bar{n}$ clusters can be compared with the reconstructed recoil variables ($p,\ \theta$).
In the event simulation the generator produces more than one γ_{ISR} per event, reflecting into the obtained recoil mass distribution; the one obtained in ISR case shows more entries in the left tail, since only one γ_{ISR} is reconstructed in the recoil, while the others are not considered.
The distribution obtained in the NO ISR case exhibits a higher peak as the recoil system reconstruction consists only of $p\pi^-$, leading to a higher efficiency than ISR case.
By selecting events around the peak, $\bar{n}$ candidates events can be identified. The recoil direction is expected to provide a good estimate of the $\bar{n}$ momentum.

5.1.4 The recoil kinematic

Before analyzing the recoil kinematic distribution, the reconstructed momentum and $cos(\theta)$ distributions of each particle composing the recoil system is examined. For particles with MC association, the MC variables can be studied for a given list of candidates as well.
For example, considering the proton candidate list, both the reconstructed momentum and the Monte Carlo (generated) momentum of the proton list -which is not affected by the reconstruction process- are available. The latter will be referred to as the "generated" momentum. Since a candidate list is the result of the reconstruction process, it also contains particles that have been mis-identified as the main particles (for example, π^+ mis-identified as p or γ mis-identified as $\bar{n}$). For example, after reconstruction, several misidentified γ may be present in the $\bar{n}$ list:

$$\bar{n}\ \text{list} = (\bar{n}, \bar{n}, \gamma, \bar{n}, ...).$$

Therefore, the "generated momentum" of this list returns the generator-level momentum for each member of the reconstructed list. This differs from using Monte Carlo values for a generator-level particle list containing only the selected particle type -in this case, anti-neutrons.
Therefore, the term "generated variable", in the following discussion, refers to the Monte Carlo value of a variable for a given reconstructed particle list, as stressed in the example above.
The relative reconstructed momentum and $cos(\theta)$ distributions of protons and pions are shown in Fig. 5.8, while the generated one are shown in in Fig. 5.9.

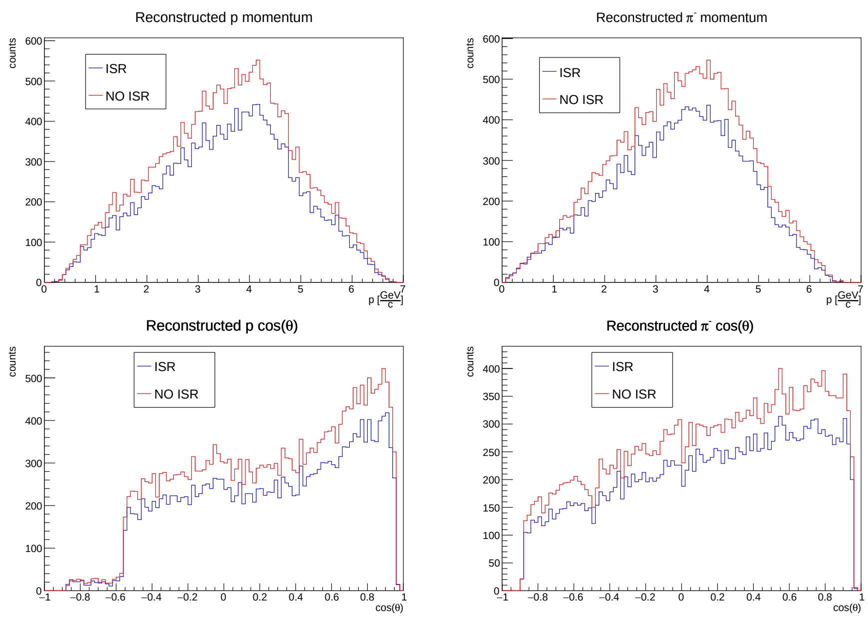


Figure 5.8: Reconstructed momentum and $cos(\theta)$ distributions of proton and pion, in both ISR and NO ISR cases.

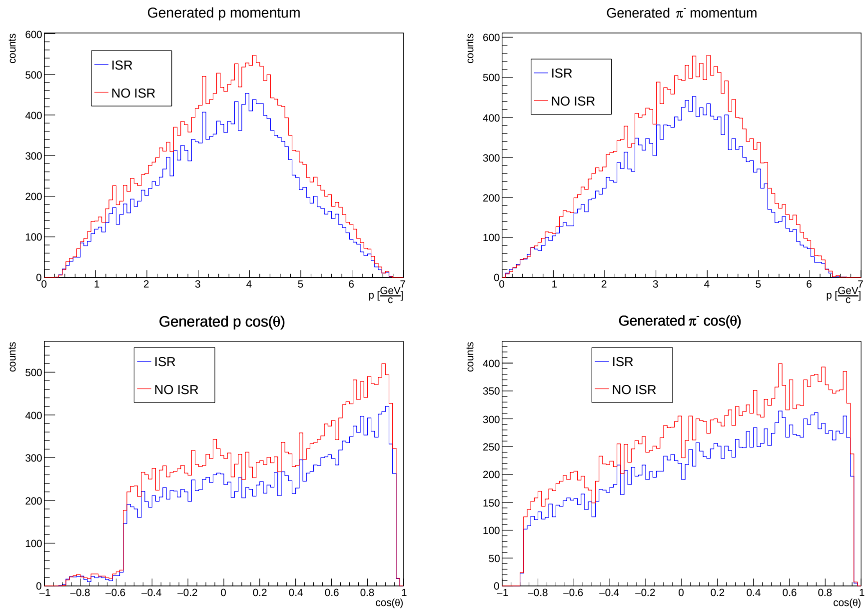


Figure 5.9: Generated momentum and $cos(\theta)$ distributions of proton and pion, in both ISR and NO ISR cases.

The distributions clearly exhibit the typical phase-space shapes, as expected from the adopted generator model. The $cos(\theta)$ distribution of protons presents low counts at $cos(\theta) \lesssim -0.6$. In this region, the lack of proper detector coverage results in insufficient proton identification performance for effective selection, failing the protonID cut. This effect is particularly pronounced for protons with momentum below 2 GeV/c, as shown in Fig. 5.10.

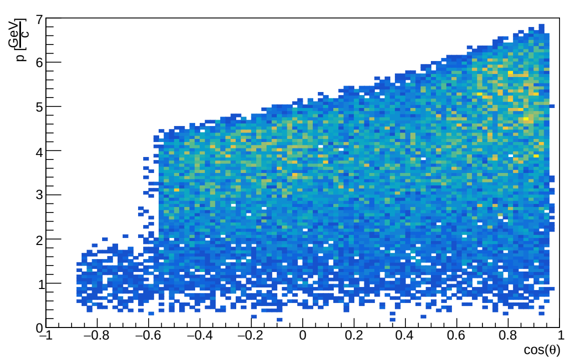


Figure 5.10: 2D distribution of proton momentum versus proton $\cos\theta$, in the NO ISR case.

The recoil momentum can be defined as:

$$\text{recoil momentum} = |\sum_i \vec{p}_i|$$

The recoil momentum distributions can be obtained in both ISR and NO ISR cases, as illustrated in Fig. 5.11. Since fewer candidates were reconstructed

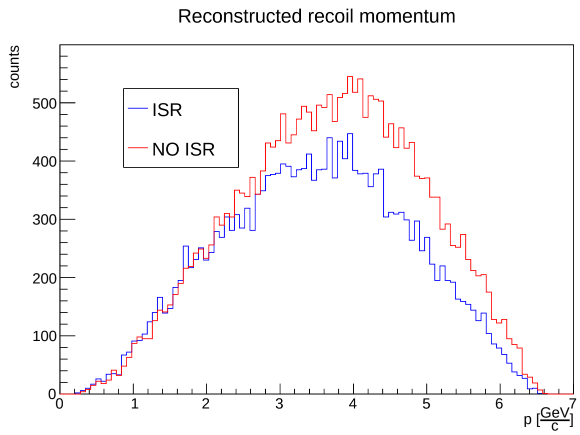


Figure 5.11: Momentum distribution of the recoil system, in both ISR and NO ISR cases.

in the ISR case than in the NO ISR case, the recoil momentum shows lower structure in the ISR case.
These distribution can be compared with that of the $\bar{n}$ candidate, both from reconstruction and from generator level.
The $\bar{n}$ candidate list originates from neutral clusters detected in the ECL. The software assumes that all shower energy is confined within the ECL. Therefore, since the calorimeter measures the energy of the clusters, the $\bar{n}$ momentum $p_{\bar{n}}$ can be defined as follows:

$$p_{\bar{n}}^2 = E_{ECL}^2 - m_{\bar{n}}^2,$$

where E_{ECL} is the energy of the $\bar{n}$ candidate cluster measured by the calorimeter and $m_{\bar{n}}$ is the mass of the anti-neutron.
The $\bar{n}$ momentum distribution is shown in Fig. 5.12.

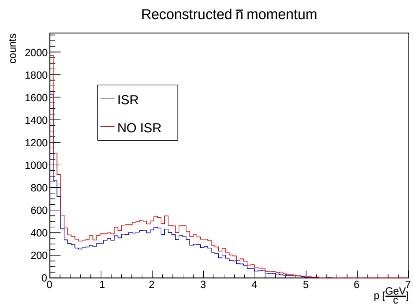


Figure 5.12: Reconstructed (left) and generated (right) momentum distribution of anti-neutron candidates, in both ISR and NO ISR cases.

Notably, the reconstructed $\bar{n}$ momentum is systematically underestimated compared to the generator-level one, indicating missing energy within neutral clusters. In fact, the reconstructed momentum distribution lies within [0, 5] GeV/c, showing a first structure at low momentum values in both the ISR and NO ISR cases. In the generator-level case, the distribution extends to [0, 6.5] GeV/c, with the same structure extending beyond 1 GeV/c.
Therefore, both the reconstructed and the generator-level distributions exhibit two structures, indicating two contributions within $\bar{n}$ candidate neutral clusters. To investigate this, the reconstructed recoil momentum is compared with the generator-level momentum of all $\bar{n}$ candidates, using MC truth matching. The neutral clusters associated to $\bar{n}$ list can be matched to true (truth = 1) $\bar{n}$ particles or to false (truth = 0) $\bar{n}$ particles. The obtained result is shown in 5.13.

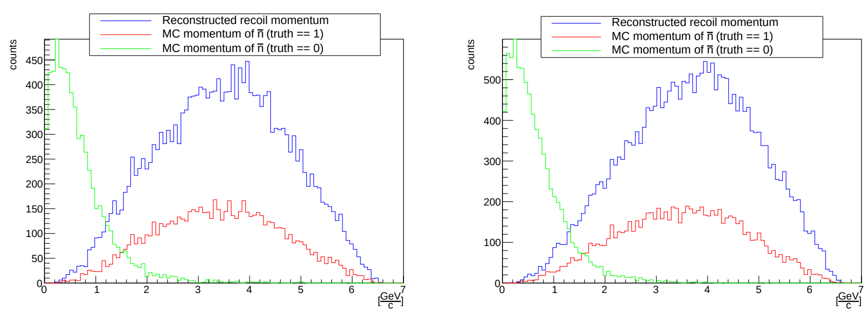


Figure 5.13: Recoil momentum and generated $\bar{n}$ list momentum. The blue distribution shows the recoil momentum, the red distribution corresponds to the generated particle associated to $\bar{n}$ by MC matching, while the green distribution corresponds to the generated particle associated to no $\bar{n}$ by MC matching.

The low-momentum structure corresponds to candidates not associated to $\bar{n}$ by the Monte Carlo truth, exhibiting significant deviation from the recoil momentum. This is likely due to cluster split-off and/or pre-showering effects, as will be discussed later. Furthermore, the distribution corresponding to the

generated $\bar{n}$ has less entries than the recoil momentum distribution, suggesting that several anti-neutrons are not correctly reconstructed. All these aspects will be discussed in the next section through analysis of $\bar{n}$ relatives (ancestors).
To obtain a measure of the recoil momentum and recoil θ resolution, these variables can be compared with those of the Monte Carlo matched $\bar{n}$ (truth = 1) candidates. The linear correlation coefficient ρ is studied, defined as:

$$\rho_{x,y} = \frac{cov[x,y]}{\sigma_x \sigma_y} \qquad \text{where } cov[x,y] = E[xy] - E[x]E[y]$$

The correlations in θ and in momentum are shown in Fig. 5.14 (generated) and in Fig. 5.15 (reconstructed).

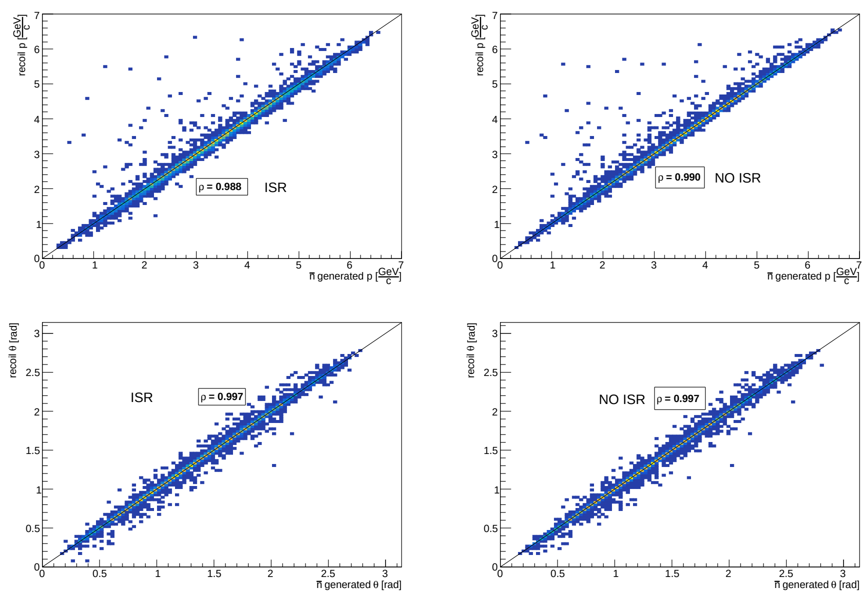


Figure 5.14: Obtained momentum and θ correlation distributions between recoil system and anti-neutron candidate, in generated case, in both ISR and NO ISR cases.

A good correlation is evident between the two θ distributions in both reconstructed and generator-level cases, indicating that the anti-neutron θ direction is properly described.
A good correlation can be observed between the reconstructed recoil momentum and the generated $\bar{n}$ one, confirming how the first is properly describing the $\bar{n}$ kinematic.
However, a poor resolution is observed in reconstructed $\bar{n}$ momentum, in both ISR and NO ISR cases, presenting an underestimation of the reconstructed $\bar{n}$ momentum as well.
The momentum residual distributions are shown in Fig. 5.16.

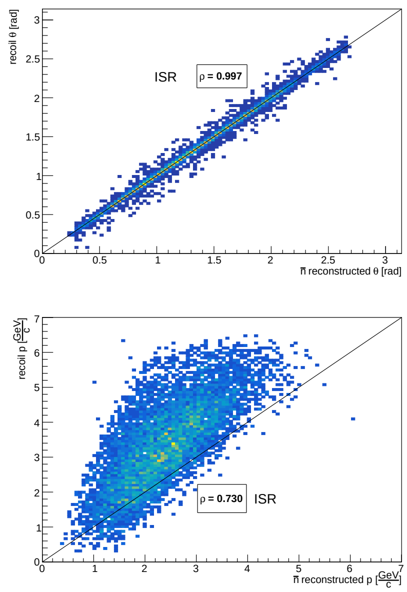


Figure 5.15: Obtained momentum and θ correlation distributions between recoil system and anti-neutron candidate, in reconstructed case, in both ISR and NO ISR cases.

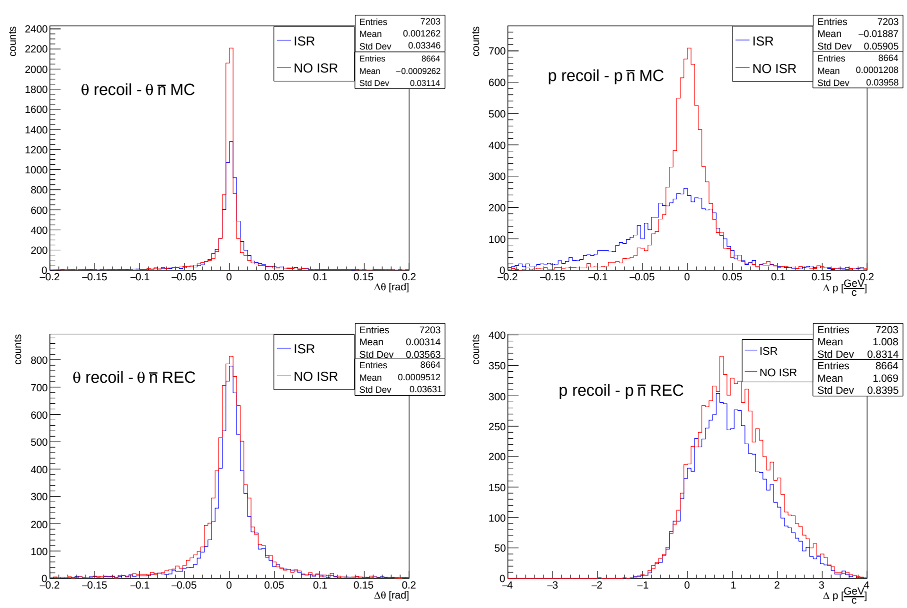


Figure 5.16: Momentum and θ angle residual distributions between recoil system and anti-neutron candidate, in both ISR and NO ISR cases.

Therefore, using only the ECL information, the $\bar{n}$ momentum is poorly reconstructed and is underestimated, as shown by the residual distributions, suggesting missing energy in the shower. Most of the anti-neutrons undergo in annihilation with nucleons such as $\bar{n}n$ and $\bar{n}p$, producing pions which escape the calorimeter. Therefore, the reconstructed momentum (or energy) in the calorimeter is not directly comparable with that of anti-neutrons, making the reconstructed $\bar{n}$ momentum an unreliable variable.
The θ distributions show good agreement, in both reconstructed and generator-level cases, suggesting that the $\bar{n}$ candidate direction is properly described by the recoil. Furthermore, considering only the θ residual distributions, the distributions in the NO ISR case exhibit higher resolution, explicitly confirming what was noted for the recoil mass distribution in Fig. 5.7.

5.2 The $\bar{n}$ relatives analysis

The origin of ECL clusters that are reconstructed as $\bar{n}$ candidates but are associated with other particles according to MC truth information is investigated in this part.
Cluster split-off and/or pre-showering effects can occur distorting the $\bar{n}$ reconstructed distributions. Cluster split-off occurs within the ECL when a portion of the primary shower produces signals in non-contiguous crystals. Pre-showering occurs when the primary interaction takes place in the TOP and its products travel towards the ECL, where they are detected as the primary signal.
Since the ISR and NO ISR results are analogous, only the ISR case is analyzed on the following section. A representation of cluster split-off and pre-showering is shown in Fig. 5.17.

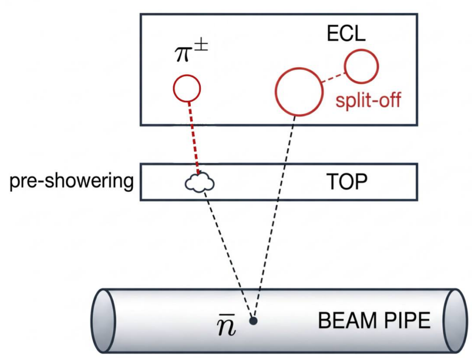


Figure 5.17: A representation displaying cluster split-off and pre-showering effects following $\bar{n}$ annihilation. Figure not to scale.

One of the secondary particles from $\bar{n}$ annihilation can be detected as the primary ECL cluster, resulting in information loss and requiring the TOP time information to improve the anti-neutron detection. Anti-neutrons, as neutral particles, do not emit Cherenkov radiation but can annihilate within the TOP volume. The resulting charged products then radiate in the bars. The Cherenkov signal is correlated with the anti-neutron time of flight and can be used to improve the energy estimation performed by the calorimeter [43].

5.2.1 The basf2 ancestor variables

Since $\bar{n}$ particles are selected along the recoil direction, there are ECL clusters in that direction that are not associated with $\bar{n}$ by MC matching. Their origin is studied, considering the possibility that $\bar{n}$ interacted with material.
In order to study the decay chain and relationsip degree of the neutral cluster associated to $\bar{n}$ candidates detected in the ECL, the basf2 ancestor variables are used. These identify whether a given particle originates from the decay of another particle or a subsequent decay chain. The following ancestor variables are employed:

- varForFirstMCAncestorOfType(type, variable): this metavariable returns the requested variable of the first ancestor of the given type -a specific particle- found in a certain point of the chain
- hasAncestor(PDG, abs): this metavariable returns a positive integer if an ancestor -a certain relatives in the chain- with the given PDG code is found, 0 otherwise. It means thats 1 = first mother, 2 = grandmother, etc. Second argument abs is optional: 1 means that the sign of the PDG code is taken into account, default is 0

A typical scenarios of $\bar{n}$ as ancestor could be the following:

0)$\bar{n}$ from IP $\rightarrow$ 1)$\bar{n}$ annihilation in TOP (pions produced) $\rightarrow$ 2) pions detected
0)$\bar{n}$ from IP $\rightarrow$ 1)$\bar{n}$ annihilation in ECL (pions produced) $\rightarrow$ 2) pions detected

The variables varForFirstMCAncestorOfType(anti-n0, p) and hasAncestor(-2112, 1) of $\bar{n}$ cluster candidates are discussed. Given the close similarity between the results, only the ISR case is discussed. The hasAncestor distribution of the selected $\bar{n}$ candidates is shown in Fig. 5.18.

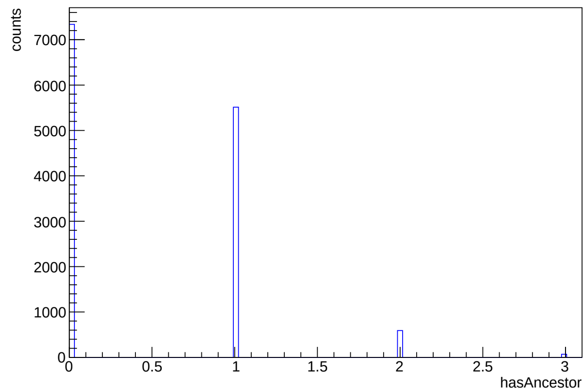


Figure 5.18: Distribution of the relative degree of anti-neutron candidates with an anti-neutron present in their ancestry chain.

From the distribution in Fig. 5.18, it is evident that the majority of candidates do not have a neutron ancestor, lacking an $\bar{n}$ in their relative chain (has_Ancestor=0), suggesting they are the primary $\bar{n}$ themselves. The second contribution is made of $\bar{n}$ candidates having $\bar{n}$ as mother etc.
In order to directly study the contribution of each term, the Monte Carlo truth is accessed via the corresponding mc_PDG. This is the PDG code of the selected candidates as associated by the Monte Carlo.
Therefore, the $\bar{n}$ candidates' mc_PDG code is studied for each HasAncestor contribution. The obtained distributions are shown in Fig. 5.19.

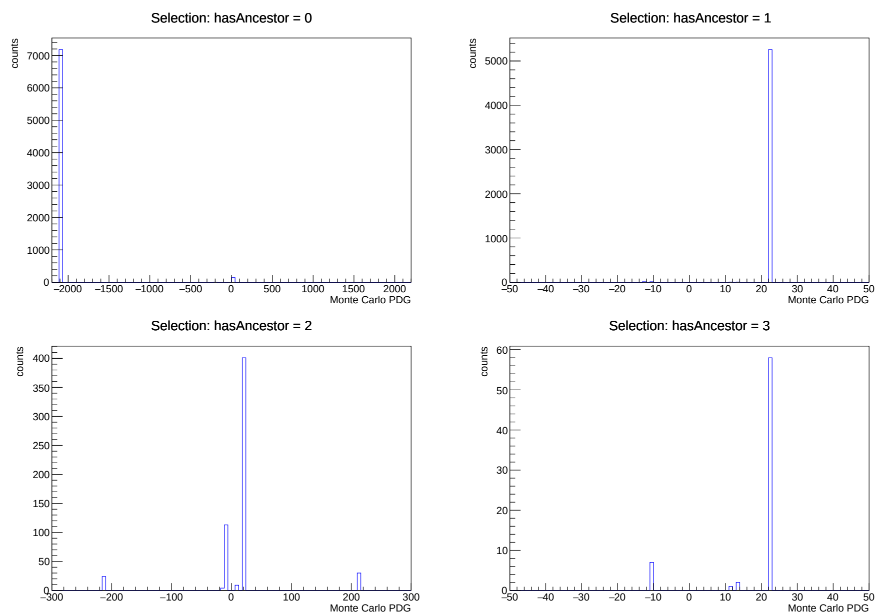


Figure 5.19: Monte Carlo PDG codes of candidates with degrees of ancestry selection.

- hasAncestor = 0 clearly shows that $\bar{n}$ candidates without an $\bar{n}$ as relatives are mainly true anti-neutrons, as their Monte Carlo PDG code is primarily -2112. This is expected since the anti-neutron comes from the IP and it is directly detected in the ECL through its own annihilation.

- hasAncestor = 1 shows that the $\bar{n}$ candidates with an $\bar{n}$ as mother are predominately gammas (Monte Carlo PDG = 22), suggesting they result from $\bar{n}$ interactions prior to $\bar{n}$ candidate ECL cluster formation.

- hasAncestor = 2 suggests that the $\bar{n}$ candidates with an $\bar{n}$ as grandmother are mainly gammas, with some spurious pions (-211 & +211) and electrons or positrons (+11 & -11). The case with hasAncestor = 3 yields analogous results.

5.2.2 Ancestor momentum

The figure Fig. 5.13 can be updated, by including the momentum of the ancestors obtained with varForFirstMCAncestorOfType(anti-n0, p). The ancestor is the primary $\bar{n}$, from which the secondary particle -mistaken for the primary- originates. The idea is to check whether the momentum information from the ancestors is consistent with that from the successfully detected primary $\bar{n}$. The obtained distributions are shown in Fig. 5.20.

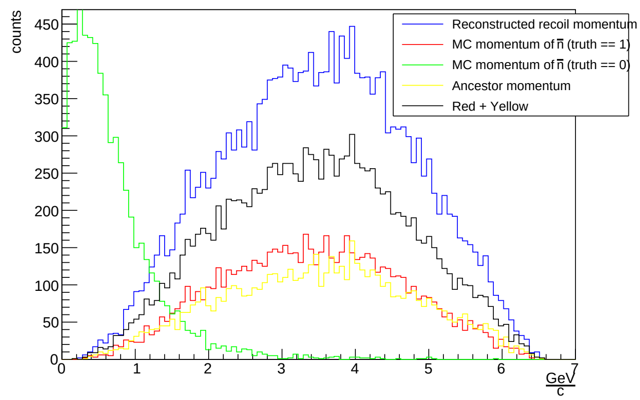


Figure 5.20: Distribution comparison of: recoil momentum (blue), generated momentum of $\bar{n}$ candidates associated to $\bar{n}$ by the Monte Carlo matching (red), generated momentum of $\bar{n}$ candidates associated to no $\bar{n}$ by the Monte Carlo matching (green) and generated momentum of $\bar{n}$ ancestors associated to $\bar{n}$ by the Monte Carlo matching (yellow). The black distribution is the sum of the second and fourth contributions. The recoil vector is the only reconstructed variable, while the others are obtained from Monte Carlo truth selection.

The recoil momentum is made of 20146 entries, corresponding with the total amount of candidates. The momentum distribution of $\bar{n}$ associated is made of 7203 entries, corresponding to 36% of total amount of candidates. The momentum distribution of no $\bar{n}$ associated is made of 6308 entries, corresponding to 31% of total amount of candidates. The momentum distribution of ancestors is made of 6173 entries, corresponding to 31% of total amount of candidates. The distribution of the sum of the second and fourth is made of 13376 entries.
This indicates that $\frac{6173}{6308} \times 100 \sim 98\%$ of the first structure shown in Fig. 5.13 is due secondary $\bar{n}$ candidates, namely particles that have a primary $\bar{n}$ as ancestor, suggesting that phenomena prior to the neutral cluster associated with $\bar{n}$ candidates can occur, such as cluster split-off and pre-showering. The quality of the ancestor momentum can be discussed by studying the residual distribution with respect to the reconstructed recoil momentum. The obtained distribution is shown in Fig. 5.21.

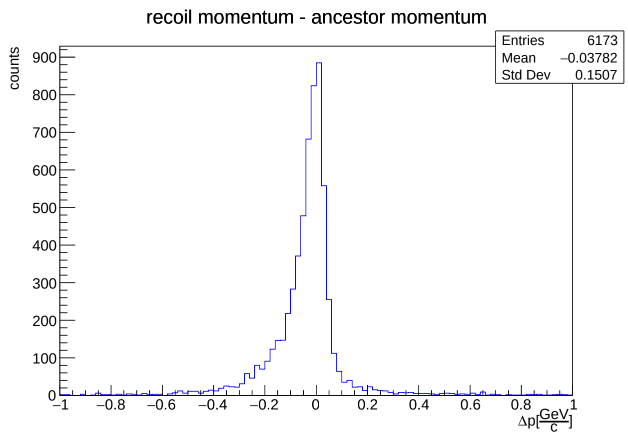


Figure 5.21: Residual distribution of primary $\bar{n}$ momentum (ancestor momentum) with respect to the recoil vector.

The reconstructed recoil momentum is a good estimator of the primary $\bar{n}$ (ancestor) momentum as well, confirming that cluster split-off or pre-showering may occur prior the ECL cluster formation.
Furthermore, clusters not associated with either primary $\bar{n}$ or their decay products lack PDG codes, indicating that the MC association has failed for these candidates. They are associated with a Not a Number (NaN) particle by the Monte Carlo matching. These are particles that did not receive a correct Monte Carlo association during MC truth matching, then they cannot be recovered.
This can be summarized using the meta-variable (mcErrors), which encodes as a bit code the reason why the Monte Carlo matching fails. This variable explicitly illustrates the behaviour shown in Fig. 5.22, where the first peak is at code 0, indicating perfect reconstruction of the particle and all its daughters, while the second peak is at code 512, confirming an MC matching error due to incorrect reconstruction of the relatives chain.

5.2.3 $\bar{n}$ production vertex

To verify the $\bar{n}$ production position, the Monte Carlo production vertex of the $\bar{n}$ is studied. To address it, the radial production distance ρ is introduced, defined as follows:

$$\rho = \sqrt{x^2 + y^2},$$

which is the Monte Carlo radial distance of $\bar{n}$ candidates, from the z-axis.
The distribution is splitted in two parts: the left part -close to the IP- ($\rho \in [0, 0.1]$ cm) and the right part -far from the IP- ($\rho \in (0.1, 220]$ cm). The obtained result is shown Fig. 5.23.

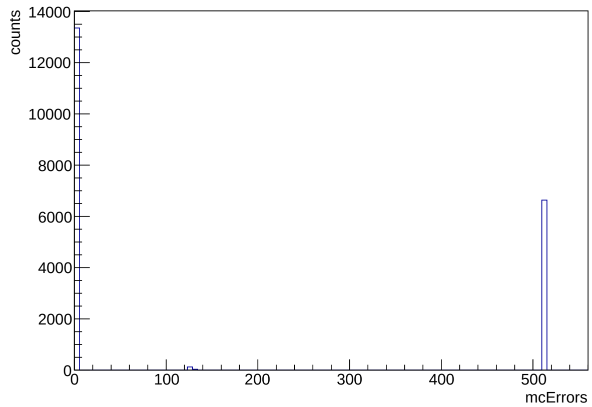


Figure 5.22: Monte Carlo Error code. 0 corresponds to a perfectly reconstructed particle, while 512 corresponds to a non valid MC match.

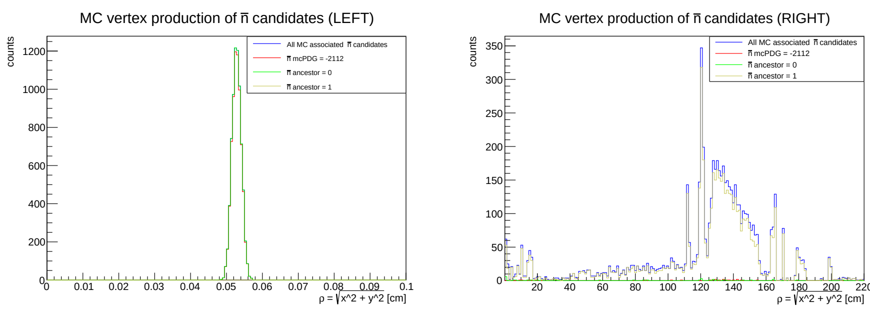


Figure 5.23: $\bar{n}$ production vertex: (left) close to the IP; (right) far from the IP. Blue distribution: all $\bar{n}$ associated by Monte Carlo matching; red: $\bar{n}$ candidates correctly associated to true $\bar{n}$; green: $\bar{n}$ candidates with no $\bar{n}$ relatives; yellow: $\bar{n}$ candidates with an $\bar{n}$ mother particle.

The correctly associated anti-neutrons overlay the $\bar{n}$ candidates without an anti-neutron ancestor, originating from the interaction point (left panel). This distribution is not centered around zero because the generator uses a default non-zero starting radial distance.
The $\bar{n}$ candidates with an $\bar{n}$ mother particle -that are mainly photons (Fig. 5.19)- are produced throughout the Belle II detector.
The first structures are in the vertex detector region ($\rho < 16$ cm) and the two peaks located at the TOP edges, mainly around 118 cm (where the TOP begins) and 125 cm (where the TOP ends and the ECL starts), explicitly showing how cluster pre-showering events occur in the events. Furthermore, an extended structure can be observed for $125\,\text{cm} < \rho < 162\,\text{cm}$, confirming that cluster

split-off effects occur within the ECL volume. Therefore, $\bar{n}$ candidates with $\bar{n}$ as mother particle ($\bar{n}$ ancestor = 1), may be produced before the ECL (pre-showering) or within the ECL (split-off).
The production vertex variable is now examined, dividing each contribute from the different ECL regions; backward region, barrel region and forward region. The obtained result is shown in Fig. 5.24.

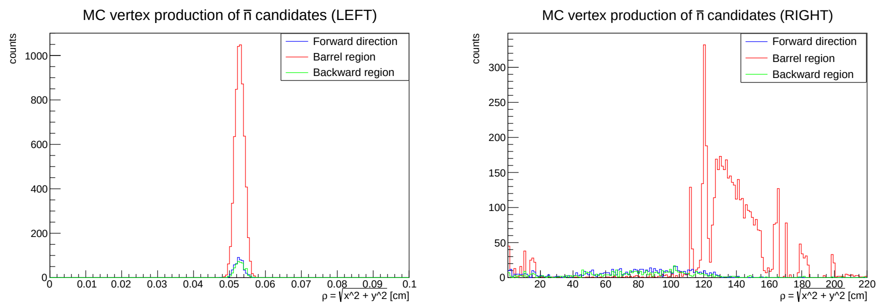


Figure 5.24: $\bar{n}$ production vertex: (left) close to the IP; (right) far from the IP. Blue distribution: $\bar{n}$ candidates detected in the forward region; red: $\bar{n}$ candidates detected in the barrel region; green: $\bar{n}$ candidates detected in the backward region.

In the forward and backward regions, the two peaks located at the TOP edges disappear, as expected, since the TOP provides no coverage in those regions. The same conclusions hold for the first structures in the vertex detector region, where the PXD and SVD are located.

5.3 Kinematic Fit effects

A Kinematic Fit (KF) is a constrained optimization technique used in particle physics to refine the measured four-momenta (or other kinematic quantities) of particles by imposing known physical constraints, such as energy-momentum conservation or recoil mass. Measured quantities like track momenta, energies from calorimeters, or decay vertices come with limited resolution from detectors. The fit adjusts these within their uncertainties to best satisfy the imposed constraint.
A one constraint 1C Kinematic Fit has been applied, constraining the recoil mass to the value of the anti-neutron mass ($\sim 0.9385\ GeV/c^2$) and then studying the consequence on the recoil kinematic variables resolution. The comparison of the residual distributions with and without the kinematic fit is shown in Fig. 5.25.

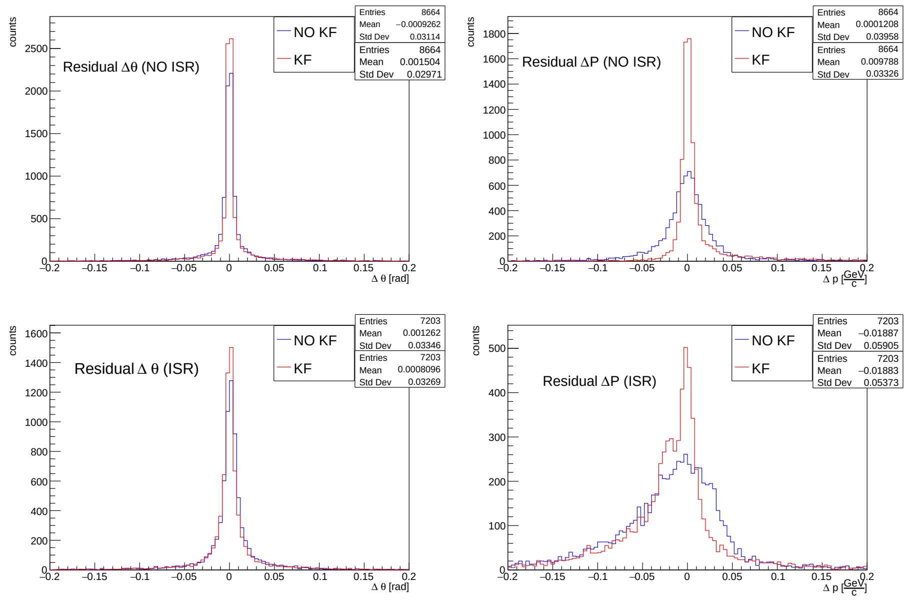


Figure 5.25: Momentum and θ angle residuals distribution between recoil system and anti-neutron candidate. Blue: before the application of the Kinematic Fit. Red: after the application of the Kinematic Fit.

Based on standard deviation value, an improvement can be observed due to the application of the kinematic fit.
In the ISR case, a 5% improvement is observed in the recoil θ resolution, while a 9% improvement is observed in the recoil momentum resolution. In the NO ISR case, a 2% improvement is observed in the recoil θ resolution, while a 9% improvement is observed in the recoil momentum resolution.
Therefore, the kinematic fit, by constraining the recoil mass, refines the resolution of the recoil kinematic variables.

5.4 Anti-neutron candidates ECL cluster variables

To isolate the contribution of neutral clusters correctly associated with $\bar{n}$ particles by the Monte Carlo association from all $\bar{n}$ candidates, the two distributions are overlaid for each cluster variable.

The obtained cluster variables in ISR case are shown in Fig. 5.26.

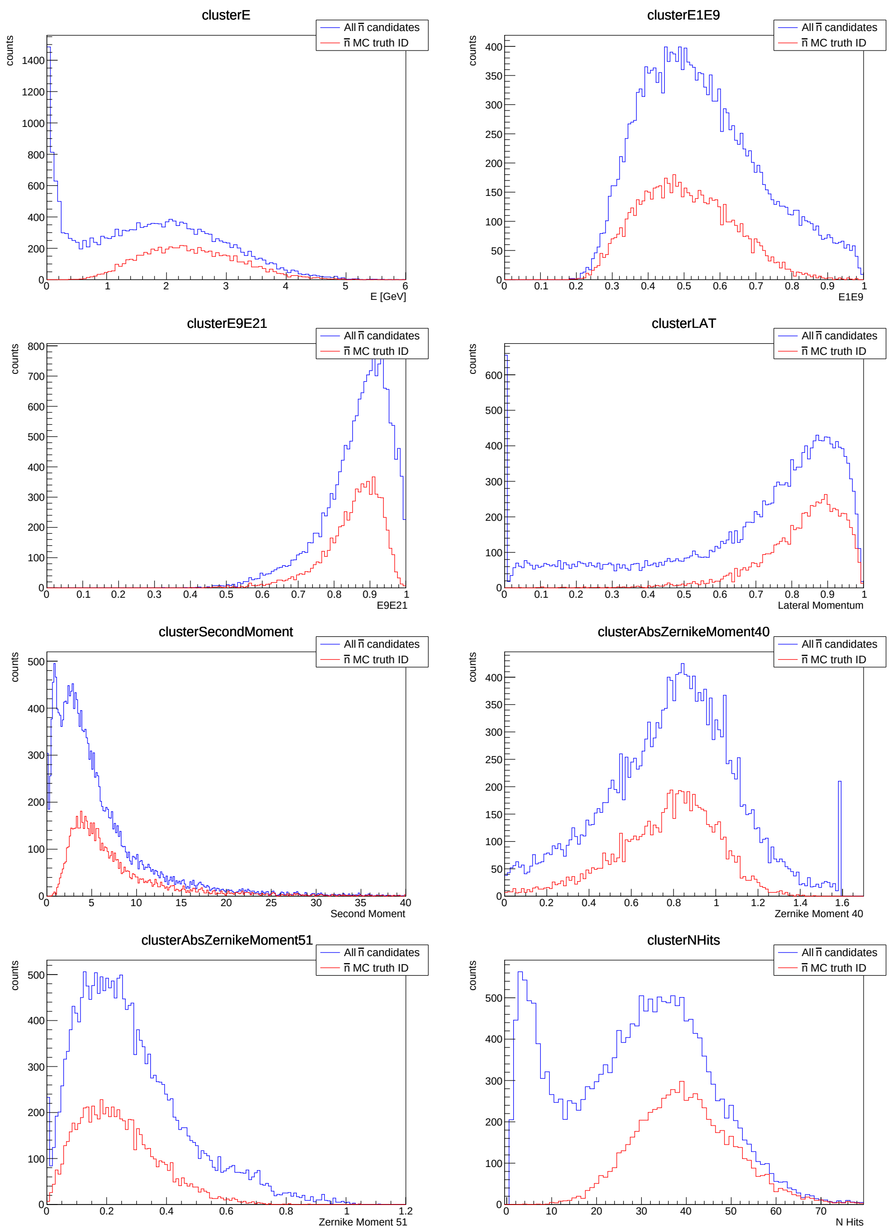


Figure 5.26: Distribution of the reconstructed cluster variables for all candidates (blue) and for clusters associated to $\bar{n}$ by MC truth, for ISR events.

The obtained cluster variables in NO ISR case are shown in Fig. 5.27.

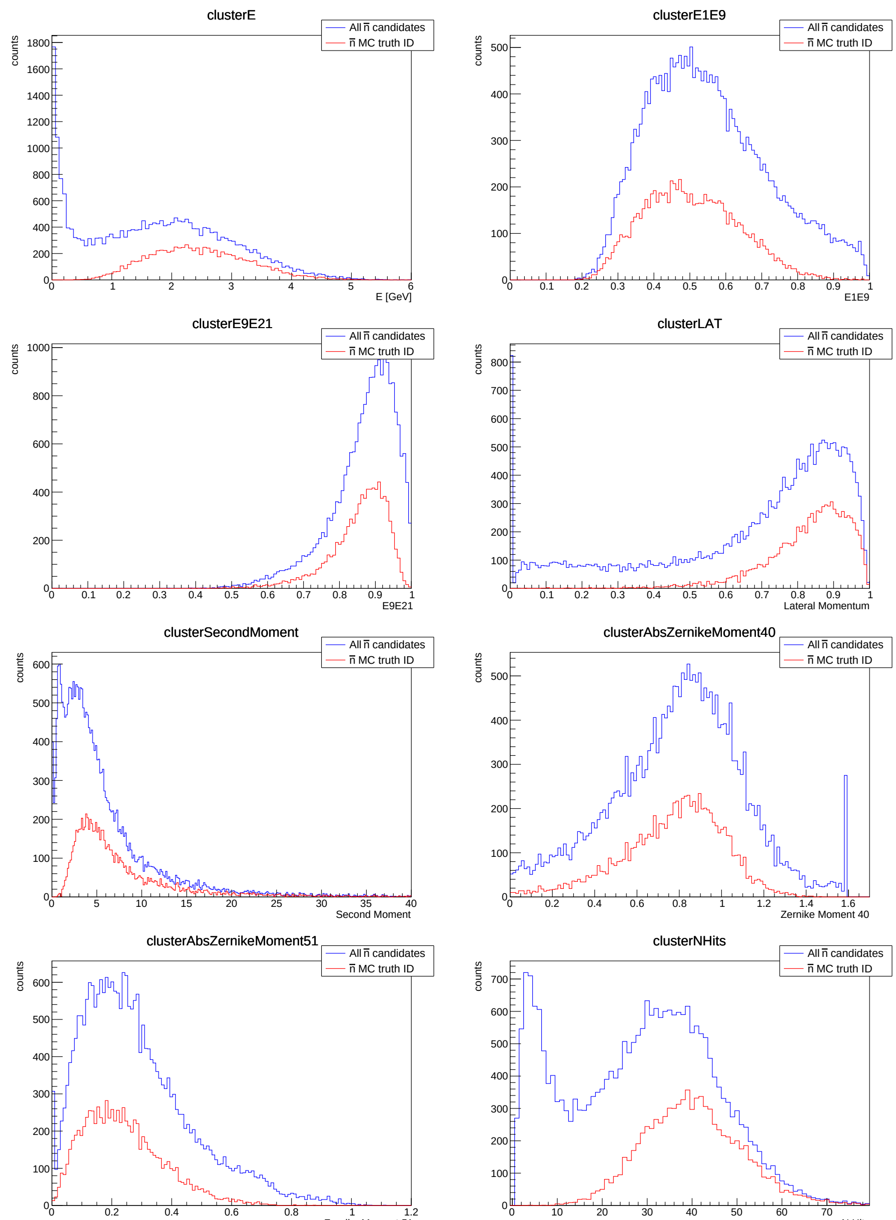


Figure 5.27: Distribution of the reconstructed cluster variables for all candidates (blue) and for clusters associated to $\bar{n}$ by MC truth, for NO ISR events.

The ISR and NO ISR cluster variables distributions presents the same information. Indeed, the measurement of cluster variables is independent of the recoil system, which could be made of either $p\pi^-\gamma_{ISR}$ or $p\pi^-$ respectively. Therefore, only the NO ISR case will be discussed in the rest of the chapter.
The cluster variable distributions of all candidates show peculiar structures compared to those correctly associated with $\bar{n}$ particles. For example, the reconstructed cluster energy for all candidates exhibits a first structure at low energy values that resembles those discussed for the anti-neutron momentum distributions. Another structure is also evident in the second moment distribution, where a structure at low values is clearly visible. The most prominent difference appears in the number of hits ($N\ Hit$) distribution, where a distinct structure at $N\ Hit < 10$ is evident.
The nature of these structures is analyzed in the following paragraph.

5.4.1 Study of cluster structures with ancestor variable

The nature of the structures not observed in the cluster variables of $\bar{n}$ associated candidates is now investigated, studying the ancestor contributions. This allows to determine whether a specific structure originates directly from primary $\bar{n}$ annihilation or from its annihilation products.
The shower shape variables with the following candidate selections are compared:

- $\bar{n}$ Monte Carlo PDG = -2112 (neutral clusters correctly associated with $\bar{n}$ particles),
- $\bar{n}$ Monte Carlo PDG = NaN (neutral clusters not associated with any particles),
- $\bar{n}$ hasAncestor = 0 (neutral clusters with no $\bar{n}$ as relative),
- $\bar{n}$ hasAncestor != 0 (neutral clusters with $\bar{n}$ as relative).

Referring to figures 5.28 and 9.3 (ISR case in appendix), the clusters correctly associated with $\bar{n}$ particles carry the same information of those without $\bar{n}$ as relative, in both ISR and NO ISR cases.
Furthermore, reconstructed neutral clusters not associated with any particles by the Monte Carlo and those with $\bar{n}$ as relative are primarily responsible for the structures observed at low values of energy, second moment, and number of crystals.
A further selection on number of crystals ($clusterNHits > 10$) can be performed to study the corresponding cluster variables, since no $\bar{n}$ signal in present below that value from Monte Carlo. The result is shown in figure 5.29 and 9.4 (ISR case in appendix).
These structures arise from small-sized clusters produced by $\bar{n}$ particles that undergo phenomena -such as cluster split-off and pre-showering- prior to neutral cluster formation for $\bar{n}$ candidates, as confirmed by ancestor studies. The requirement of at least 10 crystals eliminates these structures, without substantially reducing the signal.

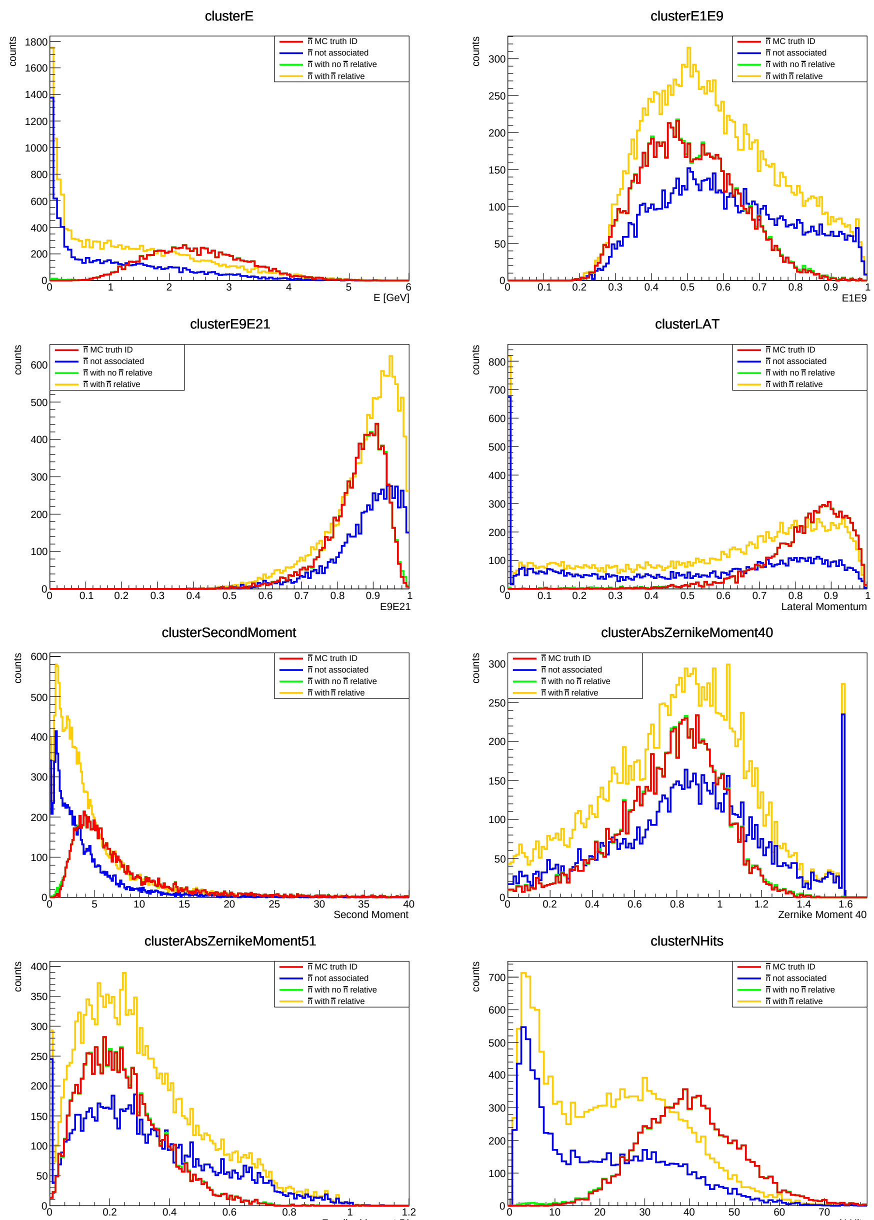


Figure 5.28: Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (NO ISR case). Red: $\bar{n}$ candidates associated with true $\bar{n}$ by Monte Carlo matching; blue: $\bar{n}$ candidates not associated with any particle (NaN); green: $\bar{n}$ candidates without $\bar{n}$ relatives; yellow: $\bar{n}$ candidates with $\bar{n}$ relative. The third contribution (green) is barely visible as it lies beneath the first distribution (red).

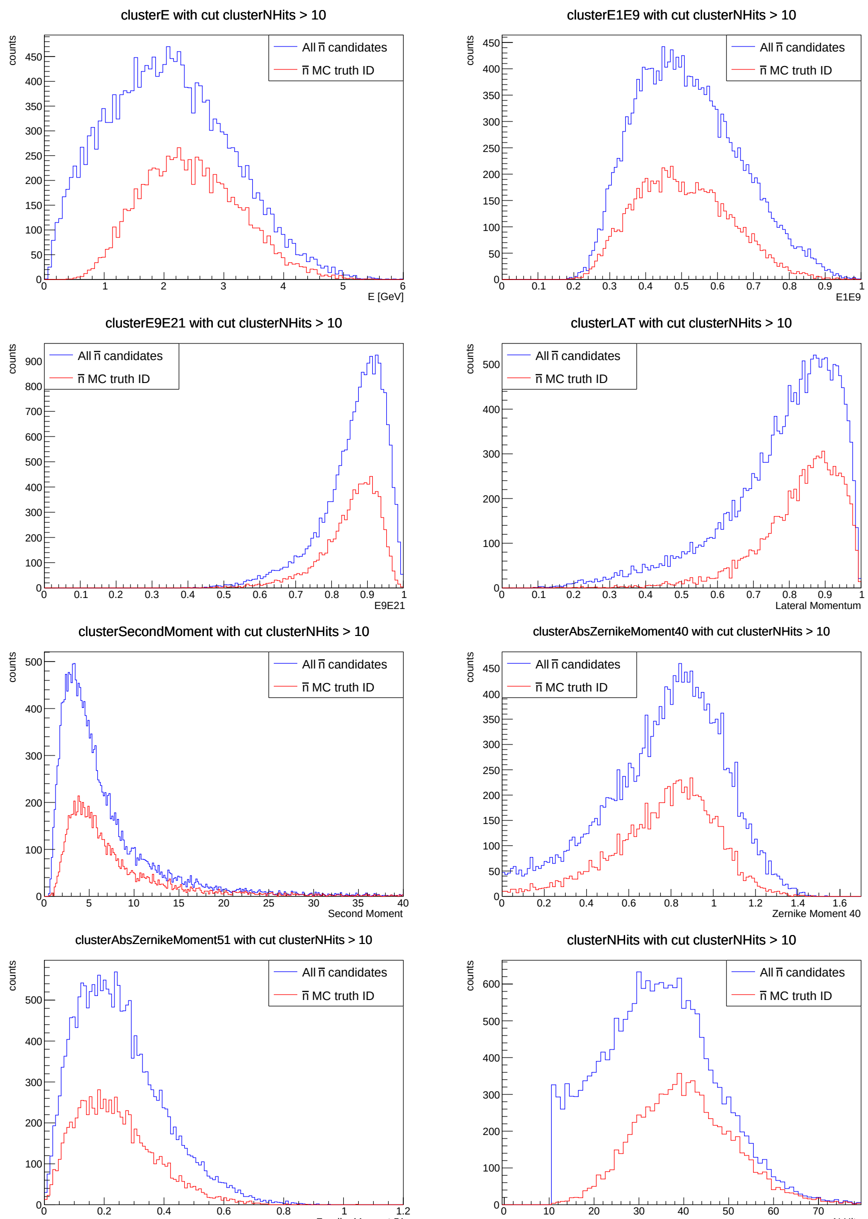


Figure 5.29: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates (blue) and those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (red) (NO ISR case). Cut $clusterNHits > 10$ is performed.

5.4.2 Cluster variables in different ECL regions

The Belle II detector is divided into barrel and forward/backward end-cap regions. The reconstructed cluster variables are now studied in each ECL region separately. Since the ISR and NO ISR cases exhibit analogous shower shapes, only the latter is discussed here, with the applied cut $clusterNHits > 10$.
The region hit distribution is shown in Fig. 5.30:

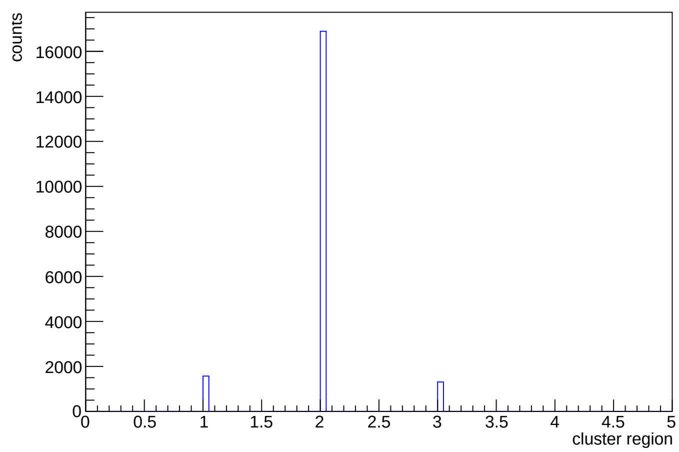


Figure 5.30: Different ECL regions contribution in neutral cluster associated to $\bar{n}$ candidates, where 1 is the forward region, 2 is the barrel region and 3 is the backward region. The barrel region is the most involved.

As observed in Fig. 5.31, the cluster variable associated to all the reconstructed $\bar{n}$ shapes are consistent across all ECL regions.
However a slight difference can be noted for cluster energy and number of hits variables. In fact, these two exhibit a shift toward lower values in the forward region, reflecting the Lorentz boost of the center-of-mass. In any case, the barrel region contains the majority of ECL clusters.

The obtained cluster variables in NO ISR case are shown below:

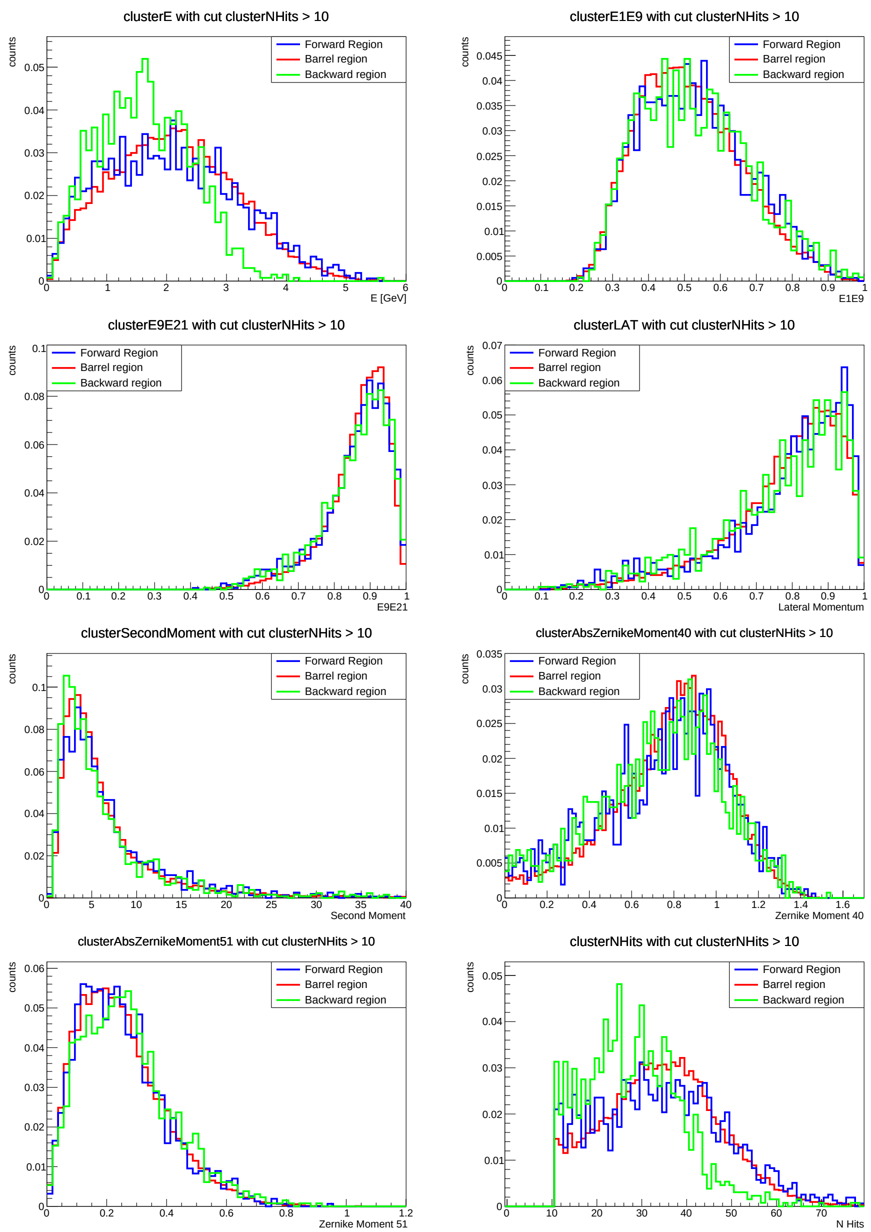


Figure 5.31: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates in different ECL regions. Blue: forward region. Red: barrel region. Green: backward region. The distributions are normalized to unit value.

5.5 Summary of Signal Channel Analysis

The channel $e^+ + e^- \to p + \pi^- + \bar{n}$ has been studied.

The recoil system $p\pi^-(\gamma_{ISR})$ is correctly reconstructed by the analysis, as evidenced by the peak above the anti-neutron mass in the recoil mass distribution. Furthermore, the kinematic distributions (p and θ) of the recoil system are consistent with those of generator-level $\bar{n}$ candidates, in both ISR and NO ISR cases. At reconstructed level, good agreement is observed in the θ distribution, while the reconstructed $\bar{n}$ candidate momentum shows underestimation and poor resolution compared to the recoil system momentum, as observed in correlations and residuals distributions, due to loss of energy leakage. Furthermore, a low-momentum structure is observed in reconstructed and generated momentum of $\bar{n}$ candidates compared to the recoil system momentum.

Energy leakages occur within ECL clusters, suggesting processes such as cluster split-off and/or pre-showering effects. Ancestor studies reveal that photons related to $\bar{n}$ annihilation are often misidentified as main $\bar{n}$ candidates, accounting for the low-momentum structure observed. In fact, the ancestors momentum can be partially recovered when correct Monte Carlo associations occur, showing good agreement with the recoil system momentum as well. A study of $\bar{n}$ production reveals that candidates with $\bar{n}$ ancestors are generated far from the IP, before the ECL (pre-showering) and within the ECL (cluster split-off).

Furthermore, a one-constraint (1C) Kinematic Fit was tested, revealing improvements in the recoil kinematic variables, such as the momentum and θ, resolution.

The reconstructed cluster variables associated with $\bar{n}$ candidates -such as cluster Energy, Lateral Momentum, Second Moment, etc.- were studied in the Monte Carlo signal sample. Similar structures to those observed in the momentum distributions were identified, particularly in the cluster Energy and Number of Hits distributions. Ancestor studies confirm the momentum distribution observations: sub-clusters from secondary particles -with $\bar{n}$ as relative- produce small-sized, low-energy structures in the cluster distributions.

In particular, a cut on the cluster number of hits yields reconstructed cluster variables closer to those expected from Monte Carlo $\bar{n}$ candidate selection, in both ISR and NO ISR cases.

Chapter 6
Study of the Monte Carlo cocktail production

The second part of the analysis of this thesis is devoted to the study of **Belle II Monte Carlo cocktail simulations**. A Belle II "cocktail" MC production (typically generated with Pythia8) simulates all relevant physics channels simultaneously. Therefore, the analyzed channel $e^+e^- \to p\pi^-\bar{n}\,(\gamma_{\mathrm{ISR}})$ is now embedded within all other background channels in the cocktail; the first task is to correctly reconstruct it amidst the surrounding backgrounds, studying the recoil mass distribution.
To achieve this, a first attempt uses the complete RUN1 $u\bar{u}$ cocktail production. Building on the signal sample discussed in the last chapter, further optimal cuts and selection criteria are developed to properly isolate the signal in the recoil mass distribution.
Afterwards, the full Belle II $q\bar{q}$ ($u\bar{u}$, $d\bar{d}$, $c\bar{c}$, $s\bar{s}$) production is analyzed, discussing the corresponding $\bar{n}$ candidates cluster variables.
The Feynman diagram of $e^+e^- \to q\bar{q}$ is shown in Fig. 6.1.

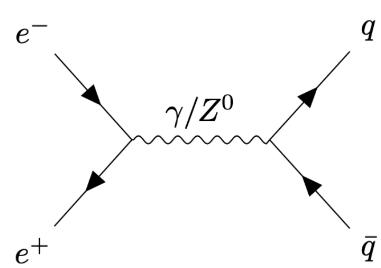


Figure 6.1: Tree level Feynman diagram of $e^+e^- \to q\bar{q}$

6.1 Types of processes at the Belle experiment enviroment

Many processes can occur during an e^+e^- collision. These may mimic the selected signal and be accidentally reconstructed as signal events, though they represent background processes.
The types of processes present depend on if beams collide on-resonance or off-resonance in reference to the $\Upsilon(4S)$ resonance. $\Upsilon(4S)$ production occurs at $\sqrt{s} = 10.58$ GeV, yielding $\Upsilon(4S) \to BB$ processes. The B^+B^- and $B^0\bar{B}^0$ final states are referred to as charged and neutral $B\bar{B}$ pairs, respectively.
The $e^+e^- \to q\bar{q}$ ($q = u, d, s, c$) processes, known as continuum production, also occur. The cross section distribution [22] of $e^+e^- \to Hadrons$ process is shown in Fig.6.2.

Process	Cross section (nb)
$e^+e^- \to \Upsilon(4S)$	1.10
$e^+e^- \to u\bar{u}(\gamma)$	1.61
$e^+e^- \to d\bar{d}(\gamma)$	0.40
$e^+e^- \to s\bar{s}(\gamma)$	0.38
$e^+e^- \to c\bar{c}(\gamma)$	1.30
$e^+e^- \to \tau^+\tau^-(\gamma)$	0.92
$e^+e^- \to e^+e^-(\gamma)$	300
$e^+e^- \to \mu^+\mu^-(\gamma)$	1.15
$e^+e^- \to \gamma\gamma(\gamma)$	4.99
$e^+e^- \to e^+e^-e^+e^-$	39.7
$e^+e^- \to e^+e^-\mu^+\mu^-$	18.9

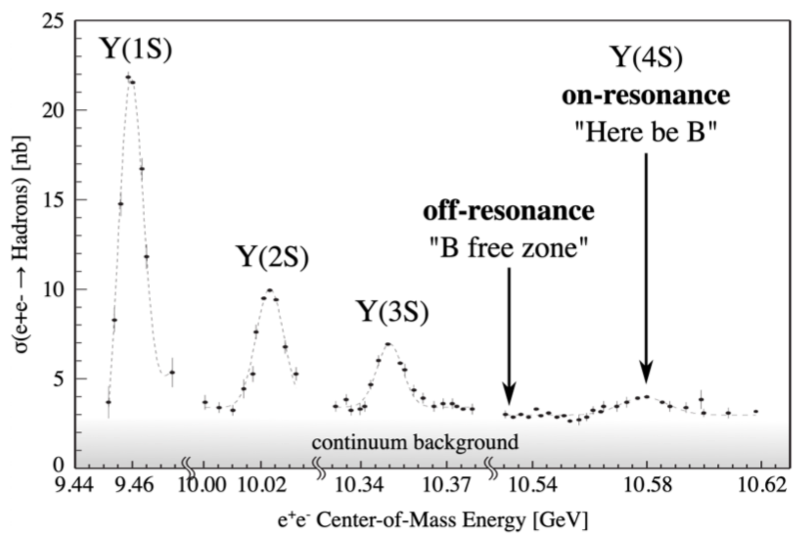


Figure 6.2: Cross section of $e^+e^- \to Hadrons$, in Υ resonances energy region. $\Upsilon(1S)$ resonance is very narrow, while $\Upsilon(4S)$ one is broader and lower.

6.2 Event selections

To follow the same approach as used in the signal sample analysis, the recoil mass is first studied using the complete $u\bar{u}$ cocktail production, which corresponds to an integrated luminosity of $\sim 1429.23\ fb^{-1}$. The following event selections have been applied:

1. Recoil system: $0\ GeV/c^2 < recoil\ mass < 2\ GeV/c^2$ and in ECL acceptance;

2. $\bar{n}$: neutral clusters from ECL and number of hits > 10 and energy < 3 GeV;

3. p: $protonID > 0.9$ and $dr < 1$ cm and $|dz| < 3$ cm;
4. π^-: $pionID > 0.1$ and $dr < 1$ cm and $|dz| < 3$ cm;
5. $\alpha < 0.20$ rad $\sim 11.5°$;
6. No further charged particles in the Rest of Event (ROE).

Four additional cuts are applied beyond those used for the signal sample, such as cluster number of hits > 10 GeV, $\alpha < 0.20$ rad, the request of no further charged particles in the Rest of Event (ROE) and the energy of the neutral cluster associated to $\bar{n}$ candidates must be less then 3 GeV.
The first cut arises from observations in the previous chapter; to achieve a proper selection of neutral clusters associated with anti-neutron candidates -reducing cluster split-off and pre-showering contributes- a requirement on the number of lighted crystals is applied.
The second cut allows a stricter selection of anti-neutron candidates through a smaller α angle. α is defined as the 3D angle between the recoil vector direction and the neutral cluster associated with the $\bar{n}$. As will be explained later, this threshold can be lowered further.
To explain the third cut, the concept of **Rest of Event** (ROE) must first be introduced. The ROE is built around the reconstructed target particles and consists of all other particles in the Rest of Event. In the present case, the reconstructed particles are $p, \pi^-, \bar{n}$ (and γ_{ISR}); therefore, the ROE includes all particles outside this reconstructed system. Consequently, the third selection criterion requires that no charged particles -i.e. tracks associated with charged particles- be present in the Rest Of Event, providing a cleaner event reconstruction. A Feynman diagram with ROE is shown in Fig.6.3.

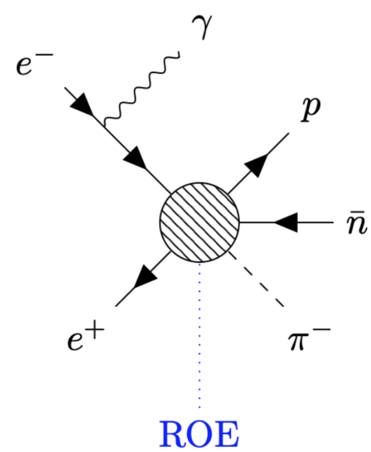


Figure 6.3: Feynman diagram of the channel of interest, with the Rest of Event ROE made of a variable amount of particles.

The fourth cut arises from cluster energy studies performed on the first chunk of $u\bar{u}$ cocktail production, corresponding to $\sim 215fb^{-1}$. The $\bar{n}$ list can be affected by several mis-identified $\gamma_{\rm ISR}$. In order to study this phenomenon, the Monte Carlo ISR flag (**mcISR**) can be used to separate the two contributions in reconstructed $\bar{n}$ list. The mcISR variable is a Monte Carlo flag that indicates whether a particle is an ISR particle or not; if mcISR $= 0$, the reconstructed particle is not an ISR photon; if mcISR $= 1$, the reconstructed particle is an ISR photon mis-identified as an anti-neutron. The obtained cluster energy distribution for $\bar{n}$ candidates, in $u\bar{u}$ chunk1 cocktail production, is shown in figure 6.4.
Two structures, at $\sim$ 4 GeV and $\sim$ 6 GeV, due to γ_{ISR} mis-identified as anti-neutrons, arise. In order to remove them, a cut on cluster energy, such as those discussed, is mandatory.

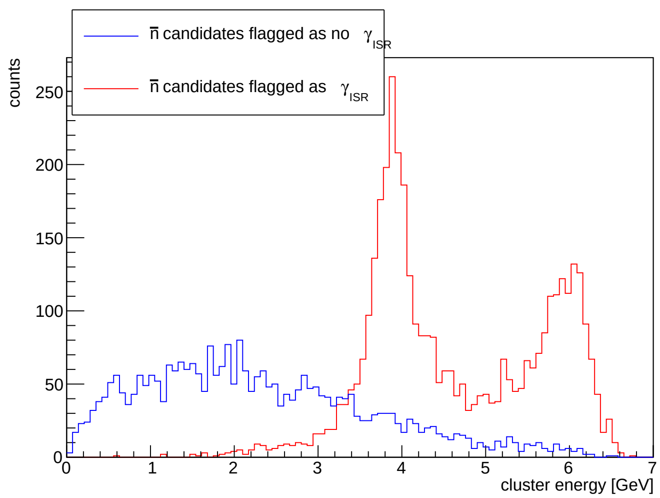


Figure 6.4: Cluster energy distribution of $\bar{n}$ candidates, divided by the Monte Carlo ISR flag (`mcISR`), in $u\bar{u}$ chunk1 cocktail production. If `mcISR` $= 0$ (blue), the reconstructed particle is not an ISR photon. If `mcISR` $= 1$ (red), the reconstructed particle is an ISR photon. `mcISR` is a Monte Carlo information.

However, γ radiation represents one of the most abundant produced particles in HEP experiments such as Belle II, making a major contribution to the background. In addition to the ISR ($E \sim$ 10 MeV - 5 GeV, Fig. 5.4) and FSR ($E <$ 10 MeV) radiation described above, bremsstrahlung ($E <$ 50 MeV) and secondary photons -such as those from $\pi^0 \to \gamma\gamma$ decays ($E <$ 400 MeV)- are also present. Therefore, NO ISR channel where the recoil system is made of p, π^- should represents a cleaner system to reconstruct than the ISR one where the recoil system is made of p, π^-, γ, since several background γ's can be mis-identified as the ISR photon, which must be accounted for in the recoil system.

In order to compare the feasibility of selecting the channel in both ISR and NO ISR cases separately, the recoil mass distribution must be first studied using the same selection criteria. The better the selection of the signal from the background, the more consistent the shower shape variables will be for neutral clusters associated with $\bar{n}$ candidates.

6.3 Recoil mass in $u\bar{u}$ cocktail production (NO ISR case)

The recoil mass distribution obtained from the $u\bar{u}$ cocktail is first studied in NO ISR case. If the recoil system p, π^- is properly reconstructed among all possible background channels, then a peak should be observed at the anti-neutron rest mass (939.5 MeV/c^2). The obtained result is shown in Fig. 6.5.

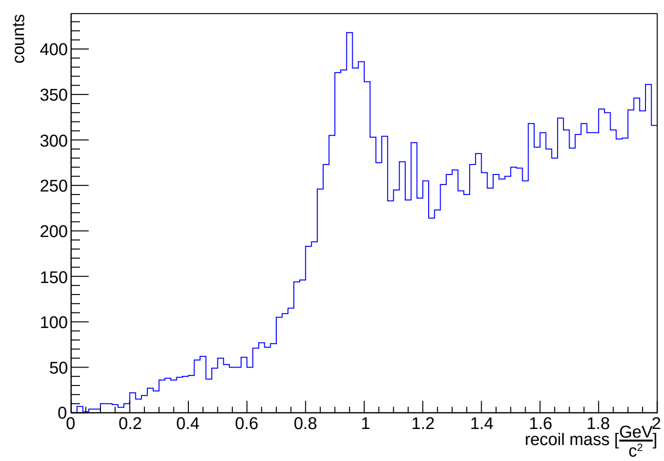


Figure 6.5: Reconstructed p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, NO ISR case.

A distinct structure is observed above the $\bar{n}$ mass value, suggesting that the selected channel is properly reconstructed. For a cleaner comparison between the signal channel and background components, the TopoAna output can be consulted in Fig. 6.6. TopoAna does not account for particle-matter interactions; thus, all displayed particles originate directly from the e^+e^- collision at the IP.

Accessing the Monte Carlo information from TopoAna, the main contribution comes from the signal channel with $p\pi^-\bar{n}$ in the final state, immediately followed by the one with $p\pi^-\bar{n}\pi^0$ in the final state. Further contribution arise from $p\pi^-\bar{n}\gamma_{ISR}$, which could be interpreted as signal, even if, in NO ISR case, the γ_{ISR} is not explicitly reconstructed within the recoil system. The fourth contribute is due to the one with $p\pi^-\bar{n}\pi^0\gamma_{ISR}$ and so on. The contributions are sorted from most to least abundant.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p$	$\pi^-\bar{n}p$	24	2405	2405
2	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	2	2317	4722
3	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	16	1980	6702
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	6	1676	8378
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	12	751	9129
6	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I\gamma^I$	3	612	9741
7	$e^+e^- \to \pi^+\pi^-\gamma^I\gamma^I$	$\pi^+\pi^-\gamma^I\gamma^I$	33	467	10208
8	$e^+e^- \to \pi^+\pi^-\gamma^I$	$\pi^+\pi^-\gamma^I$	0	381	10589
9	$e^+e^- \to \pi^-\bar{\Delta}^0 p, \bar{\Delta}^0 \to \pi^0\bar{n}$	$\pi^0\pi^-\bar{n}p$	28	313	10902
10	$e^+e^- \to \pi^0 p\bar{p}$	$\pi^0 p\bar{p}$	67	287	11189

Figure 6.6: First 10 channels of the TopoAna output in $u\bar{u}$ Monte Carlo cocktail, showing the different channel contributes, NO ISR case.

6.3.1 Recoil mass components analysis

The recoil mass distribution components can be studied by correlating the ntuple information obtained from the steering file with that produced by the TopoAna program.
Each channel is identified by its own `iDcyTr` (index of the decay tree), which allows one to represent the contribution of a given channel (or a combination of channels) to a specific variable, such as the recoil mass of the system $p\pi^-$.
The components are therefore denoted as follows:

1. $e^+e^- \to \pi^-\bar{n}p$ (iDcyTr == 24);
2. $e^+e^- \to \pi^0\pi^-\bar{n}p$ (iDcyTr == 2);
3. $e^+e^- \to \pi^-\bar{n}p\gamma_{ISR}$ (iDcyTr == 16);
4. $e^+e^- \to \pi^0\pi^-\bar{n}p\gamma_{ISR}$ (iDcyTr == 6);
5. $e^+e^- \to$ other.

The obtained stacked component to the recoil mass are shown in figure 6.7. Defining the *Signal Region* as the interval $[0.8, 1.2]$ GeV, it can be observed that contributions from the channel with iDcyTr = 2 predominantly lie to the right of the signal region. This is expected, as the recoil system computation excludes the π^0, leading to a higher recoil mass. Same conclusions can be made for the with iDcyTr = 6.
The signal channel (iDcyTr = 24) predominantly populates the signal region. The $\pi^-\bar{n}p\gamma_{ISR}$ channel (iDcyTr = 16) also mainly populates the signal region, indicating that the γ_{ISR} is negligible and has minimal impact on the recoil mass, as shown by the ISR energy distribution in Fig. 5.4. This can be directly observed from the single contribute from these two channels displayed in Fig. 6.8. The contributions from the remaining channels represent a flat background without discernible structures.

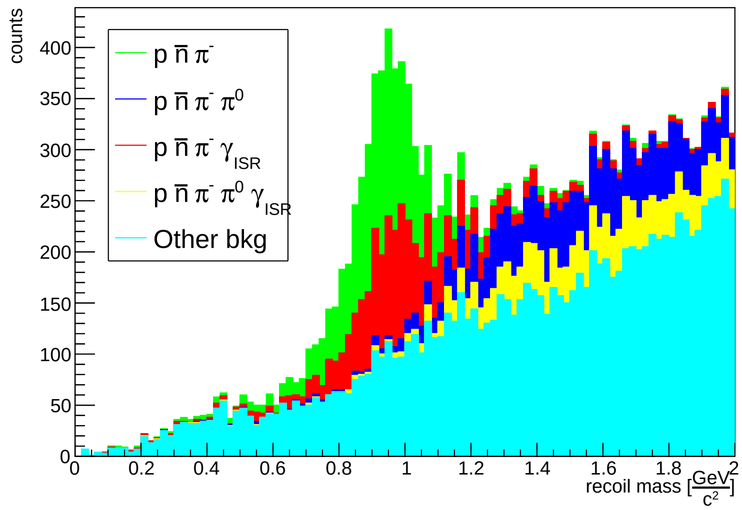


Figure 6.7: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case. Green: $\pi^-\bar{n}p$ channel. Blue: $\pi^0\pi^-\bar{n}p$ channel. Red: $\pi^-\bar{n}p\gamma_{ISR}$. Yellow: $\pi^0\pi^-\bar{n}p\gamma_{ISR}$ channel. Cyan: other channels.

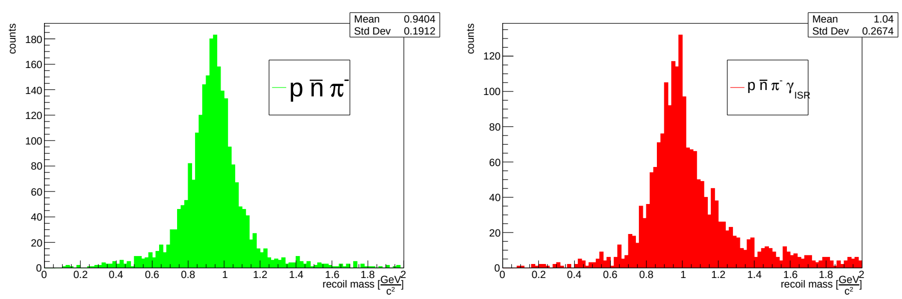


Figure 6.8: (green) p, π^- recoil mass and (red) p, π^-, γ_{ISR} recoil mas distribution in $u\bar{u}$ Monte Carlo cocktail.

Thus, despite the γ_{ISR} not being directly reconstructed, the $\pi^-\bar{n}p\gamma_{ISR}$ channel (iDcyTr = 16) can be considered a signal channel. This leads to the recoil mass composition shown in Fig. 6.9.
To further suppress the background, a tighter α cut can be studied. Therefore, the Performance Metrics and their usage are investigated to find the most appropriate cut in α.

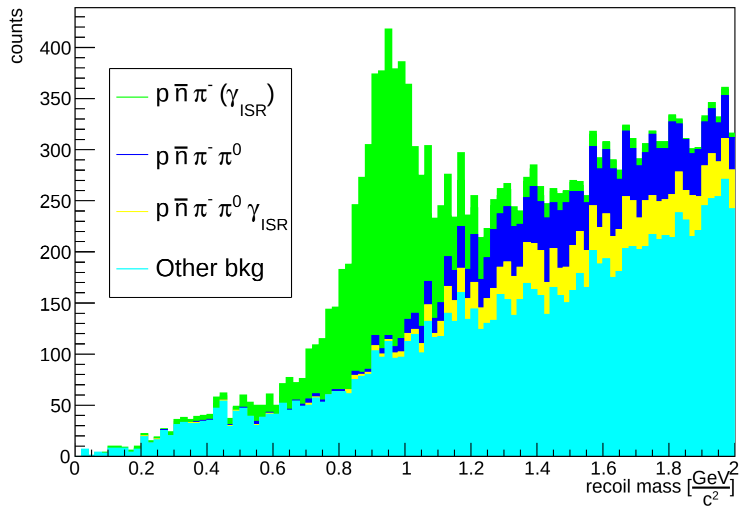


Figure 6.9: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case. Green: $\pi^-\bar{n}p$ and $\pi^-\bar{n}p\gamma_{ISR}$ channels. Blue: $\pi^0\pi^-\bar{n}p$ channel. Yellow: $\pi^0\pi^-\bar{n}p\gamma_{ISR}$ channel. Cyan: other channels.

6.3.2 Performance Metrics study on α cut

In general, a **Performance Metric** quantifies how a classification process is optimized based on a specific parameter, which could, for example, be represented by a threshold.
In this specific HEP context, a performance metric quantifies how well the selection cuts distinguish the signal from the background. In this analysis, the signal can be assumed to be the channel with $\pi^-\bar{n}p(\gamma_{ISR})$ in the final state, while the background is represented by all the other channels. Therefore, a stricter cut on the α angle is investigated to determine whether it improves signal selection over background.
To address this kind of analysis, performance metrics such as the **Figure of Merit** (FOM), the Signal Efficiency (ϵ_{sig}), the Background Efficiency (ϵ_{bkg}) and the **purity** can be employed. These are defined as follow:

- FOM $= \frac{SIGNAL(\alpha)}{\sqrt{SIGNAL(\alpha)+BACKGROUND(\alpha)}}$;
- $\epsilon_{sig} = \frac{SIGNAL(\alpha)}{SIGNAL(\alpha=0.20\ rad)}$;
- $\epsilon_{bkg} = \frac{BACKGROUND(\alpha)}{BACKGROUND(\alpha=0.20\ rad)}$;
- Purity $= \frac{SIGNAL(\alpha)}{SIGNAL(\alpha)+BACKGROUND(\alpha)}$.

SIGNAL and BACKGROUND are defined as follows:

- SIGNAL(α) is defined by the number of entries from the signal channel, depending on α;
- BACKGROUND(α) is defined by the number of entries from the other channels within the signal region, depending on α.

SIGNAL(α) + BACKGROUND (α) = TOTAL(α), the total amount of entries.

Assuming the signal region corresponds to the interval $[0.8, 1.2]$ GeV, the FOM characterizes the statistical significance of the applied threshold, namely the selected cut on α. The higher the FOM, the more signal is selected over the background (the figure is maximized). For each possible cut value of α, the Figure of Merit (FOM) is computed and the cut corresponding to the maximum FOM (or to a stable plateau) is selected. A higher FOM indicates a better trade-off between signal efficiency and background rejection, so the selection is optimized when the FOM is maximized. The obtained FOM is shown in Fig. 6.10.

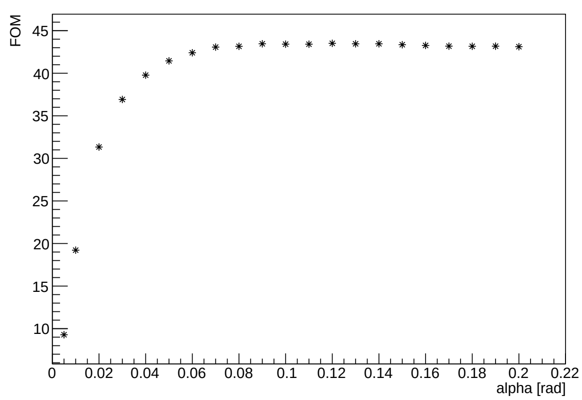


Figure 6.10: Figure of Merit (FOM) versus α angle for the $u\bar{u}$ cocktail.

One α value corresponding to the plateau can be chosen in order to maximize the FOM. The value $\alpha < 0.1$ rad ($\sim 5.73°$) is therefore chosen, corresponding to FOM = 43.4185
Furthermore, the signal efficiency ϵ_{sig} and the background efficiency ϵ_{bkg} performance metrics can be adopted to study the Receiver Operating Characteristic (ROC), a curve that shows how well a binary classifier distinguishes signal from background as the decision threshold (α cut) is varied. Each point on the curve corresponds to a different threshold (a cut on the α angle). The Area Under the ROC Curve (AUC/ROC-AUC) is a single number summarizing the overall performance: 1.0 indicates perfect discrimination, while 0.5 corresponds to no better performance than random. The obtained ROC is shown in Fig. 6.11.

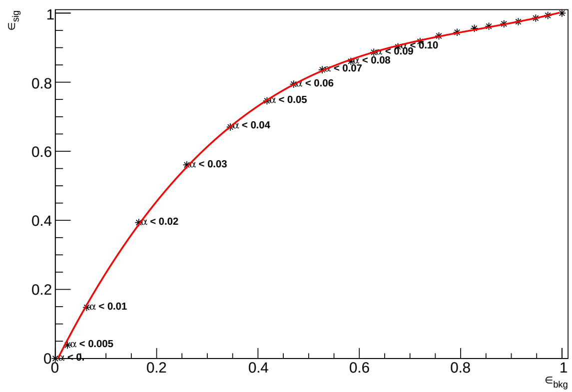


Figure 6.11: Receiver Operating Characteristic (ROC) curve constructed from the signal efficiency (ϵ_{sig}) and the Background Efficiency (ϵ_{bkg}), where each point corresponds to a different cut value on the α angle. The labels are shown only for $\alpha < 0.10$ rad.

The Receiver Operating Characteristic (ROC) curve for the α angle cut yields an area under the curve (AUC) of 0.71, indicating good discrimination power between signal and background.
Therefore, the obtained performance metrics for $\alpha < 0.1$ rad are:

$$FOM = 43.4185,\ \epsilon_{sig} = 90\%,\ \epsilon_{bkg} = 68\%,\ Purity = 63\%$$

The obtained ϵ_{sig} vs $Purity$ is shown in Fig. 6.12.

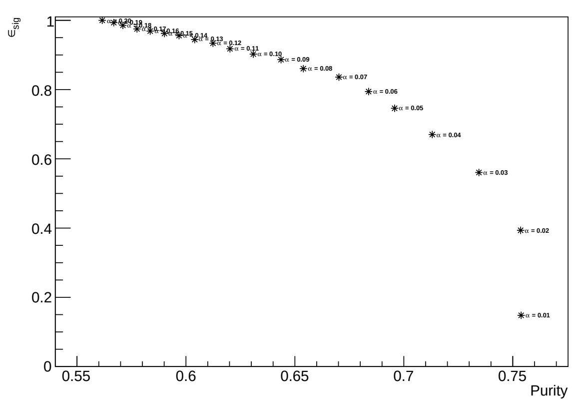


Figure 6.12: ϵ_{sig} vs $Purity$ curve for the $u\bar{u}$ Monte Carlo cocktail.

Following the described optimization, the recoil mass channel components

distribution can now be analyzed using the cut $\alpha < 0.10$ rad.
The obtained stacked component to the recoil mass are shown in Fig. 6.13.

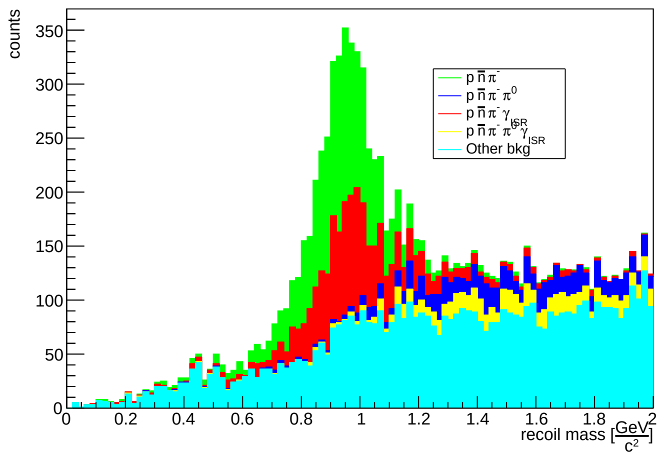


Figure 6.13: p, π^- recoil mass distribution, displaying each stacked component contribution, NO ISR case. Green: $\pi^-\bar{n}p$ channel. Blue: $\pi^0\pi^-\bar{n}p$ channel. Red: $\pi^-\bar{n}p\gamma_{ISR}$. Yellow: $\pi^0\pi^-\bar{n}p\gamma_{ISR}$ channel. Cyan: other channels. $\alpha < 10$ rad cut is applied.

As observed in the performance metrics study, the signal is more prominent relative to the background than in the previous case. Background components containing π^0 peak outside the signal region $[0.8, 1.2]$ GeV on the right side, while other sources exhibit a lower, flat distribution across the recoil mass spectrum.
The first 5 channels of $u\bar{u}$ TopoAna output with the $\alpha < 10$ rad cut is shown in 6.14.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p$	$\pi^-\bar{n}p$	14	2195	2195
2	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	10	1675	3870
3	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	2	907	4777
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	6	698	5475
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	9	598	6073

Figure 6.14: Top 5 channels from the TopoAna output for the $u\bar{u}$ Monte Carlo cocktail. NO ISR case.

After applying the $\alpha < 0.10$ rad cut, the dominant contribution remains from the signal channel $p\pi^-\bar{n}$, now immediately followed by $p\pi^-\bar{n}\gamma_{\text{ISR}}$. Both contributions can be identified as signal since they fall within the signal region $[0.8, 1.2]$ GeV.

6.4 $q\bar{q}$ cocktail production (NO ISR case)

In order to fully characterize the Monte Carlo simulation, all $q\bar{q}$ continuum components are studied. The Run 1 Monte Carlo productions for $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ are therefore analyzed.
For a cleaner comparison between the signal channel and background components, the TopoAna output can be consulted in Fig. 6.15.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p$	$\pi^-\bar{n}p$	2	2794	2794
2	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	4	2163	4957
3	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	1	1267	6224
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	9	944	7168
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	5	767	7935
6	$e^+e^- \to \pi^+\pi^-\gamma^I\gamma^I$	$\pi^+\pi^-\gamma^I\gamma^I$	10	443	8378
7	$e^+e^- \to \pi^+\pi^-\gamma^I$	$\pi^+\pi^-\gamma^I$	7	424	8802
8	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I\gamma^I$	3	315	9117
9	$e^+e^- \to \pi^0 p\bar{p}$	$\pi^0 p\bar{p}$	98	244	9361
10	$e^+e^- \to K^-p\bar{\Lambda}, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0K^-\bar{n}p$	226	216	9577

Figure 6.15: First 10 channels of the TopoAna output in $q\bar{q}$ Monte Carlo cocktail, showing the different channel contributes.

The TopoAna output reveals that the dominant contribution is from the signal channel $p\pi^-\bar{n}$ in the $q\bar{q}$ continuum. A secondary contribution comes from $p\pi^-\bar{n}\gamma_{ISR}$, which could be treated as signal despite the γ_{ISR} not being explicitly reconstructed. The next contribution comes from $p\pi^-\bar{n}\pi^0$ in the final state. The fourth contribution is due to $p\pi^-\bar{n}\pi^0\gamma_{\text{ISR}}$, and so on. The situation is very similar to that observed in the $u\bar{u}$ case.
The individual TopoAna outputs separately for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ samples are now discussed. The first 5 contributes for each sample are shown in 6.16. The main contributions to the final signal state are from the $u\bar{u}$ and $d\bar{d}$ cocktails, as expected, since no $p\pi^-\bar{n}$ system can be obtained from an $s\bar{s}$ state. A contribution from a $c\bar{c}$ state could have been expected from the following channel:

$$e- \ + e^- \to J/\Psi(c\bar{c}) \to p \ + \ \pi^- \ + \ \bar{n}$$

In order to reach the J/ψ resonance ($\sim$ 3.1 GeV) in e^+e^- collisions at Belle II, a significant lowering of the center-of-mass energy must occur due to γ_{ISR} emission. Since high-energy ISR emission is improbable (Fig. 5.4), the $c\bar{c}$ channel can be considered negligible for this analysis.
The obtained stacked components to the recoil mass, in $q\bar{q}$ cocktail, are shown in figure 6.17.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p$	$\pi^-\bar{n}p$	14	2195	2195
2	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	10	1675	3870
3	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	2	907	4777
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	6	698	5475
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	9	598	6073

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p$	$\pi^-\bar{n}p$	1	599	599
2	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	3	488	1087
3	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	0	360	1447
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	8	246	1693
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	4	169	1862

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to K^-p\bar{\Lambda}, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0K^-\bar{n}p$	3	120	120
2	$e^+e^- \to K^-p\bar{\Lambda}\gamma^I, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0K^-\bar{n}p\gamma^I$	1	102	222
3	$e^+e^- \to K^0K^-\bar{n}p, K^0 \to K^0_L$	$K^0_LK^-\bar{n}p$	9	66	288
4	$e^+e^- \to \pi^0K^-p\bar{\Lambda}, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0\pi^0K^-\bar{n}p$	15	52	340
5	$e^+e^- \to K^-p\bar{\Lambda}\gamma^I\gamma^I, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0K^-\bar{n}p\gamma^I\gamma^I$	50	48	388

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to D^{*0}p\bar{\Lambda}_c^-\gamma^I, D^{*0} \to D^0\gamma, \bar{\Lambda}_c^- \to \pi^+\pi^-\bar{\Sigma}^-, D^0 \to \pi^0K^0_S, \bar{\Sigma}^- \to \pi^0\bar{p}, K^0_S \to \pi^0\pi^0$	$\pi^0\pi^0\pi^0\pi^0\pi^+\pi^-p\bar{p}\gamma^I\gamma$	0	1	1

Figure 6.16: Top 5 channels from the TopoAna output respectively for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ Monte Carlo cocktails (from top to bottom). NO ISR case. The `iDcyTr` index values may differ across samples, since each TopoAna output is produced independently from its corresponding ROOT ntuples.

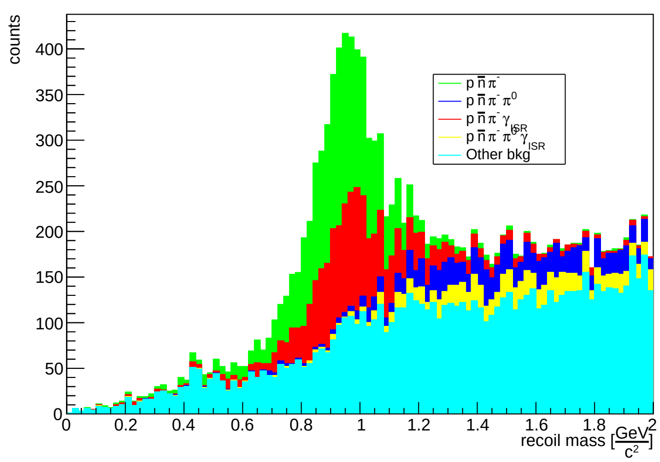


Figure 6.17: p, π^- recoil mass distribution in $q\bar{q}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case. Green: $\pi^-\bar{n}p$ channel. Blue: $\pi^0\pi^-\bar{n}p$ channel. Red: $\pi^-\bar{n}p\gamma_{ISR}$. Yellow: $\pi^0\pi^-\bar{n}p\gamma_{ISR}$ channel. Cyan: other channels. $\alpha < 10$ rad cut is applied.

A similar behaviour to the $u\bar{u}$ cocktail can be observed, where the $p\pi^-\bar{n}$ and $p\pi^-\bar{n}\gamma_{\rm ISR}$ channels mainly contribute to the signal region [0.8,1.2] GeV. Background components containing the π^0 peak appear outside the signal region [0.8,1.2] GeV on the right side, while a flat background is present from other background channels.

6.4.1 Cluster variables in $q\bar{q}$ cocktail

The $\bar{n}$ candidates reconstructed cluster variables can now be studied with the following discussed selections:

1. Recoil system: $0\ GeV/c^2 < recoil\ mass < 2\ GeV/c^2$ and in ECL acceptance;
2. $\bar{n}$: neutral clusters from ECL and number of hits > 10 and energy < 3 GeV;
3. p: $protonID > 0.9$ and $dr < 1$ cm and $|dz| < 3$ cm;
4. π^-: $pionID > 0.1$ and $dr < 1$ cm and $|dz| < 3$ cm;
5. $\alpha < 0.10$ rad $\sim 5.73°$;
6. No further charged particles in the Rest of Event (ROE).

The obtained $\bar{n}$ candidates reconstructed cluster variables are shown in Fig.6.18. Specific structures can be observed in the cluster variables distributions, compared to those obtained in Fig. 5.27.
For example, in the E1E9 distribution a second structure can be noted at a value ~ 0.85, and at value ~ 1 in the E9E21distribution as well. Similar structure appears at values close to 0 in the Second Moment distribuiton, which is partially reflected in the first structure of the N Hits distribution observed at a value of ~ 10. Another example can be seen in the Lateral Momentum distribution, where a structure at ~ 0.3 can be noted, which is typical reference of electromagnetic shower.
Therefore, a contribution from small-sized clusters -as can be seen in the Second Moment and N Hits distributions- and not spread out clusters -as can be seen in the E1E9 and E9E21 distributions- can be observed in $\bar{n}$ candidates reconstructed cluster variables. These shower shape characteristics are typical of the electromagnetic showers. This is further suggested by the structure observed in the Lateral Momentum distribution.
In order to study the nature of these singular structures, a signal and background study can be performed, isolating the individual contribution of each. If these structures are not produced by the signal, their contribution could be isolated and then removed from the signal one. In order to perform this kind of analysis, the sidebands technique is discussed in the next paragraph.

The obtained cluster variables distribution in NO ISR case, in $q\bar{q}$ cocktail.

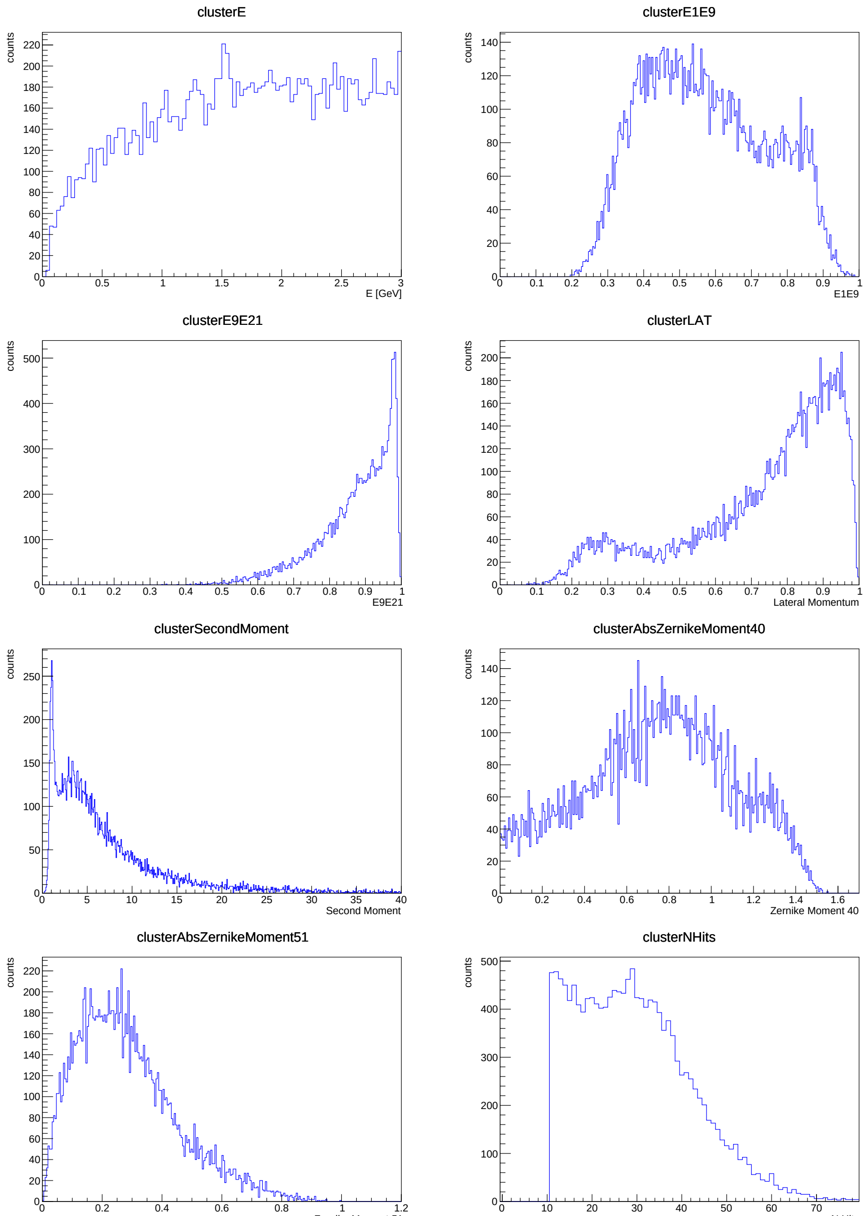


Figure 6.18: Reconstructed cluster variables distribution for $\bar{n}$ candidates in the for the recoil mass region [0, 2] GeV, in NO ISR case. $q\bar{q}$ cocktail

6.4.2 Sidebands technique in $q\bar{q}$ cocktail

In order to study separately the signal and background contributions to the cluster variables, the **sidebands technique** can be adopted.
The sidebands technique is a data-driven method used in experimental high-energy physics to estimate and subtract the background from a region of interest. It relies on identifying a discriminating variable -often the invariant mass or the recoil mass, as this case- that separates the signal from the background.
The **signal region** can be defined as the region where the signal appears in the recoil mass distribution. This region contains a mixture of signal and background. The **sidebands regions** consist of two symmetric areas (one before and one after the signal region), which may sometimes overlap, be immediately adjacent, or distant, that are assumed to contain only (or primarily) background. Therefore, in order to obtain the cluster variables distribution for the signal only, the background distribution (sidebands regions) is subtracted from the signal+background distribution (signal region) as follows:

$$N_{signal} = N_{\text{signal region}} - k \cdot N_{\text{sideband region}},$$

where:

$$k = \frac{\text{size of the signal region}}{\text{size of the sum of the sidebands regions}}$$

is a normalization factor.

In the present case, the following region have been decided:

- Signal region: [0.8, 1.2] GeV;
- Sidebands regions: [0.3, 0.7] U [1.3, 1.7] GeV;
- $k = \frac{1.2-0.8}{(0.7-0.3)+(1.7-1.3)} = \frac{1}{2}$.

The signal region and sidebands region, in the recoil mass distribution observed in Fig. 6.17, are shown in Fig.6.19.

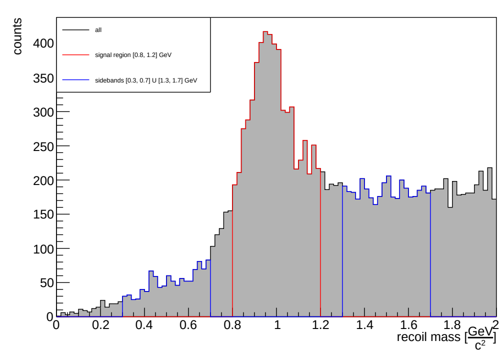


Figure 6.19: Recoil mass distribution (black), with the signal region (red) and the sidebands regions (blue), in the $q\bar{q}$ Monte Carlo cocktail. NO ISR case.

The single contributions from the signal region and the two sidebands regions to the cluster variable can be compared. Since the two sidebands regions combined are two times larger than the signal region, a scaling factor of $k = 0.5$ has been applied to the cluster variables obtained from the sidebands regions. The obtained cluster variables contribution is shown in Fig.6.20.
The cluster energy distribution shows a flat contribution from the sidebands regions, where no specific structure was observed. The E1E9 distribution shows that the structure observed in ~ 0.85 (Fig. 6.18) is mainly caused by background components, such as those coming from the sidebands regions. The same conclusion can be reached for the structure observed close to unity in the E9E21 distribution.
Furthermore, the sidebands regions provide the main contribution to the structure observed at low values in the Second Moment distribution, caused by small-sized clusters.
The cluster Zernike moment Z_4^0 presents a tail around $Z_4^0 \sim 1.2$ caused by sidebands contribution, while Z_5^1 presents no discernible structure.
Afterwards, the structure observed at a value of ~ 0.3 in the Lateral Momentum distribution is caused by sidebands contribution.
Therefore, to obtain the final $\bar{n}$ candidates cluster variables distribution, the sidebands regions contribution (background) is subtracted from the signal region contribution (signal + background). The obtained cluster variables are shown in Fig.6.21.

The obtained cluster variables distributions no longer present the singular structures observed prior to the sidebands technique application. This leads to reconstructed cluster variables more similar to those obtained from the Monte Carlo signal sample analysis (Fig. 5.27). This suggests that the applied selections and the sidebands technique properly separate the signal contribution from the background one, in q$\bar{q}$ cocktail, with no explicit γ_{ISR} reconstruction.

6.5 $q\bar{q}$ cocktail production (ISR case)

Since the same path as the NO ISR case has been followed, in the ISR case the $q\bar{q}$ cocktail is directly analyzed. The same final selections adopted in the NO ISR case have been used.
The TopoAna output of can be consulted in Fig. 6.22.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	39	10002	10002
2	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	38	7104	17106
3	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	40	5055	22161
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	47	4939	27100
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^I$	57	2572	29672
6	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I\gamma^I$	45	2268	31940
7	$e^+e^- \to \rho^-\bar{n}p\gamma^I, \rho^- \to \pi^0\pi^-$	$\pi^0\pi^-\bar{n}p\gamma^I$	62	1200	33140
8	$e^+e^- \to \pi^0\pi^0\pi^-\bar{n}p$	$\pi^0\pi^0\pi^-\bar{n}p$	49	1190	34330
9	$e^+e^- \to \pi^0 p\bar{p}\gamma^I$	$\pi^0 p\bar{p}\gamma^I$	305	1131	35461
10	$e^+e^- \to \pi^-\bar{n}p$	$\pi^-\bar{n}p$	72	990	36451

Figure 6.22: First 10 channels of the TopoAna output in $q\bar{q}$ Monte Carlo cocktail, showing the different channel contributes. ISR case.

The TopoAna output reveals a situation very similar to that obtained in the NO ISR case. In fact, now, the dominant contribution is from the signal channel $p\pi^-\bar{n}\gamma_{ISR}$. A secondary contribution comes from $p\pi^-\bar{n}\gamma_{ISR}\gamma_{ISR}$, which could be treated as signal despite the extra γ_{ISR} not being explicitly reconstructed. The next contribution comes from $p\pi^-\bar{n}\pi^0$ in the final state. The fourth contribution is due to $p\pi^-\bar{n}\pi^0\gamma_{\text{ISR}}$, and so on. The individual TopoAna outputs from $u\bar{u}$, $d\bar{d}$, $s\bar{s}$ and $c\bar{c}$ are shown in Fig. 6.23.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	2	7903	7903
2	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	0	5619	13522
3	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	11	3904	17426
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	10	3817	21243
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^I$	35	2050	23293

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	1	2099	2099
2	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	0	1485	3584
3	$e^+e^- \to \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	2	1151	4735
4	$e^+e^- \to \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	9	1122	5857
5	$e^+e^- \to \pi^-\bar{n}p\gamma^I\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I\gamma^I$	19	522	6379

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to \Lambda\bar{\Lambda}\gamma^I, \Lambda \to \pi^-p, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0\pi^-\bar{n}p\gamma^I$	36	190	190
2	$e^+e^- \to K^-p\bar{\Lambda}\gamma^I, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0K^-\bar{n}p\gamma^I$	2	163	353
3	$e^+e^- \to \Lambda\bar{\Lambda}\gamma^I\gamma^I, \Lambda \to \pi^-p, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0\pi^-\bar{n}p\gamma^I\gamma^I$	20	143	496
4	$e^+e^- \to \pi^0K^-p\bar{\Lambda}, \bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0\pi^0K^-\bar{n}p$	33	124	620
5	$e^+e^- \to K^0K^-\bar{n}p, K^0 \to K_L^0$	$K_L^0K^-\bar{n}p$	9	119	739

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \to D^{*-}D^{*0}\bar{n}p\gamma^I, D^{*-} \to \pi^0D^-, D^{*0} \to D^0\gamma, D^- \to \pi^-K_S^0, D^0 \to \pi^0\bar{K}^*,$ $K_S^0 \to \pi^0\pi^0, \bar{K}^* \to \pi^0\bar{K}^0, \bar{K}^0 \to K_L^0$	$\pi^0\pi^0\pi^0\pi^0\pi^0K_L^0\pi^-\bar{n}p\gamma^I\gamma$	0	1	1
2	$e^+e^- \to D^-\bar{n}\Lambda_c^+\gamma^I\gamma^I, D^- \to \pi^+\pi^+\pi^-\pi^-\pi^-\eta, \Lambda_c^+ \to \bar{K}^0p, \eta \to \pi^0\pi^+\pi^-, \bar{K}^0 \to K_S^0,$ $K_S^0 \to \pi^0\pi^0$	$\pi^0\pi^0\pi^0\pi^+\pi^+\pi^+\pi^-\pi^-\pi^-\pi^-\bar{n}p\gamma^I\gamma^I$	1	1	2
3	$e^+e^- \to \pi^0\bar{D}^0D_2^{*0}\gamma^I, \bar{D}^0 \to \pi^-K^+, D_2^{*0} \to \pi^0D^0, D^0 \to \pi^0\pi^0\bar{K}^*, \bar{K}^* \to \pi^0\bar{K}^0,$ $\bar{K}^0 \to K_L^0$	$\pi^0\pi^0\pi^0\pi^0\pi^0K_L^0\pi^-K^+\gamma^I$	2	1	3
4	$e^+e^- \to \Lambda_c^+\bar{\Lambda}_c^-\gamma^I\gamma^I\gamma^I, \Lambda_c^+ \to \pi^0\pi^0\Sigma^+, \bar{\Lambda}_c^- \to \pi^-\bar{\Sigma}^0, \Sigma^+ \to \pi^0p, \bar{\Sigma}^0 \to \bar{\Lambda}\gamma,$ $\bar{\Lambda} \to \pi^0\bar{n}$	$\pi^0\pi^0\pi^0\pi^0\pi^-\bar{n}p\gamma^I\gamma^I\gamma^I\gamma$	3	1	4
5	$e^+e^- \to D^{*0}p\bar{\Lambda}_c^-\gamma^I, D^{*0} \to \pi^0D^0, \bar{\Lambda}_c^- \to \pi^-\bar{\Sigma}^0, D^0 \to K_L^0\eta', \bar{\Sigma}^0 \to \bar{\Lambda}\gamma,$ $\eta' \to \pi^0\pi^0\eta, \bar{\Lambda} \to \pi^0\bar{n}, \eta \to \pi^0\pi^0\pi^0$	$\pi^0\pi^0\pi^0\pi^0\pi^0\pi^0\pi^0K_L^0\pi^-\bar{n}p\gamma^I\gamma$	4	1	5

Figure 6.23: Top 5 channels from the TopoAna output respectively for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ Monte Carlo cocktails (from top to bottom). ISR case.

As observed in the NO ISR case, the main contributions to the final signal state are from the $u\bar{u}$ and $d\bar{d}$ cocktails in the ISR case as well. The obtained stacked components to the recoil mass, in $q\bar{q}$ cocktail, are shown in figure 6.24.

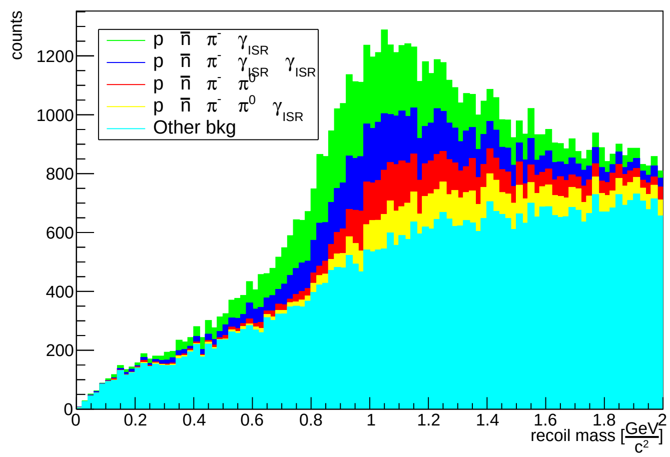


Figure 6.24: p, π^- recoil mass distribution in $q\bar{q}$ Monte Carlo cocktail, displaying each stacked component contribution, ISR case. Green: $\pi^-\bar{n}p\gamma_{ISR}$ channel. Blue: $\pi^-\bar{n}p\gamma_{ISR}$ channel. Red: $\pi^0\pi^-\bar{n}p$. Yellow: $\pi^0\pi^-\bar{n}p\gamma_{ISR}$ channel. Cyan: other channels. $\alpha < 10$ rad cut is applied.

No discernible contributions can be observed, as each channel contributes uniformly across the recoil mass distribution, both inside and outside the signal region [0.8, 1.2] GeV. The obtained $\bar{n}$ candidates reconstructed cluster variables are shown in Fig.6.18.
Since substantial background appears to the right of the signal region -suggesting that particles are missing from the recoil reconstruction- the sideband regions are redefined, for the ISR case, as follows:

- Signal region: [0.8, 1.4] GeV;
- Sidebands regions: [0.2, 0.8] U [1.4, 2.0] GeV;
- $k = \frac{1.4-0.8}{(0.8-0.2)+(2.0-1.4)} = \frac{1}{2}$.

The signal region and sidebands region, in the recoil mass distribution are shown in Fig.6.26.

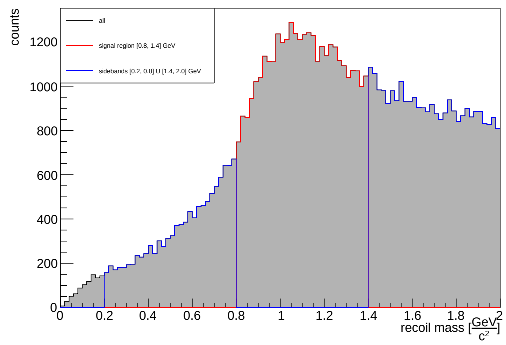


Figure 6.26: Recoil mass distribution (black), with the signal region (red) and the sidebands regions (blue), in the $q\bar{q}$ Monte Carlo cocktail. ISR case.

The obtained cluster variables contribution is shown in Fig.6.27. Even though the signal is less evident, the same conclusions as in the NO ISR case hold, observing the same kind of structures in the cluster variables distributions, but with more statistic. Therefore, the sidebands regions contribution is now subtracted from the signal region contribution. The obtained cluster variables are shown in Fig.6.28.
As observed in NO ISR case, the obtained cluster variables distributions no longer present the singular structures observed prior to the sidebands technique application, leading to reconstructed cluster variables more similar to those obtained from the Monte Carlo signal sample.
The last step consists in comparing the obtained cluster variables with those obtained from real data, performing the so called **Data/MC agreement**.

6.6 Summary of Monte Carlo cocktail Analysis

The channel $e^+ + e^- \to p + \pi^- + \bar{n}$ has been studied, with both ISR and NO ISR emission, in $q\bar{q}$ Monte Carlo cocktail.

A stricter event selection was applied using the Rest of Event (ROE) approach to exclude additional charged tracks outside the signal final state. Furthermore, the $\bar{n}$ candidates energy range was narrowed to reject the main ISR radiation background.

The recoil mass was then studied in the $u\bar{u}$ cocktail, with a components analysis performed using TopoAna output. This shows that both channels with and without explicit γ_{ISR} reconstruction mainly populate the signal region. Performance metrics were then evaluated, optimizing the α angle selection to maximize the figure of merit $\frac{S}{\sqrt{S+B}}$, achieving $\epsilon_{sig} = 90\%$ and $\epsilon_{bkg} = 68\%$ signal-background discrimination power.

The $q\bar{q}$ cocktail analysis, including recoil mass component decomposition, confirms the findings from the $u\bar{u}$ cocktail. However, in the ISR case, contributions from various channels span the entire recoil mass distribution without specific enhancement in the signal region. Thus, the signal and sideband regions were defined differently than in the NO ISR case.

Therefore, the $\bar{n}$ candidate cluster variables were studied before and after sidebands subtraction, showing good agreement with signal sample studies. The sidebands technique effectively excluded unexpected structures by removing background contributions from cluster variables, in the signal region.

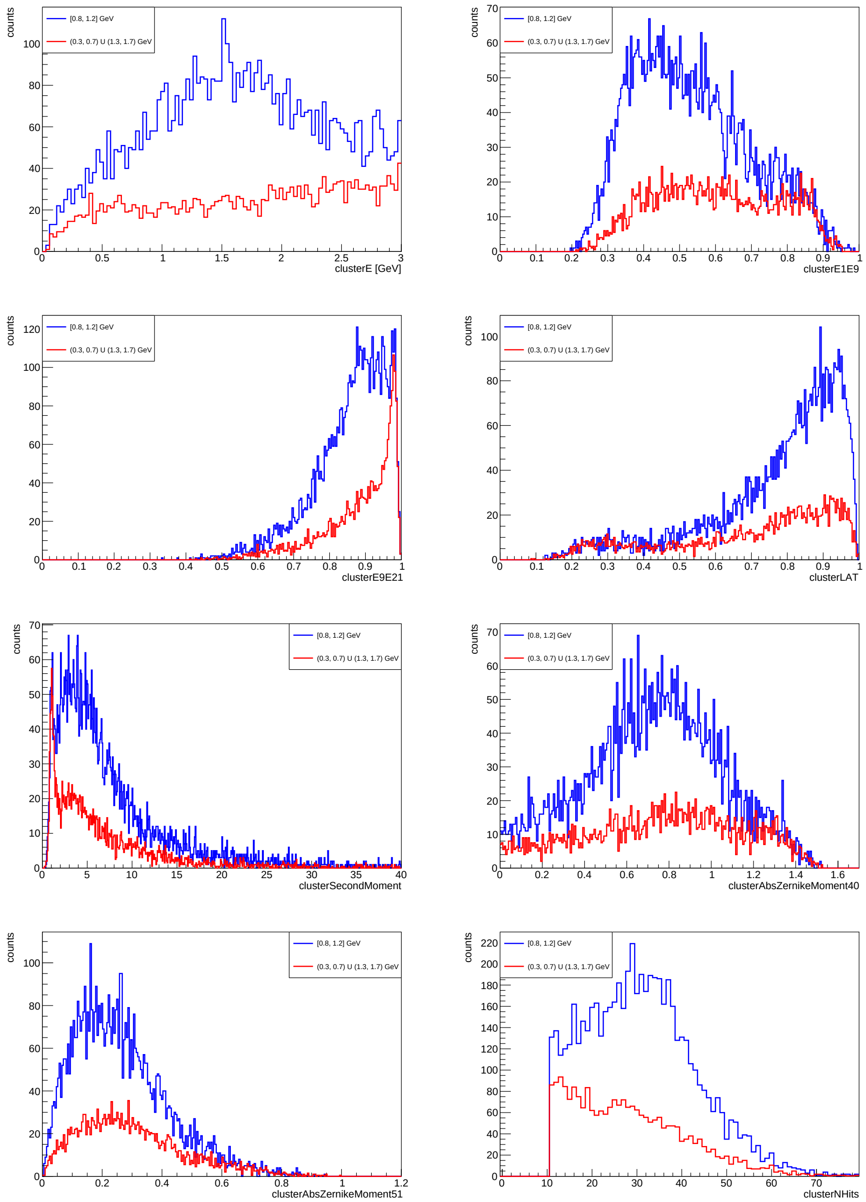


Figure 6.20: Comparison of reconstructed cluster variables for $\bar{n}$ candidates in the signal region [0.8, 1.2] GeV (blue) with $\bar{n}$ candidates in the sidebands region [0.3, 0.7] U [1.3, 1.7] GeV (red), in NO ISR case. A scaling factor of $k = 0.5$ has been applied to the cluster variables obtained from the sidebands regions. q$\bar{q}$ cocktail.

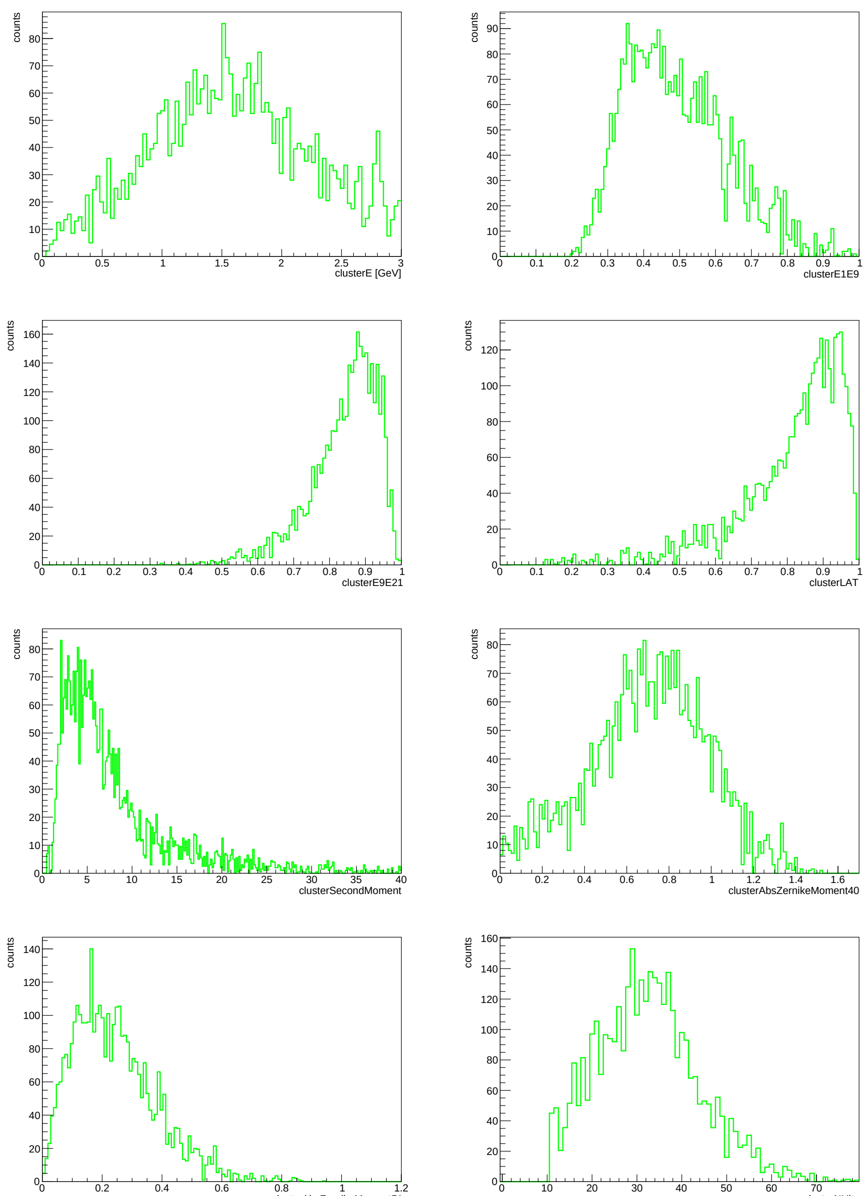


Figure 6.21: Reconstructed cluster variables for $\bar{n}$ candidates after the application of the sidebands technique, in NO ISR case. $q\bar{q}$ cocktail.

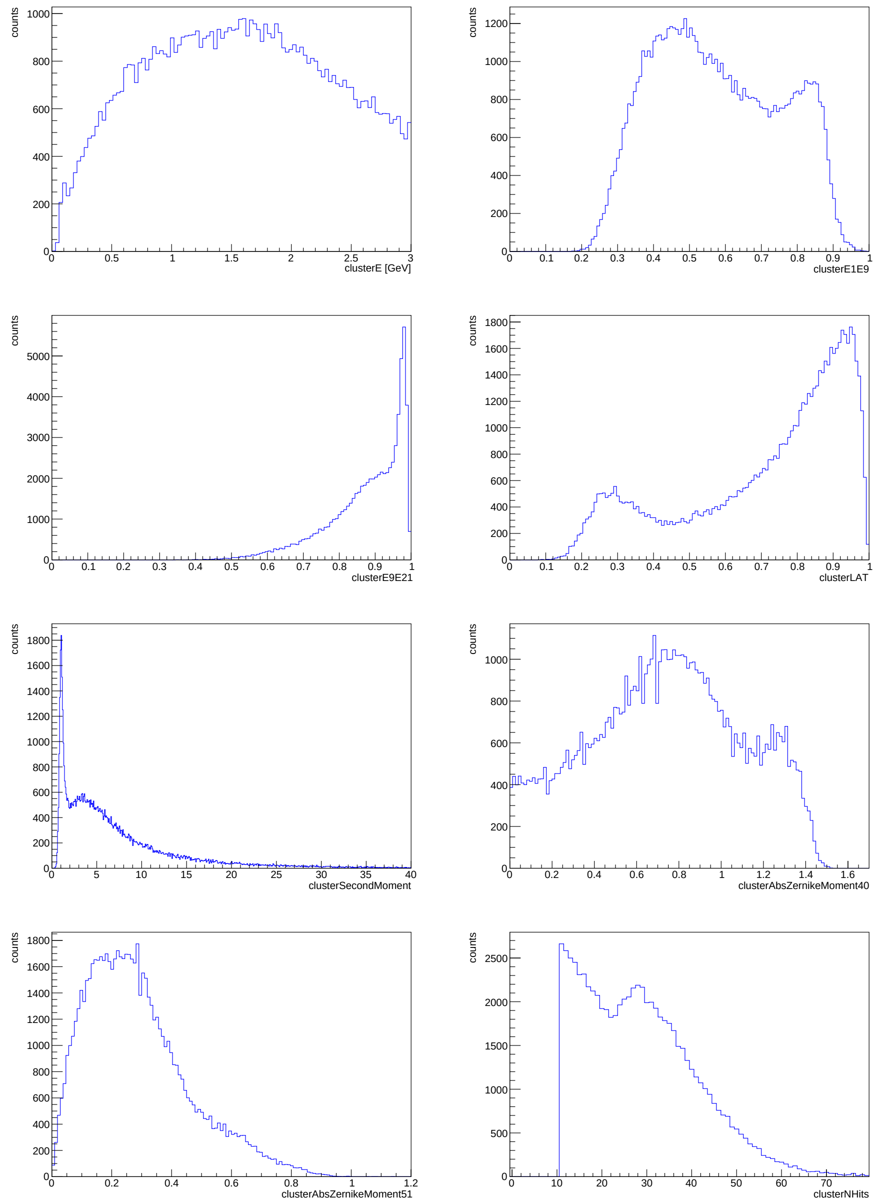


Figure 6.25: Reconstructed cluster variables distribution for $\bar{n}$ candidates in the for the recoil mass region [0, 2] GeV, in ISR case. $q\bar{q}$ cocktail

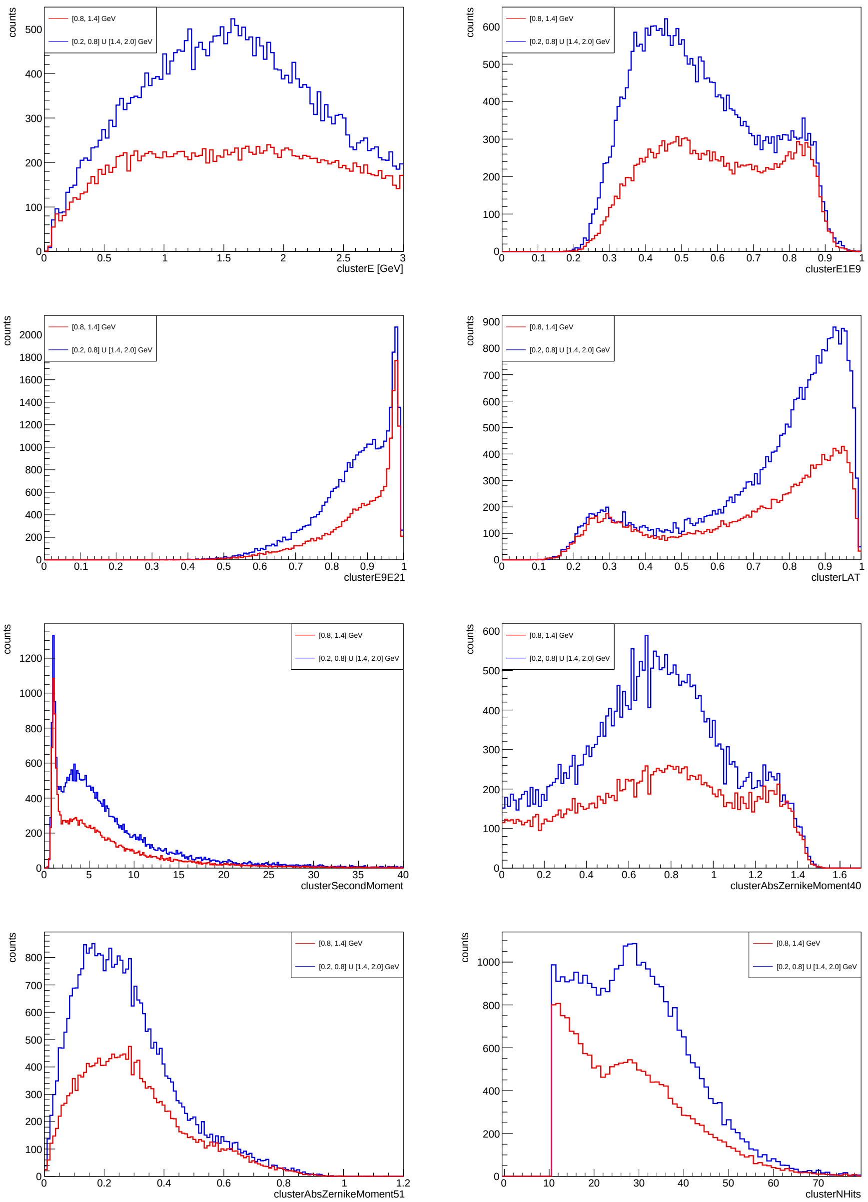


Figure 6.27: Comparison of reconstructed cluster variables for $\bar{n}$ candidates in the signal region [0.8, 1.2] GeV (blue) with $\bar{n}$ candidates in the sidebands region [0.3, 0.7] U [1.3, 1.7] GeV (red), in NO ISR case. A scaling factor of $k = 0.5$ has been applied to the cluster variables obtained from the sidebands regions. q$\bar{q}$ cocktail. NO ISR case.

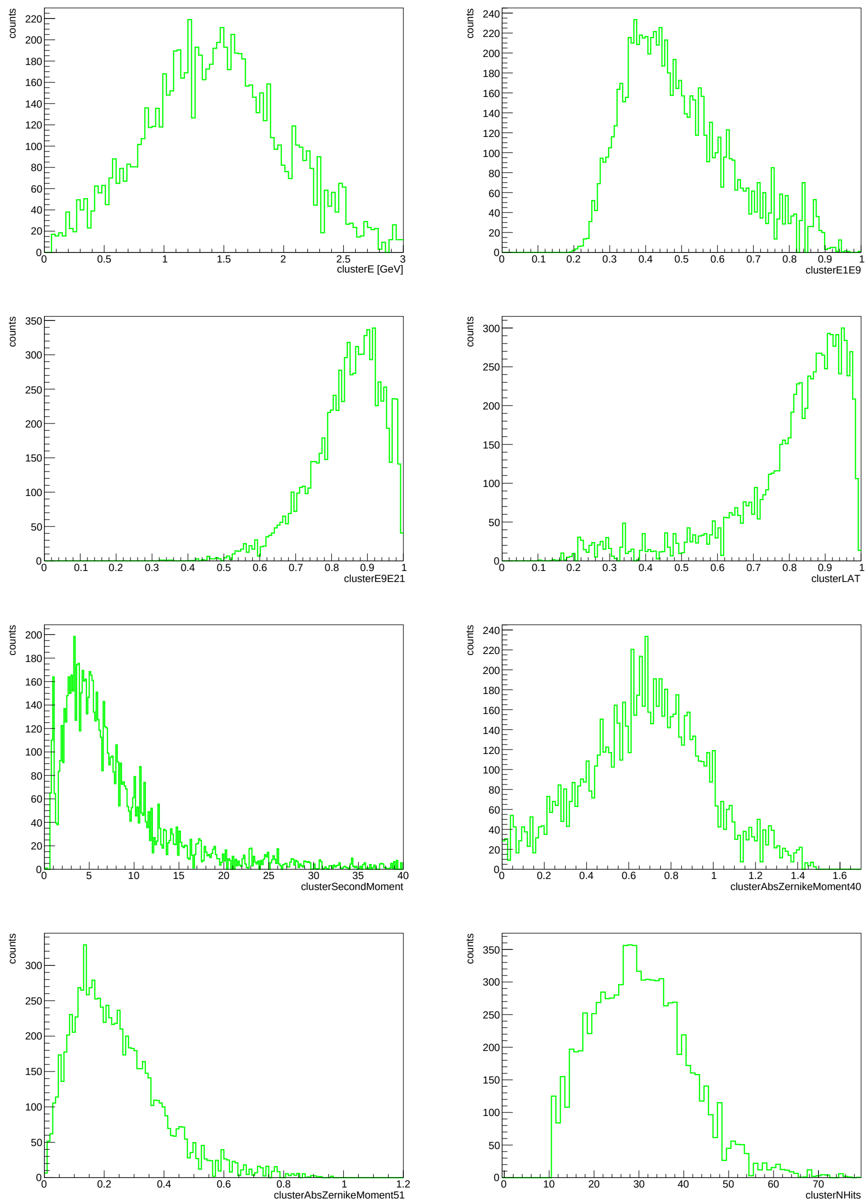


Figure 6.28: Reconstructed cluster variables for $\bar{n}$ candidates after the application of the sidebands technique, in ISR case. q$\bar{q}$ cocktail.

Chapter 7
Data - Monte Carlo comparison

This thesis concludes with a comparison of the obtained $\bar{n}$ candidates cluster variables against those obtained from data. The dataset is the latest available from Belle II data production, corresponding to an integrated luminosity of $358.27 fb^{-1}$. The same selections achieved in $q\bar{q}$ are applied, in both ISR and NO ISR cases. Since the goal is to assess Data/MC agreement for cluster variables from the analysis -rather than maximize signal selection efficiency- these variables are compared within the signal region [0.8, 1.2] GeV, without sideband subtraction.

7.1 Recoil mass in data

First, the recoil mass distribution in data is studied by analyzing the angular distributions of reconstructed protons. The resulting distribution (no ISR case) is shown in Fig. 7.1.

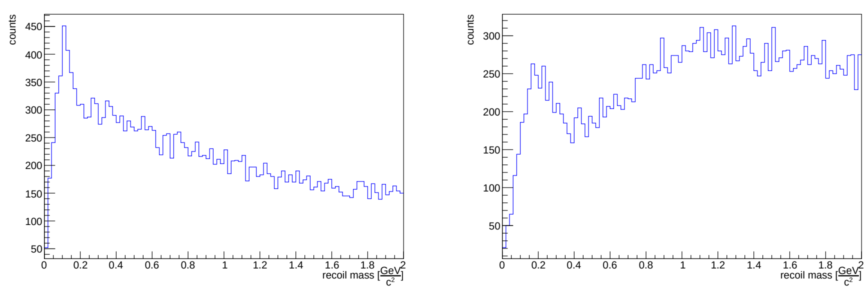


Figure 7.1: Recoil mass distribution in data. NO ISR case (left) and ISR case (right).

Unlike the Monte Carlo simulation, in data no signal peak is observed at the anti-neutron mass.A completely different behavior is observed, with a prominent structure at low recoil mass values, suggesting that some particles in the recoil system $p\pi^-$ are not properly reconstructed.
Since the proton has the strictest selection criteria ($protonID > 90\%$), it is now investigated in detail. To properly select the signal in the recoil mass distribution, the 2D distribution $\cos(\theta)$ vs momentum for reconstructed protons is studied.Several background sources not included in the Monte Carlo $q\bar{q}$ cocktail could arise. The obtained distribution is shown in Fig. 7.2.

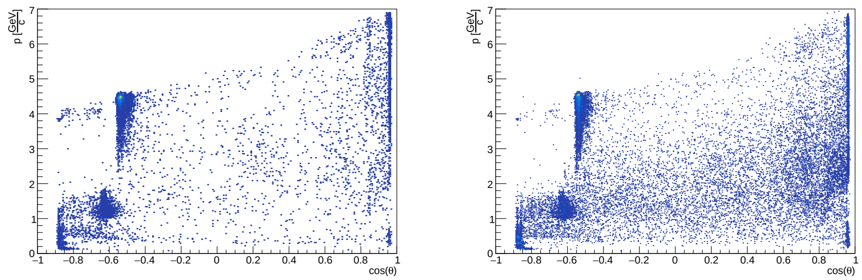


Figure 7.2: 2D cos(θ) vs momentum distribution in data. NO ISR case (left) and ISR case (right). Please, zoom in.

Two structures are observed in the backward region ($cos(\theta) < -0.45$): one at proton momentum $p < 2$ GeV/c and one at $p \sim 3{-}4$ GeV/c. Furthermore, a strip structure is observed in the forward region ($cos(\theta) > 0.85$) across the full proton momentum range.
Even if a strong selection on proton list has been performed, such as that on *protonID*, insufficient coverage in the backward and deep forward regions prevents proper proton selection, allowing multiple background sources to arise. For example background sources like the following one:

$$e^+e^- \to e^+e^- \qquad\qquad e^+e^- \to \pi^+\pi^-,$$

where the positive track could be misidentified as the reconstructed proton in the backward-forward regions.
Therefore, a further selection can be performed on reconstructed proton $cos(\theta)$ variable, such as that $-0.45 < cos(\theta) < 0.85$, in order to exclude the structures observed in Fig. 7.2. The following recoil mass distribution is shown in Fig. 7.3. A structure at the anti-neutron mass is now observed, confirming that the discussed structures caused incorrect reconstruction of the recoil system.

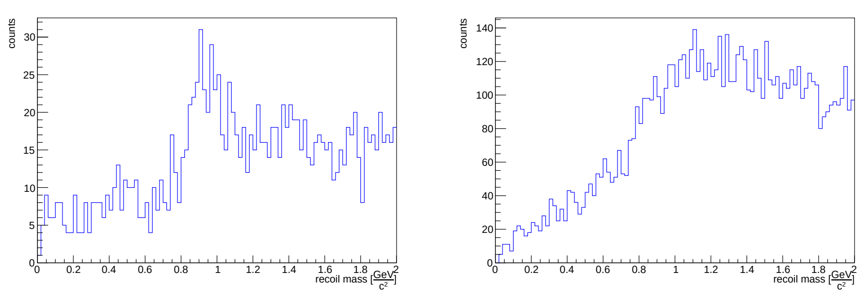


Figure 7.3: Recoil mass distribution in data. NO ISR case (left) and ISR case (right). Cut $-0.45 < cos(\theta) < 0.85$ applied, on the reconstructed protons.

Therefore, the event selections adopted for the remaining data analysis are as follows:

1. Recoil system: $0\ GeV/c^2 < recoil\ mass < 2\ GeV/c^2$ and in ECL acceptance;
2. $\bar{n}$: neutral clusters from ECL and number of hits > 10 and energy < 3 GeV;
3. p: $protonID > 0.9$ and $dr < 1$ cm and $|dz| < 3$ cm and $-0.45 < cos(\theta) < 0.85$;
4. π^-: $pionID > 0.1$ and $dr < 1$ cm and $|dz| < 3$ cm;
5. $\alpha < 0.10$ rad $\sim 5.73°$;
6. No further charged particles in the Rest of Event (ROE).

7.2 Data/MC agreement

The discussed $\bar{n}$ candidates cluster variables obtained in $q\bar{q}$ can be compared to those obtained from data. In order to perform the Data/MC agreement, the **pull variable** is adopted. It is a standardized metric used in high-energy physics to quantify discrepancies between observed data and Monte Carlo simulations. For a histogram i-th bin, the pull is defined as:

$$pull_i = \frac{N_{i,data} - N_{i,MC}}{\sqrt{\sigma^2_{i,data} + \sigma^2_{i,MC}}},$$

where $N_{i,\mathrm{data}}$ and $N_{i,\mathrm{MC}}$ represent the number of entries in the i-th bin, and $\sigma^2_{i,\mathrm{data}}$ and $\sigma^2_{i,\mathrm{MC}}$ are the corresponding uncertainties. With this definition, the pull variable corresponds to the difference between data and MC counts, weighted by the propagated uncertainty.

The resulting distributions, without sidebands subtraction , are shown in Fig. 7.4 (ISR case) and in Fig. 7.5 (NO ISR case). Those with sidebands subtraction are shown in Fig. 7.6 (ISR case) and in Fig. 7.7 (NO ISR case).
The Data/MC agreement plots reveal discrepancies between data distributions and $q\bar{q}$ Monte Carlo simulations.
The cluster energy shows good Data/MC agreement (within 2σ uncertainties) in both ISR and no-ISR cases across the full energy range, with and without sidebands subtraction.
The cluster E1E9 shows discrepancies ($> 2\sigma$) with and without sidebands subtraction, suggesting that more spread shower shapes must be accounted for. This is confirmed by discrepancies in the Zernike moments, indicating different shower shapes in data and simulation.
The cluster E9E21 shows discrepancies ($> 2\sigma$) without sidebands subtraction, but slightly better agreement afterward, confirming the trends observed in E1E9. The cluster lateral momentum shows discrepancies before sidebands subtraction but better agreement afterward (within 2σ uncertainties across nearly the

full range), without the structure at 0.3. This confirms the analysis effectively suppresses electromagnetic shower contributions, yielding an acceptable anti-neutron sample, even if the statistic is low.
The same conclusions apply to cluster second moment and N Hits variables, where better agreement is observed after sidebands subtraction, suggesting that small-sized (electromagnetic) clusters are effectively removed.
Therefore, some discrepancies in Data/MC agreement for $\bar{n}$ candidate cluster variables persist both before and after sidebands subtraction, though with clear improvement afterward. Nevertheless, low statistic -particularly in the NO ISR case- suggests that additional measurements could target different channels, such as multi-pion final states (beyond the scope of this work).
However, the obtained results agree with [44], suggesting that further studies on GEANT4 Physics lists on anti-neutron showers are needed.

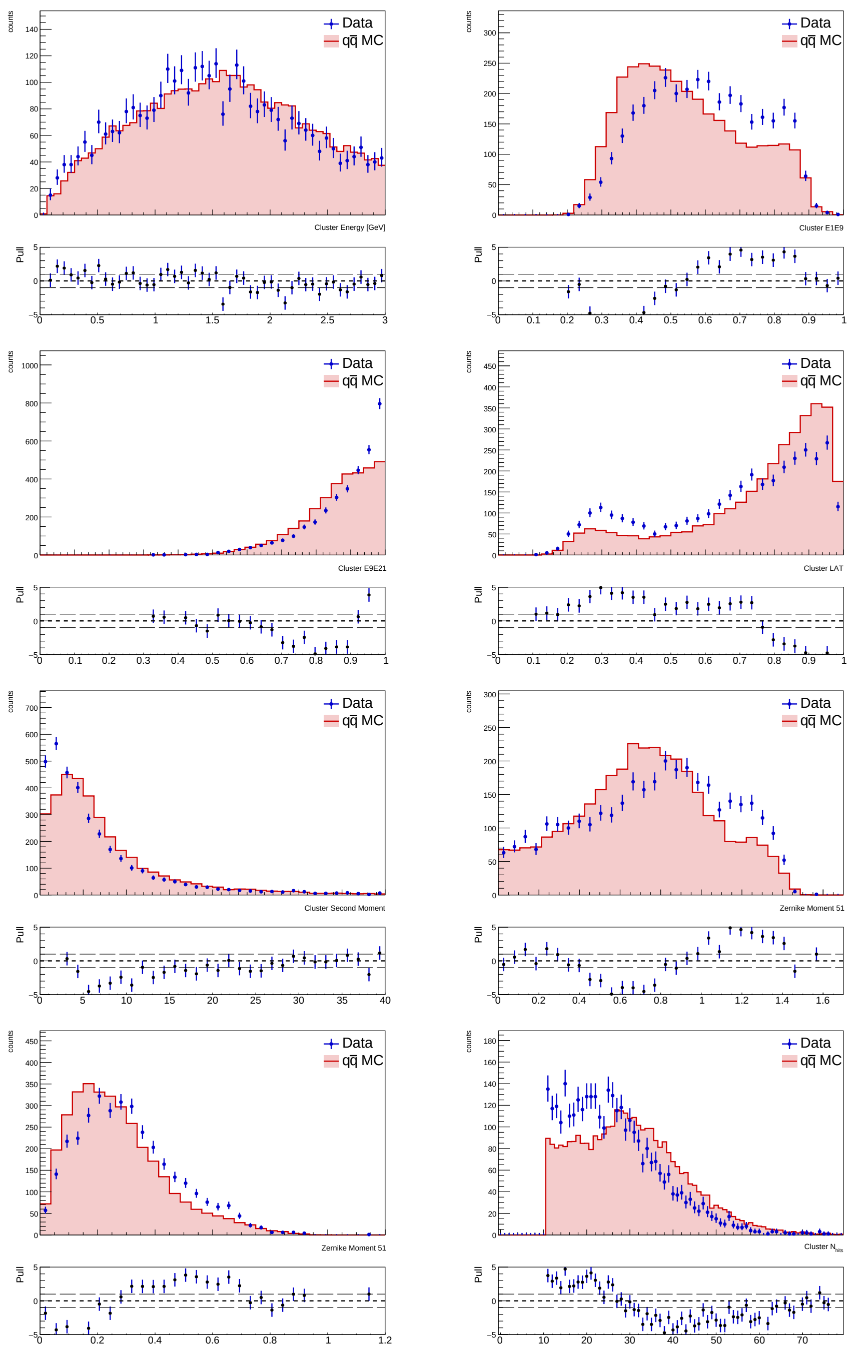


Figure 7.4: Data/MC agreement for reconstructed cluster variables for $\bar{n}$ candidates, in ISR case.

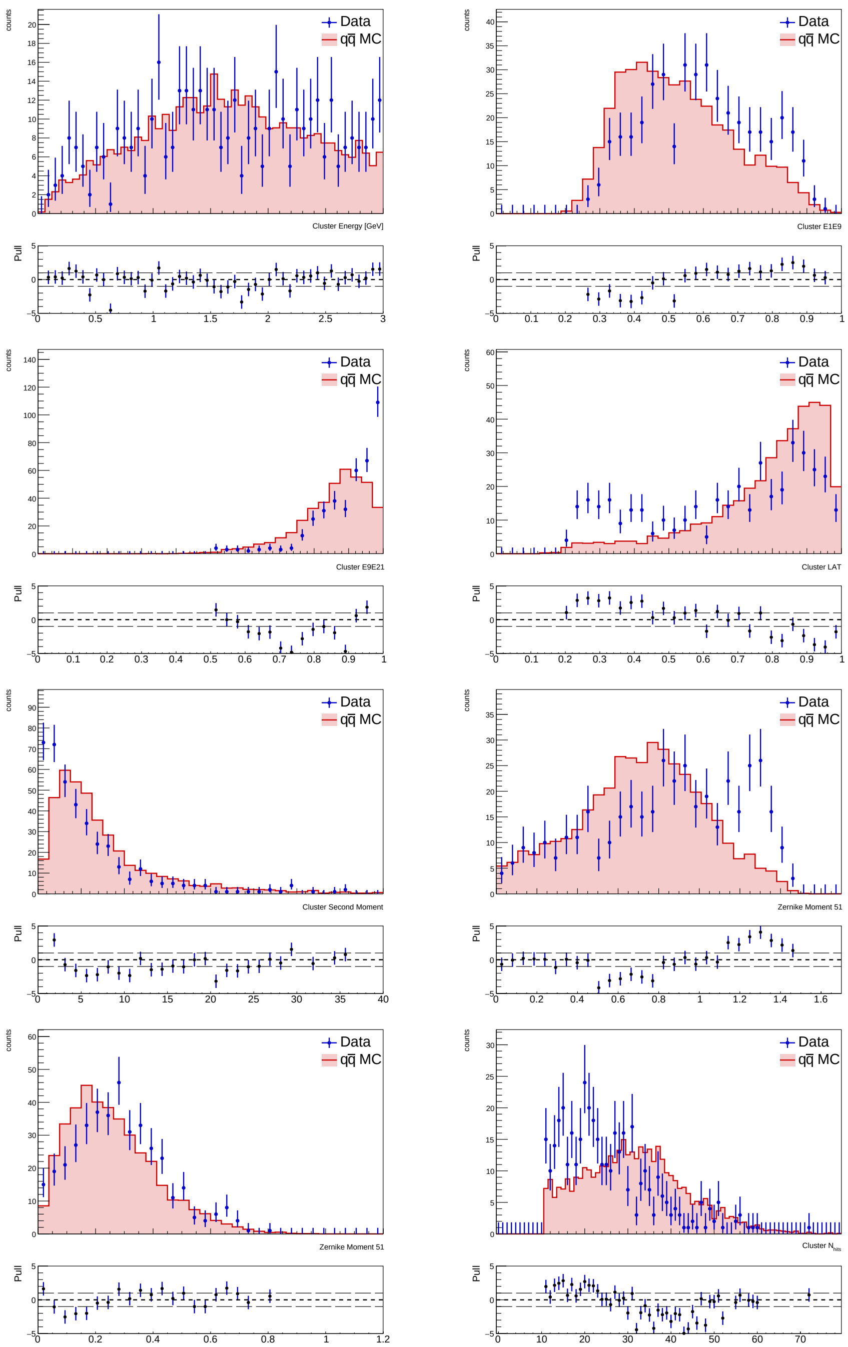


Figure 7.5: Data/MC agreement for reconstructed cluster variables for $\bar{n}$ candidates, in NO ISR case.

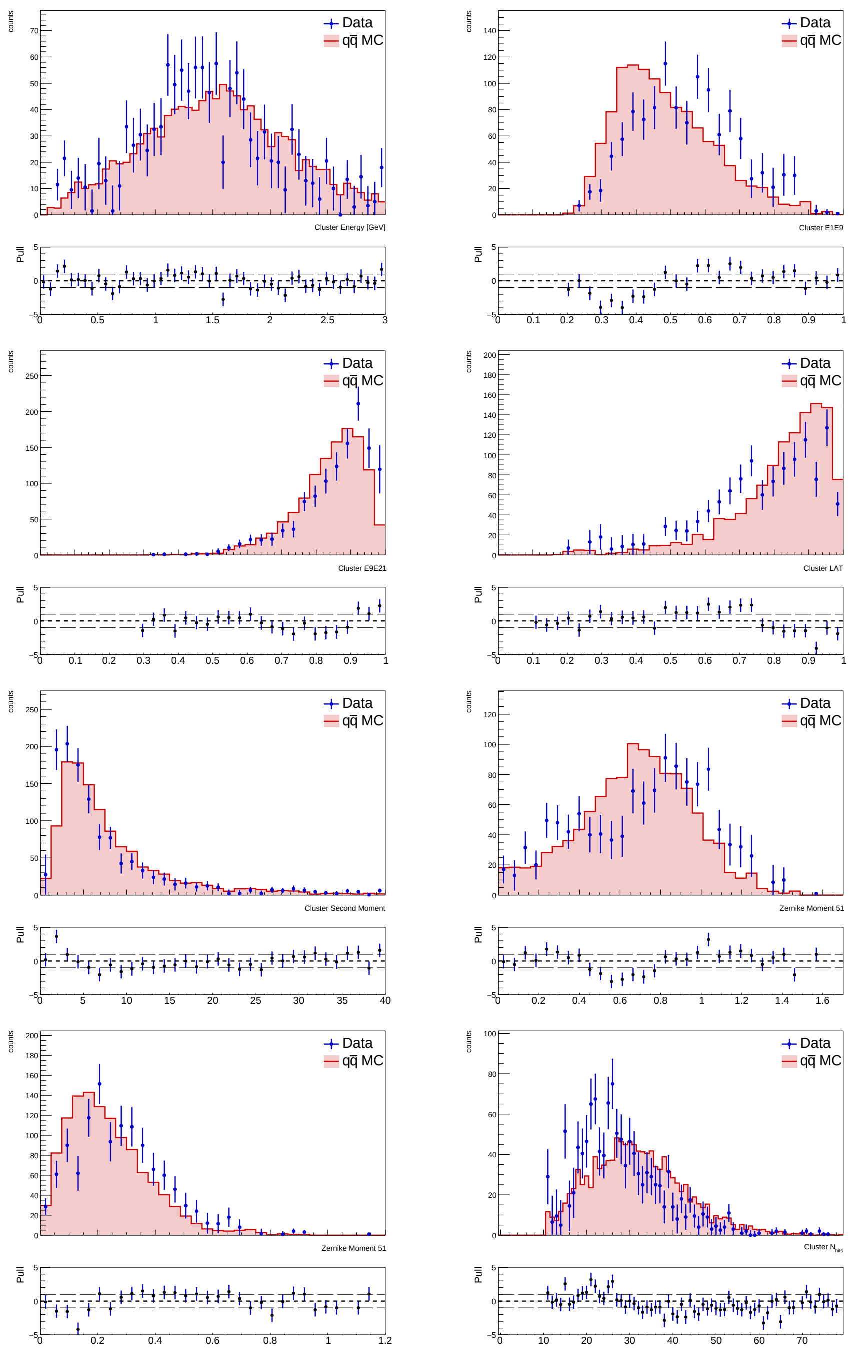


Figure 7.6: Data/MC agreement for reconstructed cluster variables for $\bar{n}$ candidates, in ISR case. Sidebands technique applied. Signal region: [0.8, 1.4] GeV; sidebands region: [0.2, 0.8] U [1.4, 2.0] GeV.

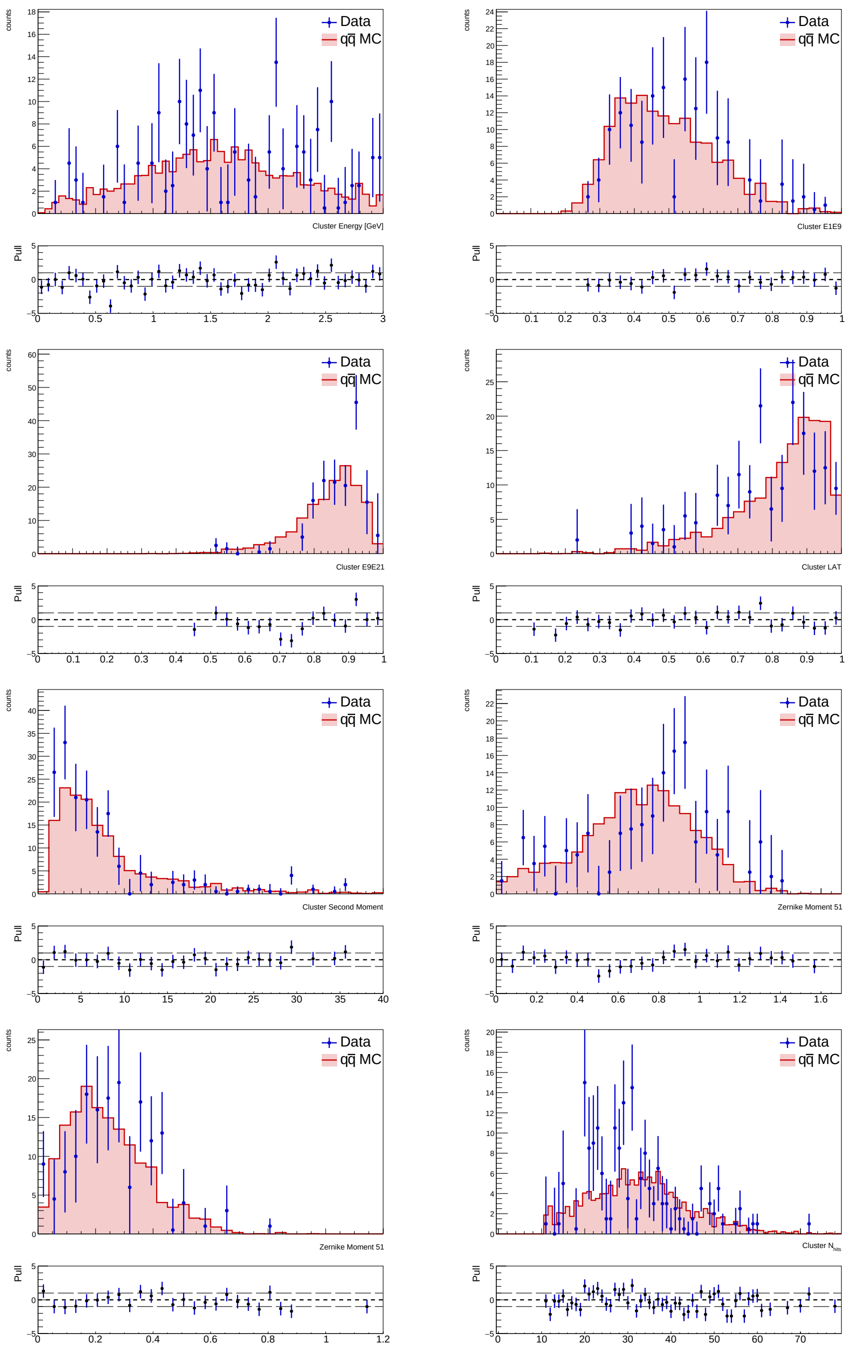


Figure 7.7: Data/MC agreement for reconstructed cluster variables for $\bar{n}$ candidates, in NO ISR case. Sidebands technique applied. Signal region: [0.8, 1.2] GeV; sidebands region: [0.3, 0.7] U [1.3, 1.7] GeV.

Chapter 8
Conclusion

In this thesis, the $\bar{n}$ hadronic showers have been studied, with particular focus on their simulation by the Belle II software. The analysis targets the channel $e^+e^- \to p\pi^-\bar{n}$, considering both ISR and NO ISR cases.

The Monte Carlo signal sample study showed good agreement between the recoil kinematic variables and the generator-level $\bar{n}$ quantities, but the reconstructed $\bar{n}$ momentum proved unreliable for estimating the true $\bar{n}$ momentum. Furthermore, a 1C kinematic fit improved the recoil resolution.
Energy loss effects in primary $\bar{n}$ candidates -such as cluster split-off within the ECL and pre-showering before the ECL- have been studied. Secondary particles (mainly photons) are often misidentified as primary clusters, creating small-sized clusters that distort cluster variable distributions. The distribution of the radial distance ρ of all $\bar{n}$ candidates was studied, confirming that secondary particles are produced throughout the Belle II detector -particularly in the TOP sub-detector (pre-showering) and ECL sub-detector (cluster split-off).

The $q\bar{q}$ cocktail study revealed the recoil mass distribution components from all cocktail channels. Therefore, additional cuts were applied, including those on the number of charged tracks in the Rest of Event (ROE) and on the α angle. A performance metric study using the Figure of Merit $\frac{S}{\sqrt{S+B}}$ showed that the optimal upper limit on α is 0.1, achieving $\epsilon_{sig} = 90\%$ and $\epsilon_{bkg} = 68\%$ signal-background discrimination.
Furthermore, the sidebands technique was applied to obtain the $\bar{n}$ candidate cluster variables from the $q\bar{q}$ cocktail. Before subtraction, several structures -such as those observed in E1E9, E9E21, cluster Lateral Momentum, and cluster Second Moment- were present. These suggested small-sized electromagnetic showers, which were excluded after sidebands subtraction, leading to cluster variables close to those obtained in the signal sample analysis.

In conclusion, Data/MC agreement was assessed using pull methods. Before sidebands subtraction, cluster variable comparisons showed discrepancies beyond 2σ uncertainties between data and Monte Carlo. After subtraction, improved agreement was observed, though low statistics suggest further studies with more prevalent channels, such as multi-pion final states. The global results agree with other Belle II studies on neutral particle hadronic interactions, suggesting that further investigation of GEANT4 Physics lists for anti-neutron hadronic interactions is needed.

Chapter 9
Appendix

9.1 Signal sample analysis scripts

The scripts related to decay files, generation files and steering files are presented in the appendix A. Only the key highlights of the complete macros are reported, in order to ensure a smoother reading.

9.2 The decay file and the generation steering file

Monte Carlo (MC) truth is widely used in High Energy Physics (HEP) experiments. It allows the generation of datasets that follow a specific physical process, propagating particles through the detector and mimicking its behaviour layer by layer. These datasets are processed like real data during the reconstruction phase, with the advantage that the true information from the MC generator is known. The more realistic the MC simulation, the better the understanding of the real dataset.
The production of a MC sample is divided in three main steps:

- The generation of the MC particles with a generator
- The simulation of the detector structure and response to a particle (propagation)
- The reconstruction of tracks, cluster etc.

The **decay file** represents the first step to generate events specifying the decay channels of interest. It is, as the name suggests, the script in which the decay channel of a specific particle is explicitly set to a desired mode. It allows to impose several constraints on particle decay modes, in order to study a specific detector response or the kinematics of a given particle. This is called in jargon "generating signal MC events". The related file has *.dec* extension.
An example from section 3 is explained below:

```
Decay vpho
1.0 p+ anti-n0 pi- PHSP;
Enddecay

End
```

Figure 9.1: Decay file example powered by Evt_Gen

The declaration of a decay begins with the keyword *Decay* and ends with *Enddecay*. In this case, the dummy particle *vpho* is costrained to decay into a proton (p+), an antineutron (anti-n0) and a negative pion (pi-) according to the Phase Space Model (PHSP), with branching ratio set to 1.0. Since the channel is the following one

$$e^+ \ + \ e^- \to p \ + \ \pi^- \ + \ \bar{n} \ (+ \ \gamma_{ISR})$$

in order to produce this specific channel with ISR emission, both *PHOKHARA* and *Evt_Gen* generators are employed. The first one is acting as ISR generator by generating:

$$e^+ \ + \ e^- \to vpho \ (+ \ \gamma_{ISR})$$

Finally, the virtual photon is decayed by Evt_Gen, according to the shown above decay file:

$$vpho \to p \ + \ \pi^- \ + \ \bar{n}$$

To assign a particle the same final states as its charge-conjugated partner (assuming charge conjugation), the *CDecay* identifier can be used, followed by the charge-conjugated state. However it is not employed in the present case. The **generation steering file** is the operative script that generates a MC sample. An example is given below:

```
9   main = b2.Path()
10
11  # Define number of events (-n number -o output in terminal)and experiment number
12  main.add_module('EventInfoSetter', expList=[0])
13
14  print("****** CHANNEL e+ e- (isr) --> vpho --> p+ pi- nbar ****** ")
15  final_state = ['p+', 'pi-', 'anti-n0']
16  decfile = "../dec_file/dec_isr.dec"
17  #Generate isr events with phokhara_evt_gen
18  ge.add_phokhara_evtgen_combination(path=main, final_state_particles = final_state, user_decay_file = decfile,)
19
20
21  main.add_module('PrintMCParticles', onlyPrimaries = False)
22
23  # Simulate the detector response
24  si.add_simulation(path=main)
25
26  # Reconstruct the objects
27  re.add_reconstruction(path=main)
28
29  mdst.add_mdst_output(path=main)
30  # Process the steering path
31  b2.process(path=main)
32
33  # Finally, print out some statistics about the modules execution
34  print(b2.statistics)
```

Figure 9.2: Usage of PHOKHARA&Evt_Gen, where simulation and reconstruction are performed

The basf2 scripts are structured as modules, which are sequentially arranged within a path, which is called *main* in this example (row9). The first module (row 12) sets the desired amount of events and the specific experiment number, which is provided via keyboard input prior to script interpretation. Then, the signal from the decay file is added (row 18), with **add_phokhara_evtgen_combination** from generators basf2 modules [1]

[1] https://software.belle2.org/development/sphinx/generators/doc/index.html

From row 24 to 27, the simulation and the reconstruction, as described in section 3.3.1, are added to the path. Then the output file is created to store the MC truth, namely the generated event with a specific signal sample. Finally the steering path is processed in row 31.

9.3 Cluster variables in ISR case

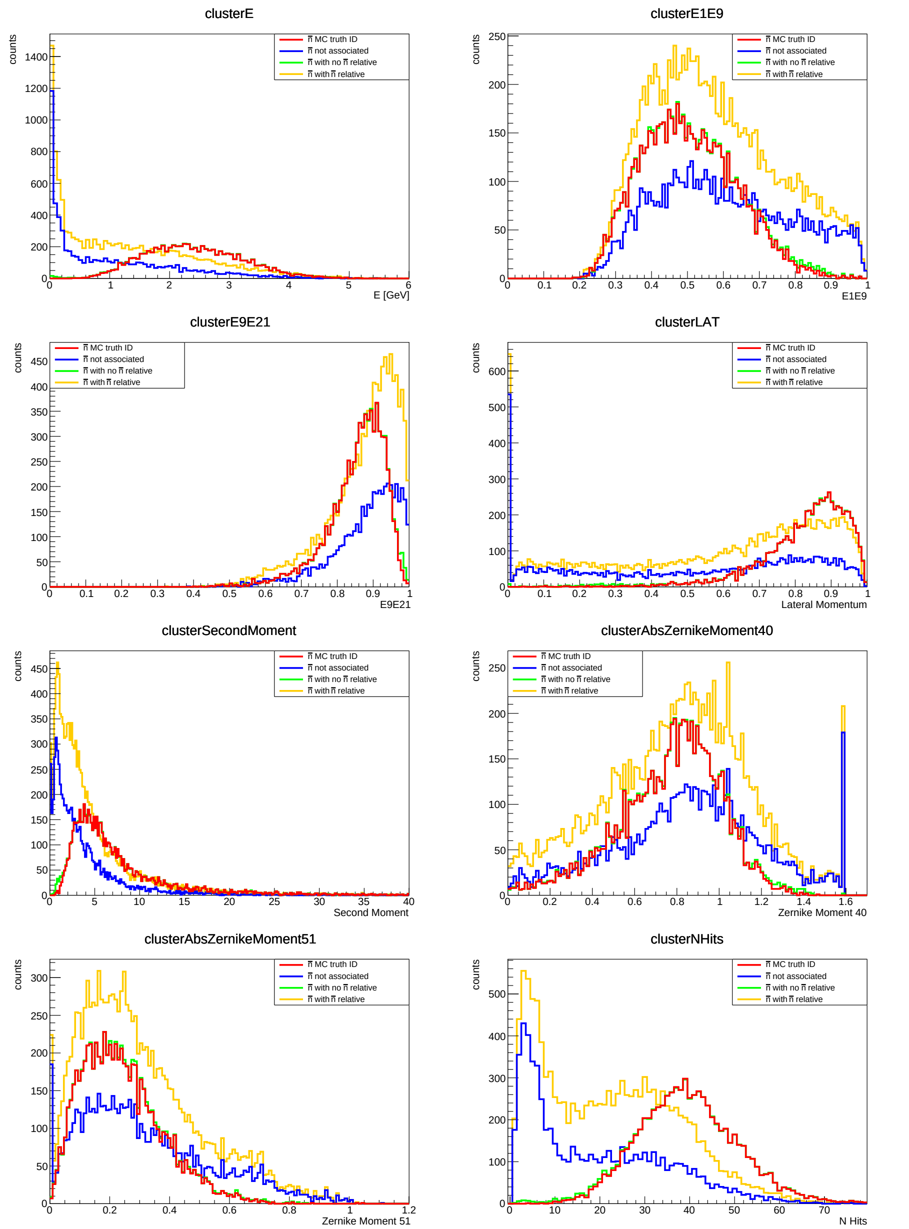


Figure 9.3: Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (no ISR case). Red: $\bar{n}$ candidates associated with true $\bar{n}$ by Monte Carlo matching; blue: $\bar{n}$ candidates not associated with any particle (NaN); green: $\bar{n}$ candidates without $\bar{n}$ relatives; yellow: $\bar{n}$ candidates with $\bar{n}$ relative. The third contribution (green) is barely visible as it lies beneath the first distribution (red). ISR case.

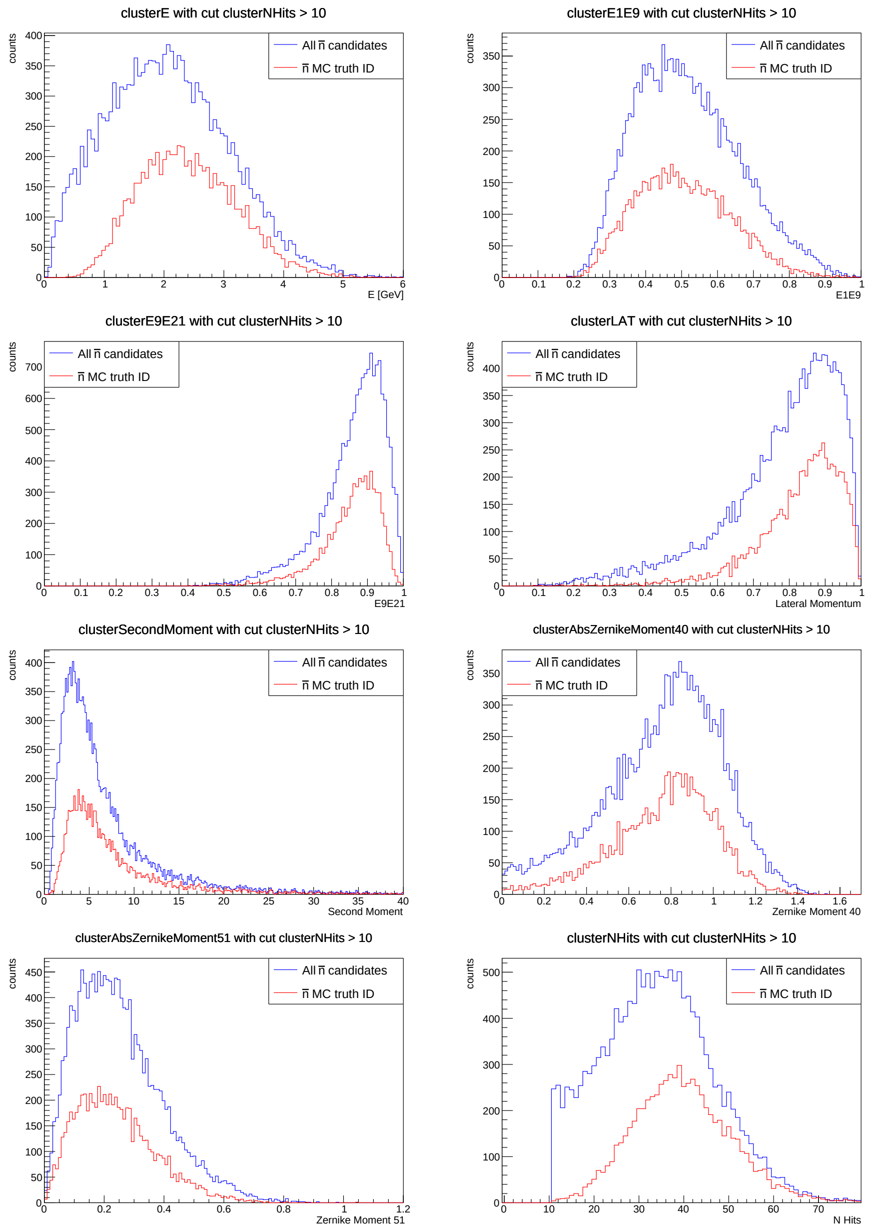


Figure 9.4: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates (blue) and those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (red) (ISR case). Cut $clusterNHits > 10$ is performed.

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